GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We study finite positive memory in prepared causal experiments and the cost frontier obtained by optimizing over every controller of a given size. For arbitrary finite-dimensional quantum failure instruments, a fixed hidden model in a compact family and positive call costs, regularization by returns to the counted initial label gives bounded global cost witnesses without adding memory or resetting the physical state. It yields a primitive continuity modulus uniform in all memory budgets, including changes of reachable support. We prove the square-root exponent optimal by an exact leading asymptotic for two scalar models and two live labels; the converse covers all stochastic controllers. We also characterize exact finite-memory attainment of the history optimum in imperfect search with any finite number of locations and arbitrary positive call costs. Such attainment occurs if and only if all logarithmic failure ratios are rational. Otherwise every finite budget has an attained strict gap. The necessity covers all stochastic controllers, including partially commensurate families, through rigidity of their projected output. Explicit least-favourable prior bounds yield exponential approximation in the label count and an effective finite synthesis cap for commensurate rational data. Classical search ordering, saddle and cyclic-search ingredients are proved in full and explicitly attributed. Exact support certificates, attained variational principles and algebraic synthesis organize the general frontier. Reported reset instruments admit a stronger scalar normal form and arbitrarily long exact resource hierarchies. Shared preparations give common permutation forms and joint Fourier memory laws. The causal comparison, predictive compression, finite-order measurement, two-stage reset, communication, rank-constrained discrimination, duality and local spectral geometry retain their separate hypotheses and complete proofs.

1. Introduction

A predictive state records the information in an experimental past that can affect future laws. Its linear dimension measures one form of compression. Implementing those laws with finite positive memory also requires stochastic transitions, a common decoder and an admissible initialization. Implementing an optimal policy introduces a further question: how much does the best attainable cost change when the device is allowed one additional persistent label? Predictive dimension, the number of physical states and the number of optimal decision values need not answer that question.

This paper relates predictive, convex and spectral structure to finite positive implementations of prepared causal experiments. A controller has one fixed parameter-independent program. It sees only the declared reports, and retains no uncounted history, clock or private seed. Persistent labels, the protected physical state, program-table bits and working arithmetic storage are separate resources. A prescribed pure initial label is counted. Freely prepared mixed starts are identified whenever that different convention is used.

The central question is now both quantitative and structural: how stable is the optimized memory frontier under perturbation of the physical experiment, and when does it reach the unrestricted-history optimum at a finite budget? Same-label regularization proves a general continuity theorem, uniform in the number of labels and valid through changes of reachable support. We prove its

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square-root exponent optimal, with an exact leading coefficient in a fully optimized two-label example. For any finite number of imperfect search locations and arbitrary positive costs, a complete arithmetic criterion distinguishes finite saturation from strict gaps at every finite budget, including families with both commensurate and incommensurate pairs. A quantitative construction and an exact rational-data test supplement this criterion. The reset subclass, shared-preparation realization theory and original causal comparison retain their stronger or independent structures; their quantifiers and mathematical scopes remain distinct.

Regularization without additional memory or physical reset. Let $\Phi^{\theta}_{a,y}$ be the completely positive nonterminal branches of a finite instrument, with commands a, reports y, and one hidden model θ fixed throughout the experiment. The initial density matrix is arbitrary. Success terminates immediately; no other stopping is allowed. Suppose the mean call-cost operators satisfy $c_*I \preccurlyeq G^{\theta}_a \preccurlyeq c^*I$ with $c_* > 0$. A common autonomous M-label controller has command rows p^k and updates $u^{k,a,y}_{\ell}$, and starts at the counted label 1. Write V_M for its least worst-model, worst-initial-state expected cost.

Suppose a controller has cost at most b in this sense. After each failure replace its update by

$$u^{k,a,y,\varepsilon}_{\ell} = (1-\varepsilon)u^{k,a,y}_{\ell} + \varepsilon \mathbf{1}_{\{\ell=1\}}.$$

The physical state is unchanged. Theorem 4.1 proves, whenever $\varepsilon b < c_*$, an initial cost bound and a global continuation bound

$$U_\varepsilon = \frac{b}{1-\varepsilon b/c_*}, \qquad K_\varepsilon = \frac{c^*}{\varepsilon} + U_\varepsilon.$$

The latter applies from every physical state at every label, even when the original controller had inaccessible dark labels. Both estimates hold uniformly over the fixed hidden model. The proof uses finite restart generations and the all-initial-state guarantee, without presuming the finiteness it is to establish.

Measure instrument distance by half the trace-norm distance of its normalized output, including success, and measure cost distance in operator norm. For two feasible compact model families at mutual distances at most (η, ζ), let B bound their common one-row fallback costs. For sufficiently small distances, with explicit conditions in Theorem 4.3,

$$\sup_{M\geq 1} |V_M(\mathcal{U}) - V_M(\mathcal{V})| \leq \frac{10B^2 + 4Bc^*}{c_*}\sqrt{\eta} + \frac{4B}{c_*}\zeta. \tag{1.1}$$

If every command has success effect at least κI, the stronger Corollary 4.5 replaces the right side by $\zeta/\kappa + c^*\eta/\kappa^2$. Scalar geometric-waiting examples prove local sharpness of both coefficients. This additional success hypothesis is not imposed on the general theorem. The same M is used on both sides. The argument transfers the bounded global witnesses of the regularized controller, rather than enlarging memory to record a longer history. Reachable supports may change and unregularized continuation potentials may diverge. The inequality also holds for unrestricted visible-history values, which are the limits of V_M with an explicit counted-prefix approximation.

A sharp singularity of the optimized frontier. The square-root exponent in (1.1) is sharp even when the physical dimension, the number of models and the memory budget are fixed. Consider two scalar models, commands a, b, unit call cost and one failure report. Their success probabilities are respectively $(1-\varepsilon, 0)$ and $(0, 1/4)$. Put

$$r_0 = \frac{\sqrt{65}-1}{32}, \qquad D = \frac{1}{4} + \frac{3}{4}r_0, \qquad c_0 = \frac{\sqrt{3}\, r_0^2}{D^2 + r_0^2/4} > 0.$$

Theorem 5.1 proves

$$V_2(\varepsilon) = \frac{1+\sqrt{65}}{2} + c_0\sqrt{\varepsilon} + o(\sqrt{\varepsilon}). \tag{1.2}$$

The primitive discrepancy is exactly ε, the costs do not change, and one fixed mixed command row has cost at most 6 throughout $0 \le \varepsilon \le 1/2$. This is an optimized two-label value, with all stochastic command and update rows admitted. The one-label value $V_1(\varepsilon) = 1/(1-\varepsilon) + 4$ is smooth at zero.

The proof first establishes an exact positive-polynomial inequality and a unique effective boundary optimizer, including the opposite orientation of the two command rows. Its quantitative rigidity forces the escape rate from a formerly inaccessible dark label to have the scale $\sqrt{\varepsilon}$. A lower bound for all six controller parameters matches an explicit repairing program. This closes the exponent question for the general primitive class. It does not assert sharpness of the general upper constant or contradict the Lipschitz law under a positive per-command success floor.

When finite memory reaches the history optimum. Let $n \ge 2$, $r_i \in (0,1)$ and $c_i > 0$. In hidden model i, only command i can succeed, with probability $q_i = 1 - r_i$; every command j costs c_j in every model. There is one failure report. The hidden model is selected once and is not given to the controller. Put $\tau_i = c_i/q_i$, $B = \sum_i \tau_i$, $u_i = -\log r_i$ and $H = \sum_i 1/u_i$. Lemma 7.1 and Theorem 7.4 prove

$$V_1 = B, \qquad (\exists M < \infty :\ V_M = V_{\mathrm{hist}}) \quad \Longleftrightarrow \quad \frac{\log r_i}{\log r_j} \in \mathbb{Q} \quad \text{for every } i, j. \tag{1.3}$$

Otherwise every finite M has an attained strict gap, although $V_M \downarrow V_{\mathrm{hist}}$. Rational primitive examples already occur for two locations with failure probabilities $1/2$ and $1/3$.

The greedy search ordering, interior preparation saddle, mixture of at most n consistent-priority words and common-power cycles are classical [95, 96]; Lemma 7.2 supplies a complete proof of the ingredients needed here. For the converse, partition commands by rational commensurability of their failure logarithms. Different classes have only finitely many cross-class ties. Every optimal word therefore has the same deterministic class suffix, with class frequency ratios that are irrational. A finite stochastic controller emitting a deterministic class suffix forces that suffix to be ultimately periodic, by a recurrence on positive label supports. This excludes all finite stochastic implementations, including randomized updates and mixtures within each class. It does not assume that an arbitrary finite Markov chain has rational frequencies.

On the positive side, finite logarithmic cosets give ultimately periodic consistent-priority words. Their optimal mixture is implemented from a counted pure initial label through the sampled command and its conditional update; every persistent branch label is charged. The sufficiency combines classical cyclic search with this resource accounting.

The quantitative conclusions do not require a known saddle prior. Lemma 7.5 gives an explicit primitive-data lower bound $\pi_i \ge \delta > 0$ for every least-favourable prior. If $C = \max_i \tau_i/(\delta \min_i \tau_i)$, $A = C \max\{C, 1/\min_i r_i\}$ and $\kappa = e^{-1/H}$, then Theorem 7.6 gives

$$0 \le V_M - V_{\mathrm{hist}} \le BA\kappa^{\lfloor M/n \rfloor}, \qquad M \ge n.$$

At most n counted finite prefixes and one common fallback realize the bound. For rational primitive data, signed prime-exponent vectors decide saturation exactly. In the affirmative case an explicit sufficient label cap reduces the history value to inherited fixed-memory algebraic synthesis (Corollary 7.7). Neither a minimal cap, an efficient elimination algorithm nor a matching lower approximation rate is claimed. The full two-location argument and its sharper normalized constants remain in Section S1 of the supplement.

Exact certificates, model quantifiers and an algebraic frontier. For fixed controller and model let T_γ^θ be its nonterminal map on M blocks of d-dimensional matrices, and let K_γ^θ be its cost operator. The inherited support theory is restated with a complete proof in Theorem 3.3. A finite cost bound b is equivalent to projections P and a positive supported potential W with

$$P_1 = I, \quad P\mathsf{T}(P)P = \mathsf{T}(P), \quad W = PWP \succeq PKP + P\mathsf{T}^*(W)P, \quad W_1 \preccurlyeq bI.$$

The exact robust quantifier order is $\exists\gamma\,\forall\theta\,\exists(P^\theta, W^\theta)$. The common controller is implementable; model-dependent proof witnesses are not additional observations. Theorem 3.4 gives attainment for every M and extends the finite-model algebraic certificate to compact semialgebraic families with unrestricted legal feedback. No general efficient elimination algorithm is claimed. The support and potential construction, finite-model synthesis and compactness mechanism are inherited from Theorems S36.2 and S46.1 and Corollary S36.3; they are not counted as separate new discoveries.

For two fixed hidden models, two commands and a single failure report, Proposition 8.1 determines every budget:

$$V_1 = 2, \qquad V_2 = \frac{1+\sqrt{5}}{2}, \qquad V_M = \frac{3}{2} \quad (M \geq 3). \tag{1.4}$$

Its two-label optimizer has a dark label that is inaccessible in one model and essential in the other. The example tests both the support quantifier and the charge for a prescribed pure initial label. A separate positive-probability example shows that deterministic report-only routing of a fixed row palette can fail for a fixed hidden model, and has a known-model nonreset quantum realization. Thus the following reset normal form has a precise structural boundary.

The complete reset memory frontier. Let $a \in \mathcal{A}$ be a command and $i \in \{1, \dots, s\}$ a failure report. The physical failure branch is

$$\mathcal{I}_{a,i}(\rho) = \operatorname{tr}(E_{a,i}\rho)\sigma_i, \qquad E_{a,i} \succeq 0, \quad \sum_i E_{a,i} \preccurlyeq I.$$

The report itself prepares the known state σ_i. The observer does not receive an additional state measurement or free reset. A successful call must terminate and failure forbids termination. The mean call cost is $\operatorname{tr}(G_a\rho)$ with $c_* I \preccurlyeq G_a \preccurlyeq c^* I$ and $c_* > 0$; a realized cost is not an extra observation. The initial density matrix is arbitrary and unknown. Write V_M for the least worst-initial-state expected cost over autonomous controllers with at most M live labels and a prescribed pure initial label.

Absence of a closed failure set is necessary and sufficient for a finite-cost controller, even with one randomized command row (Theorem 9.1). Put

$$Q_{iaj} = \operatorname{tr}(E_{a,j}\sigma_i), \qquad c_{ia} = \operatorname{tr}(G_a\sigma_i).$$

For a row palette $p^1, \dots, p^M \in \Delta(\mathcal{A})$ arising from a controller with finite worst-initial-state cost, consider the finite decision problem whose actions are exactly these rows. Its least Bellman potential is

$$x_i = \min_{1 \leq k \leq M} \sum_a p_a^k \left(c_{ia} + \sum_j Q_{iaj} x_j \right).$$

Theorem 10.1 proves that every finite-cost controller is dominated, separately at every initial density matrix, by one with the same rows, the same initial row p^1, and deterministic report-only routing $i \mapsto \kappa(i)$. The argument uses the all-initial-state requirement at an essential point: starting the original controller with its fixed initial label from each σ_i gives a finite comparison policy for every state of the row-palette problem. It does not assert transience of unreachable pairs in the original product of physical and controller states.

Let C_0 be an explicit finite worst-state cost of a full-support one-row controller. Theorem 10.2 gives

$$\begin{aligned} V_M = \min\{ b : {} & p^k \in \Delta(\mathcal{A}), \quad \kappa : \{1,\dots,s\} \to \{1,\dots,M\}, \quad x \in [0, C_0]^s, \quad b \in [0, C_0], \\ & x_i \geq \sum_a p_a^{\kappa(i)} (c_{ia} + \sum_j Q_{iaj} x_j) \quad (1 \leq i \leq s), \\ & \sum_a p_a^1 (G_a + \sum_i x_i E_{a,i}) \preccurlyeq bI \}. \end{aligned} \tag{1.5}$$

The minimum is attained and each feasible point constructs an actual controller. Conversely, every cost bound below C_0 realized by any stochastic controller gives such a point. Only s scalar continuation potentials are needed, independently of M; no sM-state product resolvent is imposed. Singular effects and arbitrary Bellman ties are allowed. For exact algebraic primitive data, this compact finite union of semialgebraic sets decides every cost threshold and gives an exact minimizer. No polynomial running time or executed general elimination is asserted.

Proposition 10.3 proves continuity of each V_M in the admissible primitive instrument. The lower argument uses the bounded potentials and compact row palettes; the upper argument uses stability of a fixed proper realizing controller. Thus a strict finite resource gap persists under sufficiently small perturbations, including perturbations that remove commutation among effects and cost operators.

There are exact growing-dimensional instances. For any nonempty proper action subsets $A_1, \dots, A_s \subset \{1, \dots, q\}$, the inherited rational construction has $E_{a,i} = (r/s)I$ and costs in $[1, 4]$. Theorem 10.4 computes V_M by a finite minimum over at most $M - 1$ continuing actions and a nonempty initial support. Its optimum has pure continuing rows and an initial row uniform on that support. This purity conclusion is for this displayed family; the general normal form preserves arbitrary mixed command rows. For $A_i = \{i\}$, $s = q = n \geq 2$ and $r = 1/2$, the full formula becomes

$$V_M = 4 + \frac{1}{n} + \frac{(n-1)\max\{n - M + 1, 0\}}{n^2}, \qquad M \geq 1. \tag{1.6}$$

It gives $V_1 > V_2 > \cdots > V_{n+1} = V_*$ in dimension $2n$, with one exact decrement $(n-1)/n^2$ at every step. In this family the full gap is exactly twice the older first-residual certificate D_M. The distinction between a lower certificate and the true constrained optimum is therefore quantitative, not only terminological.

The optimum threshold and positive exposure. The unrestricted reset optimum is the spectral minimum $V_* = \min_p \lambda_{\max}(\sum_a p_a L_a)$, where $L_a = G_a + \sum_i v_i E_{a,i}$ and v is the primitive Bellman optimum. When one attaining dual preparation ρ_* satisfies $\operatorname{tr}(\rho_* E_{a,i}) > 0$ for every command a and report i, the optimal-action sets $\mathcal{A}_i^*$ give a simpler exact memory threshold. Write $\mathcal{P}_*$ for the optimal initial rows, $I(p) = \{i : \operatorname{supp} p \subseteq \mathcal{A}_i^*\}$, and $t(J)$ for the least number of actions meeting all $\mathcal{A}_i^*$ with $i \in J$. The inherited Theorem 11.1 states

$$m_* = 1 + \min_{p \in \mathcal{P}_*} t(\{1, \dots, s\} \setminus I(p)). \tag{1.7}$$

Positive definite effects are a direct sufficient exposure hypothesis. The explicit semidefinite example has a one-label optimum that this cover formula would overcount without exposure. It remains a valid counterexample to removing that hypothesis; it is also correctly handled by the new general frontier. The arbitrary-cover realization and its NP-hardness consequence are inherited unchanged. Their unrestricted value may be simple even though memory synthesis is hard.

Shared preparations and causal response realization. Fix a finite group G and finite response laws $b_i : G \to \Delta(C)$. All preparations must use one command program and one decoder; the decoder does not receive the preparation identity for free. For closed convex constraints on every command word, Theorem 14.1 gives a common permutation core, including exact charges for prescribed pure initializers. For the distinct target tables $\mathcal{B}$, let P range over G-invariant polytopes in the full response simplex containing $\mathcal{B}$, and let $v(P)$ count their vertices. Then Theorem 14.2 gives

$$\kappa_G(\mathcal{B}) = \min_P v(P), \qquad m_G(\mathcal{B}) = \min_P \{v(P) + |\mathcal{B} \setminus \operatorname{vert} P|\}. \tag{1.8}$$

The polytope may extend beyond the target span. These are different resource conventions: a freely prepared mixed row and a prescribed pure device label have different initialization costs.

Horizontal concatenation of the preparation Fourier blocks gives the joint, potentially noncyclic response module. With a common group mean, its permutation multiplicity cover is the exact

sufficiently small positive-contrast memory, with an explicit simultaneous positive decoder. Weighted singular values give robust lower certificates. Exact families on the fixed group C_2 require $2q$ shared labels although every preparation separately needs two, or have mixed memory two and unbounded pure initialization overhead. Unequal means have a nonnegative-rank obstruction. The full finite-group normal form, nonabelian multiplicity, quotient duality, and the displayed complete symmetric-group mean–contrast law retain their proofs and scopes.

Operational approximation and the qubit boundary. Finite causally prepared heralds followed by delayed terminal tests give a robust nonnegative-factorization obstruction without requiring a physical density matrix at each hidden simulator label. The decoder may have arbitrary stochastic test columns, but cannot see an uncounted copy of the earlier history. Theorem 15.2 first controls joint herald and test events, and only then conditions. For the original qubit phase $z = (3 + 4i)/5$, Theorem 15.3 implies

$$\delta \le \frac{25\zeta}{128 \cdot 200^J J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge \min\{m : \tbinom{m}{\lfloor m/2 \rfloor} \ge J\}, \qquad \zeta = \min\{1, 7B/8\}. \tag{1.9}$$

At each fixed $B > 0$ the unrestricted lower is of order $\log\log(1/\delta)$; the retained constructive upper is logarithmic. The matching leading logarithm for physical density dictionaries keeps its Born-decoder and badly-approximable-phase hypotheses. These rates concern stopped response approximation, not the original general-M squared-sign decision-risk frontier. That separate problem remains open and is not replaced by the exact reset cost hierarchy.

Classical ingredients, priority and the proof architecture. Bellman verification and stochastic-shortest-path theory are classical [81]. Finite stochastic controller search, bounded policy iteration and nonlinear optimization of fixed controller size have an established literature [82, 83, 86]. Parameter synthesis, robust undiscounted goal costs and fixed hidden models with common finite-state controllers are also established [87, 88, 93]; recent algebraic complexity results further delimit effectivity claims [94]. Quantum reachable supports and expected-cost observables have their own prior theory [89, 90, 91, 92]. Continuity under model perturbation has an established discounted control theory [98]. The present estimates concern undiscounted success stopping with a fixed counted memory budget and possibly changing reachable support. Neither writing a nonlinear program nor invoking real algebraic elimination is claimed as a new discovery. The general step here is the same-label regularization and its uniform-in-memory stability across changing reachable supports. The new sharpness theorem establishes the exact worst-case exponent within that class, rather than restating its upper estimate. The inherited compact risk enclosure in Theorem S32.3 uses a finite-prefix completion whose memory can grow; it does not supply this fixed-memory comparison. For the reset subclass, the structural content is simultaneous domination with the same counted rows, removal of all stochastic routing for the all-state reset problem, the s-potential compact certificate in the undiscounted model, and the full exact resource law and stable hierarchy.

Residual automata, positive realization, representation theory, equivariant factorizations and antichains retain their classical attribution [44, 79, 48, 49, 54, 61, 62, 75, 67, 85]. In classical search, an infinite pure search sequence and a finite autonomous stochastic implementation are different strategy classes [97]. The search saddle and common-power cyclic ordering are attributed to their existing theory. The additional claims submitted here are all-stochastic necessity through projected commensurability classes, explicit primitive-data resource bounds, and counted implementation with an effective cap. Set-cover hardness is used through its explicit physical reduction [84]. Targeted primary-source comparisons distinguish these ingredients from the theorem statements; they do not certify exhaustive priority or independent editorial significance.

The general frontier, stability and sharpness chain uses primitive feasibility, reachable-support cost verification, and same-label restart. It needs no group or phase-arithmetic assumption. The

search saturation proof uses the stated scalar family, a classical greedy saddle and a finite support-set obstruction; it is not a universal instrument classification. The reset frontier depends on the experiment semantics and the finite reset existence theorem. It does not require the group realization or qubit arithmetic results. The shared-preparation chain uses the finite-group convex normal form and its Fourier premises. The approximation chain uses operational factorization and phase separation. The general actual-history comparison and paid construction remain in the main article; their lower witnesses use the competitor's actual histories. The supplement preserves full independent proofs for finite-order quantum measurement, two-stage reset, memory and communication, rank-constrained discrimination, performed-law compactness, variational and dual attainment, spectral memory and local acquired geometry. The R58 companion remains independently compilable. No model-specific value formula is promoted to a solution of arbitrary causal feedback, global geometry, or the independent A1/A2 programme.

2. Prepared kernels, acquired task laws and observers

All hidden-state, action and report spaces are standard Borel. A fixed unknown index $\theta \in \Theta$ specifies an actual preparation and positive normalized causal kernels. Positivity means nonnegative measures, not an everywhere strictly positive density. Dependence on the entire physical past may be included in the hidden state. A legal policy sees past visible reports and its admitted randomization, but not θ or a future audit. Iteration of these kernels defines its finite-prefix laws; the consistent countable path law is their kernel extension.

Each call has a cost $c_i \geq c_* > 0$. The decision to make the next call is predictable. In the geometric route, the declared productive-call flag is also predictable; in the primitive-instrument route, productive reports are specified outcomes and can be random. Commitment is permitted only after a productive report, and occurs exactly once almost surely in each model. At that time a specified future audit $Z \in H$ is scored, where $H \subset \mathbb{R}^k$ is nonempty, compact and convex. The prediction is committed before that audit is observed. A prospective joint report–audit kernel is defined at every productive call, including rejected calls. The stopping coin is independent of the audit conditional on the entire current visible report and the past. Write $L = \operatorname{diam}(H)^2$; the case $L = 0$ is trivial whenever a legal rule exists.

The core cost C_τ includes initial preparation charged to the core, failed calls and all productive calls through commitment. A separately executed final audit must be added to the raw budget. This convention is fixed before taking a risk infimum. Physical time, stored program constants and temporary computation are not identified with call count.

Three resource interfaces. A checkpoint procedure may use its whole precommit history, but must output one element of a fixed dictionary of at most M values. This counts its output width only; it is not a claim of $\log_2 M$ total working bits.

An autonomous observer has an alphabet S, $|S| \leq M$, and one fixed parameter-independent program. At an intercall boundary its only persistent observer memory is its label $s \in S$. Its next command distribution is a fixed function of s. A physical port name, elapsed time, subprogram index, random seed or former report is not an extra free input. During a call its current command, current report and fresh coins are available. Whatever persists to another call must be encoded in S. A fixed decoder $r : S \to H$ reads only the committed label.

In the *event-stopped* interface a post-report kernel emits $(s', e) \in S \times \{0, 1\}$. An event $e = 1$ ends the experiment and retains s' for the decoder; no subsequent command is executed. This public commitment event is not a stored bit used in continuing computation. The same label may be used live on one path and terminal on another. In the *state-stopped* interface a fixed map $h_* : S \to \{0, 1\}$ determines halt status. The initial label is live, and type-one labels are globally absorbing terminal labels with no executed command rows. Entry into a terminal label is allowed only after a productive

report. Thus live and terminal types are disjoint in this interface, and only in this interface by definition.

Write $R_i^{\Theta}(N, M)$, $i \in \{\mathrm{cp}, \mathrm{ev}, \mathrm{st}\}$, for the infimum of worst-case expected squared loss over programs with robust almost-sure commitment and $\sup_\theta \mathbb{E}_\theta C_\tau \le N$. An empty legal class has value $+\infty$. A program chosen once at random is not a free persistent seed. The ideal measurable class may have fixed real constants; effective implementation is stated separately in Section S42.

A physically acquired comparison law. Execute a declared prepared comparison experiment and obtain its joint report–audit probability $K_0(dx, dz)$ on $X \times H$. Its report marginal is ν. The law is genuinely preparable and its specification has its own resource account. It need not be a member of Θ. In particular no finite set of calibration data is silently promoted to a uniform certificate. Set

$$f(x) = \int z K_0(dz \mid x), \quad w(x) = \int \|z - f(x)\|^2 K_0(dz \mid x), \quad d_r(x) = w(x) + \|f(x) - r\|^2. \tag{2.1}$$

Disintegration gives Borel versions with $f \in H$; null sets may be completed arbitrarily. The profile is

$$\mathcal{Q}_m(b) = \min_{\substack{C \subset H,\ 1 \le |C| \le m \\ 0 \le h \le b,\ \int h\, d\nu = 1}} \int [w + \mathrm{dist}(f, C)^2] h\, d\nu, \qquad b \ge 1,\ m \ge 1. \tag{2.2}$$

We use $+\infty$ for an infeasible cap or width. The quantities f, w are task coordinates of acquired predictive information, not a complete substitute for the predictive quotient. Their derivation and the quotient construction in Appendix S43 precede any use of local dimension.

Lower certificates on actual histories. Fix *one* actual model $\theta_* \in \Theta$, independent of the competing program. At each actual reached productive pre-call history h and legal command a, write $P_{h,a}(dy, dz)$ for its conditional full current report–audit law. Let $P^0_{h,a}$ be a measurable comparison row on the same report space, with a Borel mark $x = \pi_{h,a}(y)$ of law ν and

$$\mathbb{E}^0[Z \mid y] = f(\pi_{h,a} y), \qquad \mathbb{E}^0[\|Z\|^2 \mid y] = w(\pi_{h,a} y) + \|f(\pi_{h,a} y)\|^2. \tag{2.3}$$

These are conditional identities given the *full* report, not just its mark. Versions and inequalities are required on actual reachable histories, not only almost everywhere under an unrelated comparison path law.

Fix $0 < l_- \le 1 \le u_- < \infty$. For all Borel $q : Y \to [0, 1]$ and all finite-image Borel $r : Y \to H$, suppose predictable nonnegative defects $a_-(h, a), e_-(h, a)$ obey

$$\int q\, d(P_{h,a})_Y \le u_- \int q\, d(P^0_{h,a})_Y + a_-(h, a), \tag{2.4}$$

$$\int q(y)\|z - r(y)\|^2 P_{h,a}(dy, dz) \ge l_- \int q(y)\|z - r(y)\|^2 P^0_{h,a}(dy, dz) - e_-(h, a). \tag{2.5}$$

Independent current randomization can be adjoined to both rows: integrating inequalities for each fixed random value proves the same inequalities with the same defects. All command-conditioned lower inequalities hold before averaging the chosen command.

Let A_τ count productive calls. At this same actual model require, for every legal competing procedure,

$$\mathbb{E}_* C_\tau \ge \kappa_- \mathbb{E}_* A_\tau, \qquad \kappa_- > 0. \tag{2.6}$$

At budget N suppose, uniformly over all robustly feasible competitors,

$$\mathbb{E}_* \sum_{j \le \tau:\ j \text{ productive}} a_-(H_{j-1}, a_j) \le A_N, \qquad \mathbb{E}_* \sum_{j \le \tau:\ j \text{ productive}} e_-(H_{j-1}, a_j) \le E_N. \tag{2.7}$$

These are actual occupation certificates, not properties merely of the proposed upper program. Uniform row defects give $A_N \le (N/\kappa_-)\bar{a}_-$ and $E_N \le (N/\kappa_-)\bar{e}_-$. Lemma S40.6 supplies a nonuniform alternative.

One common upper construction. Independently, let $\mathcal{G}$ be a parameter-independent program which, from each of its start histories, reaches one productive report. Its conditional expected attempt cost is at most $\kappa_+ > 0$ on every surviving history in every $\theta \in \Theta$. It uses B live roles and returns to its start label on rejection; no reset of the physical state is assumed. It forms a virtual mark x, a current choice J and a permitted export $T_J r$ of its task readout. For every finite dictionary C assume

$$|\{T_j r : j \in J,\ r \in C\}| \le D_0|C|, \qquad D_0 \in \mathbb{N}. \tag{2.8}$$

Every continuing phase and coin needed for this construction is included in the B live roles. For fixed $0 < l_+ \le 1 \le u_+ < \infty$ and $a_+, e_+ \ge 0$, assume for all Borel $q : X \to [0,1]$ and finite-image Borel $r : X \to H$, conditionally on every such history,

$$\mathbb{E}[q(x)] \ge l_+ \int q\, d\nu - a_+, \tag{2.9}$$

$$\mathbb{E}[q(x)\|Z - T_J r(x)\|^2] \le u_+ \int q(x) d_{r(x)}(x)\nu(dx) + e_+. \tag{2.10}$$

The expectation here includes the within-attempt choice J. Unlike the lower, the upper is not required to hold for each choice separately. The choices retained after this call are nevertheless charged through the actual export or live role. These hypotheses express one common causal construction, not modelwise optimizers.

3. The finite-memory frontier beyond reported resetting

Reported resetting permits the scalar and memory-preserving reduction of Theorem 10.1. The next result does not assume resetting. It permits arbitrary finite-dimensional failure maps and an unknown model which remains fixed during the experiment. The appropriate finite certificate retains the counted controller labels and their reachable quantum supports. In particular, the quantum state need not be determined by the latest report.

The support construction and its potential equations are inherited from Lemma S36.1, Theorems S36.2 and S46.1; finite-model algebraic synthesis is Corollary S36.3. The primitive fallback is already supplied by Theorem S26.1 and Lemma S32.1. We give their positive-cost form in full to make the new stability argument self-contained. The extension here permits a compact semialgebraic model family with unrestricted legal feedback. Reachability and cost observables are established ingredients [89, 90, 91, 92], as are classical bounded-controller and robust synthesis [86, 87, 88, 93]. The next section supplies the new comparison that preserves the exact memory budget.

3.1. Primitive instruments and counted controllers.

Fix a Hilbert space $\mathcal{H}$ of dimension $d \ge 1$, a nonempty finite command set $\mathcal{A}$, and a finite nonterminal report set $\mathcal{Y}$. Let Θ be a nonempty compact parameter space. For every $\theta \in \Theta$ and $a \in \mathcal{A}$, the nonterminal branches are completely positive maps $\Phi^\theta_{a,y} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ satisfying

$$B^\theta_a = I - \sum_{y \in \mathcal{Y}} (\Phi^\theta_{a,y})^*(I) \succeq 0.$$

The effect B^θ_a is the probability of success; success requires immediate stopping and no other stopping is permitted. The mean call-cost operator is G^θ_a, with common constants

$$0 < c_* I \preccurlyeq G^\theta_a \preccurlyeq c^* I < \infty. \tag{3.1}$$

The maps and costs depend continuously on θ. All commands are legal, and the only observations are the sampled command, the declared report, and success. The realized cost is not an additional observation. The physical implementation of these maps and costs is part of the instrument specification, as in Section 9.

The adversary chooses one pair (θ, ρ), consisting of a model and an initial density matrix. This pair is not observed by the controller. The same θ governs every subsequent call. Thus the model is not reselected after a report, and the observed history can contain information about it.

An autonomous controller with at most $M \geq 1$ live labels is specified by command rows and update rows

$$p^k \in \Delta(\mathcal{A}), \qquad u^{k,a,y} \in \Delta([M]), \qquad 1 \leq k \leq M. \tag{3.2}$$

Its prescribed initial label is 1. At label k it samples a from p^k; after nonterminal report y it writes label ℓ with probability $u_\ell^{k,a,y}$. These are fresh random choices, not an uncounted persistent seed. Unused labels and duplicated rows are allowed. Denote this compact product of simplexes by $\mathcal{C}_M$.

For a controller $\gamma = (p, u)$, work in the direct-sum matrix algebra

$$\mathfrak{A}_M = \bigoplus_{k=1}^{M} \mathcal{B}(\mathcal{H}), \qquad \langle W, X \rangle = \sum_k \operatorname{tr}(W_k X_k).$$

The nonterminal evolution and call-cost operator are

$$(\mathsf{T}_\gamma^\theta X)_\ell = \sum_{k,a,y} p_a^k u_\ell^{k,a,y} \Phi_{a,y}^\theta(X_k), \tag{3.3}$$

$$K_{\gamma,k}^\theta = \sum_a p_a^k G_a^\theta. \tag{3.4}$$

The map T_γ^θ is completely positive and trace-nonincreasing. Let $\iota_1(\rho) = (\rho, 0, \ldots, 0)$. The total expected cost and its robust value are

$$J_\gamma(\theta, \rho) = \sum_{n=0}^{\infty} \langle K_\gamma^\theta, (\mathsf{T}_\gamma^\theta)^n \iota_1(\rho) \rangle, \tag{3.5}$$

$$J(\gamma) = \sup_{\theta \in \Theta,\ \rho \succeq 0,\ \operatorname{tr}\rho = 1} J_\gamma(\theta, \rho), \qquad V_M = \inf_{\gamma \in \mathcal{C}_M} J(\gamma). \tag{3.6}$$

All summands are nonnegative. Consequently this definition is unambiguous even for an improper controller. Finite cost implies finite expected duration by (3.1).

3.2. **A primitive feasibility criterion.** For a fixed model, choose any Kraus representations $\Phi_{a,y}^\theta(X) = \sum_\mu A_{a,y,\mu}^\theta X (A_{a,y,\mu}^\theta)^*$. A nonzero subspace $S \subseteq \mathcal{H}$ is *closed and dark* if

$$A_{a,y,\mu}^\theta S \subseteq S \quad \text{for all } a, y, \mu, \qquad B_a^\theta|_S = 0 \quad \text{for all } a. \tag{3.7}$$

The condition is representation-independent: it says that every command has zero success probability on S and all its failure branches preserve S.

Theorem 3.1 (Feasibility from arbitrary initial quantum states)**.** *The following conditions are equivalent.*

1. *There is one visible-history policy with finite uniform expected cost for every (θ, ρ).*
2. *No model $\theta \in \Theta$ has a nonzero closed dark subspace.*
3. *Every fixed full-support command distribution $\bar{p}$, used independently on every call with one live label, has finite uniform expected cost for every (θ, ρ).*

In particular, feasibility for unrestricted history and feasibility at every finite live-label budget coincide. For such a $\bar{p}$ set

$$\mathsf{F}^\theta = \sum_{a,y} \bar{p}_a \Phi_{a,y}^\theta, \qquad \bar{G}^\theta = \sum_a \bar{p}_a G_a^\theta.$$

Then $r(\mathsf{F}^\theta) < 1$ for every θ, and an explicit uniform cost bound is

$$C_0 = \max_{\theta \in \Theta} \lambda_{\max}\Big[(I - (\mathsf{F}^\theta)^*)^{-1} \bar{G}^\theta\Big] < \infty. \tag{3.8}$$

Proof. If (3.7) holds, an initial density matrix supported on S remains there after every failure and has zero success probability at every call, regardless of the policy. Its cost is infinite by the positive floor. This proves the necessity of the second condition.

For sufficiency fix θ and write $\mathsf{F} = \mathsf{F}^\theta$. It is positive and trace-nonincreasing, so its spectral radius is at most one. One direct justification is that, for Hermitian X, $-\|X\|I \preccurlyeq X \preccurlyeq \|X\|I$ and positivity bounds $\mathsf{F}^n(X)$ by $\|X\|\mathsf{F}^n(I)$ in order; the latter has trace at most d. Decomposing a general matrix into its real and imaginary Hermitian parts shows that the powers are bounded.

Suppose its spectral radius is one. The numbers $\operatorname{tr}(\mathsf{F}^n(I))$ decrease. Their limit is positive: otherwise the same order bounds would make F^n tend to zero on every matrix, contradicting the existence of an eigenvalue of modulus one. The positive Cesàro averages

$$Z_N = \frac{1}{N}\sum_{n=0}^{N-1} \mathsf{F}^n(I)$$

have bounded trace and a subsequence converging to $Z \succeq 0$ with $\operatorname{tr} Z > 0$. Since $\mathsf{F}(Z_N) - Z_N = (\mathsf{F}^N(I) - I)/N$, one has $\mathsf{F}(Z) = Z$.

Put $S = \operatorname{ran} Z$. Taking traces in this equality gives $\sum_a \bar{p}_a \operatorname{tr}(B_a^\theta Z) = 0$. Every term is nonnegative and every $\bar{p}_a$ is positive, so $B_a^\theta|_S = 0$ for every a. The identity $\mathsf{F}(Z) = Z$ is a sum of positive Kraus images. Pairing it with a vector in $\ker Z$ shows, term by term, that every Kraus image has range in S; equivalently $A_{a,y,\mu}^\theta S \subseteq S$. This constructs a nonzero closed dark subspace. Therefore its absence forces $r(\mathsf{F}^\theta) < 1$.

The geometric operator series converges and gives the cost operator in (3.8). The primitive maps and costs are continuous, and the inverse is continuous wherever it exists. Compactness of Θ therefore makes the displayed maximum finite. This single row works for all models and all initial density matrices, proving the third condition and hence the first. □

The all-initial-state quantifier is material to the necessity: a particular supplied initial state might never reach a closed dark subspace. The theorem also uses the fixed finite alphabet and the availability of every command at every call.

3.3. **Reachable corners and exact cost witnesses.** For fixed θ, γ, abbreviate $\mathsf{T} = \mathsf{T}_\gamma^\theta$ and $K = K_\gamma^\theta$. A tuple of orthogonal projections $P = (P_1, \dots, P_M)$ defines the matrix corner $P\mathfrak{A}_M P = \bigoplus_k P_k \mathcal{B}(\mathcal{H}) P_k$. Call it invariant if

$$P_1 = I, \qquad P\mathsf{T}(P)P = \mathsf{T}(P). \tag{3.9}$$

Multiplication in this formula is blockwise. By positivity, this identity implies that T maps the entire corner into itself. Indeed every positive element of the corner is bounded above by a scalar multiple of P, and general elements follow by linear decomposition.

Lemma 3.2 (Finite reachable-support construction)**.** *The least invariant corner containing the initial label is obtained by taking the blockwise support of*

$$R = \sum_{n=0}^{Md-1} \mathsf{T}^n \iota_1(I). \tag{3.10}$$

In particular it is determined after finitely many linear operations and support computations, without an infinite-history enumeration. If $J_\gamma(\theta, \rho)$ is finite for every initial density matrix, the restriction of T to this corner is transient and its cost operator is finite on every block of the corner.

Proof. Let $R_n = \sum_{j=0}^n \mathsf{T}^j \iota_1(I)$ and let $P^{(n)}$ be its blockwise support. These supports increase, beginning with total rank d and having total rank at most Md. If no rank is added at a step, then $\mathsf{T}(R_n)$ is supported on $P^{(n)}$. The operator R_n is positive definite on this support, so $P^{(n)}$ is bounded by a finite multiple of R_n. Positivity now implies (3.9), and all later supports coincide.

There can be at most $Md - d$ strict enlargements, so (3.10) has already reached the least invariant corner. Every invariant corner with first block I contains all the summands, proving minimality.

For the last assertion, finite cost for every initial density matrix implies finite cost for $\iota_1(I)$. For each fixed n, the total cost starting from $\mathsf{T}^n\iota_1(I)$ is a tail of that nonnegative series and is finite. Hence the total cost starting from R is finite. On its support there is an $\varepsilon > 0$ with $R \succeq \varepsilon P$. Positivity makes total cost from P finite as well.

The positive cost floor then gives $\sum_{n\geq 0} \operatorname{tr}(\mathsf{T}^n(P)) < \infty$. Order domination by P shows that the restricted powers tend to zero on every element of the corner. In finite dimension their spectral radius is strictly less than one: an eigenvalue of modulus at least one would contradict this convergence. Thus the restricted geometric series and its adjoint converge. The finite positive cost operator is

$$W = \sum_{n=0}^{\infty} (\mathsf{T}_P^*)^n(PKP), \qquad \mathsf{T}_P^*(Z) = P\mathsf{T}^*(Z)P.$$

This proves the assertion on every reachable block, including those whose exposure probability is small. □

Theorem 3.3 (Exact finite support and Lyapunov certificate)**.** *Fix M, a controller γ, a model θ, and $b < \infty$. Its cost is at most b for every initial density matrix if and only if there exist block projections P and Hermitian block operators W satisfying*

$$P_k = P_k^* = P_k^2, \qquad P_1 = I, \qquad P\mathsf{T}_\gamma^\theta(P)P = \mathsf{T}_\gamma^\theta(P), \tag{3.11}$$

$$W = PWP \succeq 0, \qquad W \succeq PK_\gamma^\theta P + P(\mathsf{T}_\gamma^\theta)^*(W)P, \tag{3.12}$$

$$W_1 \preccurlyeq bI. \tag{3.13}$$

The projections may be chosen as the least reachable supports in Lemma 3.2. With that choice the minimal positive cost witness satisfies equality in (3.12)*, and* $\operatorname{tr}(W_1\rho)$ *is the actual expected cost from ρ.*

Proof. Suppose first that the certificate holds. Invariance keeps every nonterminal state, starting from $\iota_1(\rho)$, inside its corner. Pair (3.12) with such a state and iterate. For every N,

$$\sum_{n=0}^{N-1} \langle K_\gamma^\theta, (\mathsf{T}_\gamma^\theta)^n \iota_1(\rho)\rangle + \langle W, (\mathsf{T}_\gamma^\theta)^N \iota_1(\rho)\rangle \leq \operatorname{tr}(W_1\rho) \leq b.$$

Drop the nonnegative remainder and use monotone convergence. This gives the asserted total-cost bound. Its positive cost floor also bounds the expected number of calls by b/c_*, so success occurs almost surely.

Conversely, the assumed uniform initial bound makes the construction in Lemma 3.2 applicable. Take its least support P and its convergent cost series W. The series is supported on P and solves the Lyapunov equation there. Pairing it with any initial density matrix gives exactly (3.5); hence $\operatorname{tr}(W_1\rho) \leq b$ for every ρ, which is equivalent to (3.13). Its minimality among positive supersolutions follows by iterating the inequality from zero. This establishes both directions without postulating finite costs at inaccessible labels or subspaces. □

3.4. **Attainment and exact synthesis at every memory budget.**

Theorem 3.4 (Robust quantum finite-memory frontier)**.** *Assume the equivalent feasibility conditions of Theorem 3.1. Then for every $M \geq 1$ the value V_M in* (3.6) *is finite and attained by a controller in $\mathcal{C}_M$. It is nonincreasing in M. For every finite b, the following is a necessary and sufficient condition for $V_M \leq b$:*

$$\exists\, \gamma \in \mathcal{C}_M \quad \forall\, \theta \in \Theta \quad \exists\, P^\theta, W^\theta \quad (3.11)\text{–}(3.13) \text{ hold.} \tag{3.14}$$

The controller variables are common to all models. The certificate variables may depend on the fixed model; they are proof witnesses and are not supplied as information to the controller.

Suppose in addition that $\Theta \subseteq \mathbb{R}^r$ is a compact semialgebraic set defined over the real algebraic numbers, and the real and imaginary matrix coefficients of the primitive maps and costs are polynomial or rational functions of θ with algebraic coefficients and no poles on Θ. For each fixed M, feasibility at any algebraic budget is decidable by finite real-algebraic elimination. The value V_M is algebraic, and one attaining controller can be chosen with algebraic command and update probabilities and computed by a finite exact procedure.

Proof. The one-row policy of Theorem 3.1 can be represented in $\mathcal{C}_M$, giving $V_M \le C_0$. For fixed θ, ρ, N, the truncated sum in (3.5) is a continuous function of the finitely many controller probabilities. The full robust objective J is the supremum of these continuous functions over θ, ρ, N. It is therefore lower semicontinuous on the compact space $\mathcal{C}_M$. Its finite minimum is attained. This argument does not require a uniform upper bound for the potential on every label. Padding by unused labels gives monotonicity.

The fixed-model certificate theorem applies to the same controller at every θ. This proves both directions of (3.14); attainment identifies the budget condition with $V_M \le b$, including $b = V_M$.

For the effective statement, express all complex matrices in real and imaginary coordinates. Projection identities, corner invariance and controller normalization are polynomial identities or inequalities. Positive semidefiniteness is expressed by the nonnegativity of all principal minors of each Hermitian block. Rational coefficients introduce no difficulty, since their denominators have no zeros on the specified model set and may be handled by polynomial equations for their inverses. Thus (3.14) is a finite first-order formula over the real algebraic numbers. Its quantifier order is exactly the displayed order, so elimination does not give the controller access to θ.

Elimination produces the feasible budget set. It is the nonempty ray $[V_M, \infty)$ by the budget equivalence and attainment. A finite endpoint of a one-dimensional semialgebraic set defined over the real algebraic numbers is algebraic. Substituting this endpoint in the quantified formula leaves a nonempty semialgebraic set of controller probabilities defined over the same field. Exact sampling supplies algebraic probabilities in that set. This is a finite exact procedure; no polynomial complexity, numerical tie certification, or previously executed general elimination is claimed. □

For a finite model set, (3.14) is simply a finite family of support and cost certificates with shared controller variables. For a compact infinite set its universal quantifier is essential. The theorem does not assert that a least-favorable model attains the outer supremum, that V_M has a general closed formula, or that the unrestricted-history optimum is reached at a finite label budget. It also does not identify finite algebraic synthesis with efficient computation.

With the single constant terminal decoder, event stopping keeps the same total-label frontier V_M. State stopping uses one disjoint terminal label, giving the live-label frontier V_M at $M + 1$ total labels. These counts include the prescribed initial label.

3.5. **Why fixed models and reachable supports cannot be removed.**

Example 3.5 (Information remains after a physical reset)**.** Let $d = 1$, let there be a single nonterminal report, and let $\Theta = \{0, 1\}$. Each call costs one. The success probabilities of commands a, b are respectively

$$(1/2, 1/10) \quad \text{under model 0}, \qquad (1/10, 1/2) \quad \text{under model 1}.$$

All success and failure probabilities are strictly positive, and there is only one physical reset state.

A one-label controller uses one command row, choosing a with probability p. Its two mean costs are

$$\frac{1}{1/10 + (2/5)p}, \qquad \frac{1}{1/2 - (2/5)p}.$$

Their maximum is minimized at $p = 1/2$, with value $10/3$. Now use three counted rows: an initial uniform row, a pure a row, and a pure b row. After the first failure choose the row for the opposite

command, and thereafter alternate the two pure rows. For model 0, starting with a gives mean cost $(1+1/2)/(1-(1/2)(9/10)) = 30/11$; starting with b gives $38/11$. Model 1 exchanges these numbers. Thus the prescribed uniform initial row followed by counted alternation has robust cost $34/11 < 10/3$, an exact improvement of $8/33$.

In contrast, a deterministic update depending only on the common report must always select one of these same three rows. Selecting the uniform row gives cost $10/3$; selecting either pure row after the first call gives worst-model cost $1 + (7/10)10 = 8$. Therefore no such report-only update dominates this controller with its same rows. The palette theorem for one known reset instrument has not failed: its physical resetting hypothesis does not erase information about a fixed hidden model, and that theorem does not include this uncertainty quantifier.

The same example has a known-model, genuinely nonreset quantum realization. On $\mathbb{C}^2$ use single failure Kraus operators

$$A_a = \operatorname{diag}(1/\sqrt{2}, 3/\sqrt{10}), \qquad A_b = \operatorname{diag}(3/\sqrt{10}, 1/\sqrt{2}), \qquad G_a = G_b = I.$$

Both failure operators are invertible, so their posterior states depend on the input. The success effects are $I - A_a^* A_a$ and $I - A_b^* A_b$. All expected-cost operators in the displayed controllers are diagonal. The two basis states therefore give exactly the two scalar cost calculations above, and arbitrary coherences do not change them. In this known instrument the all-initial-state one-label value is again $10/3$, while the three-label construction attains $34/11$. This supplies an explicit full-rank nonreset instance of the larger certificate class.

Example 3.6 (Finite initial cost with unbounded continuation witnesses)**.** Let $d = 1$, $\Theta = [0, 1]$, and keep one nonterminal report and unit cost. The failure probabilities of a, b are

$$f_a(\theta) = \theta(1-\theta), \qquad f_b(\theta) = 1 - \theta.$$

Consider the two-label controller which starts with a and, after failure, uses b forever. At $\theta > 0$ its cost is $1 + \theta(1-\theta)/\theta = 2 - \theta$; at $\theta = 0$ its first call succeeds surely, so its cost is 1. Its robust cost is therefore 2, although no parameter attains that supremum.

For $0 < \theta < 1$, the second label is reachable and its true continuation cost is $1/\theta$, which is unbounded as $\theta \downarrow 0$. At $\theta = 0$ that label is both unreachable and dark. A Lyapunov inequality on the entire two-label space would then demand $W_2 \geq 1 + W_2$, which is impossible. The correct support certificate instead has $P_1 = 1, P_2 = 0$ at this endpoint. At $0 < \theta < 1$ it has both projections equal to one and the exact cost witnesses $W_1 = 2 - \theta, W_2 = 1/\theta$. At $\theta = 1$ the second label is again unreachable; either its zero support or the larger transient support gives a valid certificate.

Every primitive model in this example satisfies the feasibility theorem: command a succeeds with probability at least $3/4$. The obstruction is therefore neither infeasibility of the model family nor a zero cost. It is the unjustified requirement that every unused controller state possess a global finite potential, or that all model-dependent continuation witnesses have a uniform bound merely because the initial cost does.

The product evolution, positive cost operators, and Lyapunov verification have classical antecedents in finite-state controller synthesis and quantum program analysis. The present statement keeps the counted controller, the arbitrary initial state, the fixed-model quantifier, and support changes in one necessary-and-sufficient certificate. Its role is to extend the finite resource frontier beyond reported resetting while preserving the stronger scalar normal form, closed cover solutions and rigidity results in their previously proved reset class. It does not silently extend the report-only normal form or solve the separate squared-sign minimax risk and optional-commitment problems.

4. Memory-preserving regularization and stability

Reachable supports can change at a parameter boundary, and the exact potentials in Example 3.6 need not be uniformly bounded. We now regularize the controller without adding a label or resetting

the physical system. The operation is an occasional return to its already counted initial label. The all-initial-state cost quantifier makes this useful even when the physical posterior and the fixed hidden model remain unknown.

The primitive fallback, centered effect estimate (S32.3), and compact-family completion in Theorem S32.3 are retained predecessors. The additional requirement here is that regularization and perturbation use exactly the existing labels; no finite-history expansion is allowed in the fixed-memory comparison. Classical robust FSC synthesis [88, 93] and quantum cost verification [91, 92] provide the surrounding theory, but the bounds below are proved directly.

4.1. **Returning to the initial controller label.**

Theorem 4.1 (Regularization with the same live-label budget)**.** *Suppose an M-label controller $\gamma = (p, u)$ has robust cost at most $b < \infty$ for the compact model family of Section 3. For $0 < \varepsilon \le 1$ replace only its post-failure update rows by*

$$u_\ell^{k,a,y,\varepsilon} = (1-\varepsilon)u_\ell^{k,a,y} + \varepsilon \mathbf{1}_{\{\ell=1\}}. \tag{4.1}$$

The command rows and prescribed initial label are unchanged. If $\varepsilon b < c_*$, the resulting controller γ_ε has robust initial cost at most

$$U_\varepsilon = \frac{b}{1-\varepsilon b/c_*}. \tag{4.2}$$

Moreover it has uniform cost at most

$$K_\varepsilon = \frac{c^*}{\varepsilon} + U_\varepsilon \tag{4.3}$$

from every physical density matrix at every controller label, for every fixed model. Its exact cost operators can therefore be chosen on the full product algebra and satisfy

$$W^\theta = K^\theta_{\gamma_\varepsilon} + (\mathsf{T}^\theta_{\gamma_\varepsilon})^*(W^\theta), \qquad 0 \preccurlyeq W^\theta_k \preccurlyeq K_\varepsilon I, \qquad W^\theta_1 \preccurlyeq U_\varepsilon I. \tag{4.4}$$

Proof. Realize (4.1) by a fresh independent coin after every failure: with probability ε write label 1, and otherwise use the original update. No physical operation is performed by this coin. The label update already belongs to the declared controller interface.

Start from label 1 with any physical state, and stop a segment at success or at the first successful restart coin. Up to that coin its command and report law is the original controller's law cut at an additional stopping time. Its expected cost is at most b. The original controller has expected duration at most b/c_*. The probability of a restart before success is consequently at most

$$r_\varepsilon = \varepsilon b/c_*.$$

For example this follows by summing, over failure calls, the probability of reaching that call without an earlier restart and then obtaining a successful restart coin; deleting the restriction on earlier restart coins only increases the sum to ε times the original expected number of failures.

After a restart the physical state can be arbitrary, but the controller again has label 1 and the same fixed model. Conditional on every such segment history, the all-initial-state and all-model bound applies again. Thus the probability of reaching segment $j+1$ is at most r_ε^j, and its conditional expected cost is at most b. Summing finitely many segments and then using monotone convergence gives $b\sum_{j\ge 0} r_\varepsilon^j$. This proves (4.2) without assuming the finiteness of the regularized cost in advance.

Now start at any other label and any physical state. Before its first success or restart, the probability of surviving n calls without either event is at most $(1-\varepsilon)^n$: each nonterminal call offers an independent restart coin. The expected number of calls in that first segment is at most $1/\varepsilon$, and its expected cost is at most c^*/ε. Any following segment starts at label 1, whose total remaining cost is at most U_ε. This gives (4.3).

All blocks are now proper at finite cost. The positive cost floor and the order argument in Lemma 3.2 make the full product map transient. Its geometric cost series gives (4.4), and the cost

bounds just proved give the two operator inequalities. The estimate is uniform in the fixed model and does not require the observer to know that model. □

The state of the physical experiment is unchanged at a controller restart. In particular, (4.1) does not add a freely available reset operation to a nonreset instrument. Nor does it add a clock, a stored seed, or another live label.

4.2. **Perturbations of the primitive instrument.** To compare two primitive models θ, ϑ with the same physical and report spaces, include the scalar success outcome and form the trace-preserving output map

$$\mathsf{N}_a^\theta(\rho) = \big((\Phi_{a,y}^\theta(\rho))_{y\in\mathcal{Y}}, \operatorname{tr}(B_a^\theta\rho)\big).$$

The norm on this direct sum is the sum of the block trace norms. Define

$$d_{\mathrm{inst}}(\theta,\vartheta) = \max_a \sup_{\rho\succeq 0,\ \operatorname{tr}\rho=1} \tfrac{1}{2}\|\mathsf{N}_a^\theta(\rho) - \mathsf{N}_a^\vartheta(\rho)\|_1, \tag{4.5}$$

$$d_{\mathrm{cost}}(\theta,\vartheta) = \max_a \|G_a^\theta - G_a^\vartheta\|. \tag{4.6}$$

These distances compare the declared physical maps and mean costs. They make no measurement available to the controller.

Lemma 4.2 (Perturbation of a bounded cost witness)**.** *Fix one counted controller and suppose $0 \preccurlyeq W_k \preccurlyeq KI$ at every live label. If $d_{\mathrm{inst}}(\theta,\vartheta) \le \eta$, then*

$$-\eta K I \preccurlyeq ((\mathsf{T}_\gamma^\vartheta)^* - (\mathsf{T}_\gamma^\theta)^*)(W)_k \preccurlyeq \eta K I \qquad (1\le k\le M). \tag{4.7}$$

If in addition $d_{\mathrm{cost}}(\theta,\vartheta)\le\zeta$ and W is a global supersolution for model θ, then, provided $\delta = \zeta + \eta K < c_*$,

$$\frac{W}{1-\delta/c_*} \tag{4.8}$$

is a global supersolution for model ϑ, whose call costs also satisfy the lower bound $c_* I$.

Proof. Fix a current label k and an initial density matrix ρ. After a sampled command a, regard the block observable on report y as $\sum_\ell u_\ell^{k,a,y} W_\ell$, and put the observable zero on success. Every block lies between zero and KI. The two normalized output states have a difference D of trace zero. Hence, for such an observable Z,

$$|\operatorname{tr}(ZD)| = |\operatorname{tr}((Z - \tfrac{K}{2}I)D)| \le \tfrac{K}{2}\|D\|_1 \le K\eta.$$

Average over the same command row and then range over ρ. The resulting scalar inequalities are exactly (4.7). Including success in the normalized output is what justifies the factor $1/2$; an unnormalized failure-output difference need not have trace zero.

The source supersolution and the cost difference now give

$$W - (\mathsf{T}_\gamma^\vartheta)^*(W) \succeq K_\gamma^\vartheta - \delta I.$$

Put $\alpha = (1-\delta/c_*)^{-1}$. Since $K_\gamma^\vartheta \succeq c_* I$ and $(\alpha-1)c_* = \alpha\delta$, multiplying this inequality by α gives $\alpha W - (\mathsf{T}_\gamma^\vartheta)^*(\alpha W) \succeq K_\gamma^\vartheta$. This proves (4.8). □

For compact families $\mathcal{U},\mathcal{V}$ of primitive models, write $V_M(\mathcal{U})$ and $V_M(\mathcal{V})$ for their respective frontiers. A directed comparison from $\mathcal{U}$ to $\mathcal{V}$ means that for every $\vartheta\in\mathcal{V}$ there is a $\theta\in\mathcal{U}$ with both $d_{\mathrm{inst}}(\theta,\vartheta)\le\eta$ and $d_{\mathrm{cost}}(\theta,\vartheta)\le\zeta$. The same paired models must satisfy both bounds.

Theorem 4.3 (Uniform stability of the complete memory frontier)**.** *Suppose the source family $\mathcal{U}$ is feasible and has common cost bounds c_*, c^*. Suppose the target family $\mathcal{V}$ has the same lower cost bound and admits the directed comparison just specified. Put $b = V_M(\mathcal{U})$. For any $0<\varepsilon\le 1$ satisfying*

$$\varepsilon b < c_*, \qquad \zeta + \eta K_\varepsilon < c_*,$$

where $U_\varepsilon, K_\varepsilon$ are (4.2)– (4.3)*, one has*

$$V_M(\mathcal{V}) \le \frac{U_\varepsilon}{1-(\zeta+\eta K_\varepsilon)/c_*}. \tag{4.9}$$

The target controller has the same M labels as the source optimizer.

In particular assume both families are feasible, have the common cost bounds, have a one-row cost bound B, and admit the comparison in both directions. If $0<\eta\le 1$ and

$$\sqrt{\eta} \le \frac{c_*}{2B}, \qquad \zeta + c^*\sqrt{\eta} + 2B\eta \le \frac{c_*}{2}, \tag{4.10}$$

then, simultaneously for every $M \ge 1$,

$$|V_M(\mathcal{U}) - V_M(\mathcal{V})| \le \frac{10B^2+4Bc^*}{c_*}\sqrt{\eta} + \frac{4B}{c_*}\zeta. \tag{4.11}$$

For $\eta = 0$ the same bound holds with the first term zero whenever $0 \le \zeta \le c_*/2$. Consequently the entire family of finite-memory values is locally uniformly continuous in the primitive model family, including at changes of reachable support.

Proof. Take an attaining source controller, supplied by Theorem 3.4, and apply the label restart of Theorem 4.1. It is one common controller for every source model and has global witnesses bounded by K_ε. For a target model ϑ, choose its paired source model θ only in the proof. The preceding lemma transfers that source witness to a target supersolution. Its initial block is bounded by the right-hand side of (4.9). The controller probabilities do not depend on either member of this pair. Applying the certificate theorem separately to every fixed target model gives one uniform robust bound and proves the directed inequality. This is an offline comparison of complete fixed models; it does not allow a model to change during the experiment or reveal a model to the observer.

For the explicit two-sided estimate set $\varepsilon = \sqrt{\eta}$. Since $b \le B$, the first smallness condition gives

$$U_\varepsilon \le 2B, \qquad U_\varepsilon - b \le \frac{2B^2}{c_*}\sqrt{\eta}, \qquad \delta := \zeta + \eta K_\varepsilon \le \zeta + c^*\sqrt{\eta} + 2B\eta \le c_*/2.$$

It follows that

$$\begin{aligned}\frac{U_\varepsilon}{1-\delta/c_*} - b &\le \frac{2B^2}{c_*}\sqrt{\eta} + \frac{4B}{c_*}\delta \\ &\le \frac{2B^2+4Bc^*}{c_*}\sqrt{\eta} + \frac{8B^2}{c_*}\eta + \frac{4B}{c_*}\zeta.\end{aligned}$$

Use $\eta \le \sqrt{\eta}$ and then apply the same argument in the opposite direction. This proves (4.11). For $\eta = 0$, take $\varepsilon \downarrow 0$ in the directed inequality; the resulting difference is at most $b\zeta/(c_*-\zeta) \le 2B\zeta/c_*$ in each direction, which implies the stated bound.

Finally fix any feasible compact source family. Its full-support one-row maps in Theorem 3.1 have invertible resolvents at every source model. Continuity and compactness give a common neighborhood in which the same row remains proper with a bounded cost, and the positive cost bounds persist locally within the class of valid instruments. Thus a common B, c_*, c^* can be used for all sufficiently close compact families. The quantitative estimate proves the asserted local uniform continuity. No continuity of the optimizing supports, labels, or individual unregularized potentials is required. □

The modulus is a proved upper estimate. No optimality of its square-root exponent is asserted. Its constants do not grow with the controller label budget; their dependence on the physical experiment is displayed through the positive cost bounds and the one-row cost cap. The reset-specific continuity theorem remains available with its stronger scalar structure.

4.3. **Relation to unrestricted visible history.** Let $V_{\mathrm{hist}}(\mathcal{U})$ be the infimum of the robust cost over all visible-history policies with model-independent causal command kernels. Success stopping and the physical observation port are unchanged. The following completion argument identifies the limit of the counted-memory frontier; it does not assert finite saturation.

Corollary 4.4 (Finite-memory approximation to unrestricted control)**.** *For a feasible compact model family with one-row bound C_0,*

$$V_M \downarrow V_{\mathrm{hist}} \quad \text{as } M \to \infty. \tag{4.12}$$

More explicitly, for every integer $T \geq 1$ let

$$M_T = 1 + \sum_{t=0}^{T-1} (|\mathcal{A}||\mathcal{Y}|)^t.$$

Then

$$V_{\mathrm{hist}} \leq V_{M_T} \leq V_{\mathrm{hist}} \left(1 + \frac{C_0}{c_* T}\right). \tag{4.13}$$

The initial label, every retained finite history, and the fallback row are all included in M_T. The uniform primitive stability bound (4.11) *also holds for V_{hist}.*

Proof. Take any visible-history policy of robust cost $b < \infty$. Follow its causal kernels for at most T calls, recording the preceding command/report pairs in the counted labels. On failure at the Tth call, switch to the common one-row policy of Theorem 3.1. A rooted history tree has at most $\sum_{t=0}^{T-1} (|\mathcal{A}||\mathcal{Y}|)^t$ nonterminal prefixes; one additional label implements the fallback row. Thus this is an actual controller with at most M_T labels and a pure initial label. The depth is stored in the history label rather than supplied by a free clock.

The expected prefix cost is at most b. The original policy has expected duration at most b/c_*, so its probability of surviving T calls is at most $b/(c_* T)$. On that event the actual physical posterior is some density matrix and the model remains one fixed member of the family. The common all-initial-state fallback bound therefore makes the additional cost at most $C_0 b/(c_* T)$. The constructed controller has robust cost at most $b(1 + C_0/(c_* T))$.

Take the infimum over the original policies; an attaining unrestricted policy is not assumed. Every finite controller induces model-independent visible-history command kernels. Indeed, conditional on a fixed command/report history, the physical branch likelihood is common to every compatible private label path and cancels from its conditional distribution; only the controller probabilities remain. Arbitrary definitions on zero-probability histories cause no change to a performed law. Thus finite controllers belong to the comparison class, yielding the reverse inequality in (4.13). Monotonicity in the label budget and this approximating sequence prove (4.12). Finally pass to the infimum over M in the two-sided bound (4.11), whose constants are independent of M. This proves its asserted extension to V_{hist}. □

The finite-history truncation and fallback are a completion device within the inherited causal framework. The new stability argument uses a different operation: randomizing the existing label updates to obtain global cost witnesses at the same fixed memory. Neither operation grants the controller a quantum reset, an unknown-state oracle, or knowledge of the fixed hidden model.

4.4. **A uniform success floor: a Lipschitz frontier.** The general modulus allows commands which can fail with probability one at some physical states. A uniform lower success effect gives a stronger estimate, directly from the classical resolvent argument. The statement below records its exact dependence on the primitive success margin and retains the same live-label budget throughout.

Corollary 4.5 (Positive-success refinement)**.** *Let $\mathcal{U}, \mathcal{V}$ be compact model families on the same interface and the positive-cost conventions of Section 3, with common ceiling c^* and paired primitive discrepancy bounds η, ζ in both directions. Assume that for some $0 < \kappa \le 1$,*

$$B_a^\theta \succeq \kappa I \qquad \text{for every command and every model in } \mathcal{U} \cup \mathcal{V}. \tag{4.14}$$

Then, simultaneously for every $M \ge 1$,

$$|V_M(\mathcal{U}) - V_M(\mathcal{V})| \le \frac{\zeta}{\kappa} + \frac{c^*\eta}{\kappa^2}. \tag{4.15}$$

The same estimate holds for the unrestricted visible-history values. There is no small-error restriction in this inequality.

For either individual family, with its one-row fallback bound $C_0 \le c^*/\kappa$ and the counted M_T of Corollary 4.4,

$$0 \le V_{M_T} - V_{\mathrm{hist}} \le C_0(1-\kappa)^T \qquad (T \ge 1). \tag{4.16}$$

This rate is exponential in the history horizon. When $|\mathcal{A}||\mathcal{Y}| > 1$, the displayed history construction uses exponentially many labels in that horizon, so the statement does not give exponential convergence in the label count.

For $0 < \kappa < 1$, the universal local coefficients $1/\kappa$ for cost perturbations and c^*/κ^2 for instrument perturbations in (4.15) *cannot be reduced. One-dimensional, one-command, one-model experiments suffice to show this, at every live-label budget.*

Proof. For any counted controller γ, normalization of its update rows and (4.14) give the block inequality

$$(\mathsf{T}_\gamma^\theta)^*(I) \preccurlyeq (1-\kappa)I.$$

Positivity implies that this adjoint contracts the Hermitian block operator norm by at most $1-\kappa$. Thus every controller, from every label and every physical state, is proper and has exact cost potential

$$W^\theta = (I - (\mathsf{T}_\gamma^\theta)^*)^{-1} K_\gamma^\theta, \qquad 0 \preccurlyeq W^\theta \preccurlyeq (c^*/\kappa)I.$$

The Neumann inverse has norm at most $1/\kappa$ on Hermitian blocks. No label restart or additional controller state is needed.

For paired models θ, ϑ, subtract the two cost equations to obtain

$$W^\vartheta - W^\theta = (I - (\mathsf{T}_\gamma^\vartheta)^*)^{-1}\big[K_\gamma^\vartheta - K_\gamma^\theta + ((\mathsf{T}_\gamma^\vartheta)^* - (\mathsf{T}_\gamma^\theta)^*)W^\theta\big].$$

Lemma 4.2 bounds the bracket by $\zeta + \eta c^*/\kappa$ in Hermitian operator norm. Therefore

$$\|W^\vartheta - W^\theta\| \le \zeta/\kappa + c^*\eta/\kappa^2.$$

This bounds the initial-block expected costs for every density matrix. Taking the worst model using the declared pairings, then the infimum over the same controller class, proves one direction of (4.15); exchange the families for the other direction. The bound is independent of M. Passing to the limit in Corollary 4.4 proves the history assertion.

For the approximation rate, every visible-history policy has conditional success probability at least κ at each call, irrespective of its history, unknown physical state, or fixed model. Its probability of surviving T calls is at most $(1-\kappa)^T$. Truncate a finite-cost competing policy at that horizon and execute the common fallback on its open branches. The expected prefix cost is at most the original cost, and the added cost is at most $C_0(1-\kappa)^T$. The same history tree and one fallback label give exactly the previous count M_T. Taking the infimum over competing policies proves (4.16); an attaining unrestricted policy is not required.

Finally take a scalar experiment with one command, one failure report and mandatory success stopping. Let the source success probability be $\kappa + \eta$ and its deterministic call cost be $C > 0$, while the target success probability is κ and its call cost is $C + \zeta$, where $0 \le \eta \le 1 - \kappa$ and $\zeta \ge 0$.

The instrument distance is exactly η, the cost distance is exactly ζ, and the common ceiling may be taken to be $c^* = C + \zeta$. No controller can affect these geometric waiting times. Their exact difference, at every memory, is

$$\frac{C+\zeta}{\kappa} - \frac{C}{\kappa+\eta} = \frac{\zeta}{\kappa} + \frac{C\eta}{\kappa(\kappa+\eta)}. \tag{4.17}$$

Setting $\eta = 0$ attains the cost coefficient exactly. Setting $\zeta = 0$ and letting $\eta \downarrow 0$ gives the ratio $C/\kappa^2 = c^*/\kappa^2$ for instrument error. This proves the two stated local sharpness assertions. □

The extra hypothesis (4.14) is stronger than absence of a common dark subspace: a jointly feasible detector family may have a zero success effect on some state for each individual command. The corollary applies to the stated positive success class only. Its matched Lipschitz constants do not prove sharpness of the square-root modulus in the general class, and the resolvent method itself is not claimed as a new perturbation principle.

5. Sharpness of the uniform primitive modulus

The exponent in Theorem 4.3 cannot be increased on its stated class. This failure of a Lipschitz estimate already occurs for two scalar models, two commands and two live labels. The values remain finite, the call costs are identically one, and the same one-row fallback works throughout the perturbation. Thus the phenomenon is not caused by a loss of feasibility or by an increasing physical dimension. It comes from the cost of repairing a label which was dark and unreachable in one of the models.

Put $\beta = 1/4$ and $q = 1 - \beta = 3/4$. Let $\mathcal{U}_\varepsilon = \{0_\varepsilon, 1\}$, where $0 \le \varepsilon < 1$, be scalar instruments with unit call cost, one failure report, and the following success probabilities:

	a	b
0_ε	$1-\varepsilon$	0
1	0	β.

(5.1)

The model is fixed throughout the run and is unobserved. Success requires immediate stopping; the sampled command is known to the controller. For a scalar instrument, the displayed probabilities specify all the nonterminal maps. These are valid instruments for every stated parameter, with positive deterministic paid costs and no additional observation.

The controller convention is that of (3.2): a prescribed pure initial live label, stochastic command and update rows, and no free clock or persistent private seed. We write $V_2(\varepsilon)$ for its two-live-label robust optimum. Define

$$r_0 = \frac{\sqrt{65}-1}{32}, \qquad B_0 = \frac{1}{r_0} = \frac{1+\sqrt{65}}{2}, \qquad D = \beta + qr_0, \qquad c_0 = \frac{2\sqrt{q}\, r_0^2}{D^2 + \beta r_0^2}. \tag{5.2}$$

In particular $c_0 > 0$; its approximate value is 0.4564191442.

Theorem 5.1 (An exact square-root singularity at fixed memory)**.** *For the primitive family* (5.1)*,*

$$V_2(0) = B_0, \qquad \lim_{\varepsilon \downarrow 0} \frac{V_2(\varepsilon) - B_0}{\sqrt{\varepsilon}} = c_0. \tag{5.3}$$

The minimum defining each $V_2(\varepsilon)$ is attained. The instrument discrepancy between the paired families $\mathcal{U}_0$ and $\mathcal{U}_\varepsilon$ is exactly ε, and their cost discrepancy is zero. For $0 \le \varepsilon \le 1/2$ they have unit cost floor and ceiling and a common one-row cost bound 6*.*

Consequently no local uniform bound with exponent strictly greater than $1/2$ can replace the instrument term in Theorem 4.3, even at the single fixed budget $M = 2$. In contrast, the one-label frontier of the same family is

$$V_1(\varepsilon) = \frac{1}{1-\varepsilon} + \frac{1}{\beta}. \tag{5.4}$$

The exponent obstruction therefore concerns an optimized memory frontier, rather than a scalar waiting-time singularity.

We prove the theorem below. Two elementary facts about scalar two-label chains will be useful. If the success probabilities at the labels are h_1, h_2 and the probabilities of a nonterminal transition to the other label are t_1, t_2, respectively, the finite cost from label 1 is

$$C_1 = \frac{h_2 + t_2 + t_1}{h_1 h_2 + h_1 t_2 + h_2 t_1}. \tag{5.5}$$

This formula applies when the denominator is positive. When the second label is inaccessible, its possibly infinite cost must instead be discarded: the initial cost is $1/h_1$. These conclusions follow directly from $C_1 = 1 + (1 - h_1 - t_1)C_1 + t_1 C_2$ and its second-row equation. Whenever both costs are finite, their ordering is the reverse of the ordering of h_1, h_2. Indeed

$$(h_2 - h_1)C_1 = (h_2 + t_1 + t_2)(C_1 - C_2).$$

Thus routing a given failure to the label with larger success probability cannot increase the cost when that routing does not affect the other model. This last statement can also be checked by substituting the old positive cost vector into the new Bellman inequality; finite-prefix iteration proves the comparison without an unproved properness assumption.

5.1. **The unperturbed optimum and its rigidity.** Specify every two-label controller by six numbers in $[0, 1]$:

$$\begin{array}{c|l} r, s & \text{probabilities of command } a \text{ at labels } 1, 2, \\ \alpha, \gamma & \text{probabilities of writing label 2 after, respectively, an } a\text{- or } b\text{-failure at label 1,} \\ \xi, \delta & \text{probabilities of writing label 1 after, respectively, an } a\text{- or } b\text{-failure at label 2.} \end{array} \tag{5.6}$$

Label 1 is the initial label. There is no restriction on any of these probabilities. The description includes one-label controllers by allowing the second label to remain unused.

Lemma 5.2 (The unique effective optimal program at zero perturbation)**.** *At $\varepsilon = 0$, every robust optimal two-label controller has*

$$r = r_0, \qquad s = 0, \qquad \alpha = 1, \qquad \gamma = \delta = 0. \tag{5.7}$$

The value of ξ is immaterial because command a is never used at label 2. The optimum is B_0. Every controller with $r \le s$ has robust cost at least

$$B_{\mathrm{opp}} = 2(\sqrt{19} - 2) > B_0. \tag{5.8}$$

The lower bound in (5.8) *is attained within that orientation.*

Proof. In model 0_0, an a call always succeeds. Changing updates after an a-failure therefore leaves this model's cost unchanged.

First suppose $r \ge s$. In model 1 the second label has at least as large a success probability as the first. By the preceding two-label comparison, routing every a-failure to label 2 can only improve model 1. Hence it gives a lower bound for an arbitrary controller. This routing sets $\alpha = 1, \xi = 0$. The endpoint $r = 0$ makes model 0 improper. At $r = 1$ the first call in model 1 is certainly a failure and at least $1/\beta$ further calls are required in expectation, giving cost at least 5. Both endpoints are therefore harmless for a lower bound strictly below 5; in the following division take $0 < r < 1$. Let C_0, C_1 denote the two initial costs after this improvement. For their explicit expressions put

$$\begin{aligned} N_0 &= s + (1 - s)\delta + (1 - r)\gamma, \\ D_0 &= rs + r(1 - s)\delta + s(1 - r)\gamma, \\ N_1 &= \beta(1 - s) + q(1 - s)\delta + r + q(1 - r)\gamma, \\ D_1 &= \beta(1 - s)[\beta + qr + q(1 - r)(\delta + \gamma)]. \end{aligned}$$

When the relevant denominators are positive, $C_0 = N_0/D_0$ and $C_1 = N_1/D_1$. Define

$$f(t) = \frac{\beta + t}{\beta(\beta + qt)}, \qquad 0 \le t \le 1. \tag{5.9}$$

It is strictly increasing. The inequality

$$C_1 \ge f(1/C_0) \tag{5.10}$$

has the following explicit positive-polynomial certificate:

$$\begin{aligned} & N_1\beta(\beta N_0 + qD_0) - D_1(\beta N_0 + D_0) \\ &= \beta q s^2(1-r)^2\gamma^2 + 2\beta q s r(1-s)(1-r)\gamma\delta \\ &\quad + \beta(1-r)[\beta^2(r-s) + s^2(\beta + 2qr)]\gamma \\ &\quad + \beta q r^2(1-s)^2\delta^2 + \beta s r(1-s)(\beta + 2qr)\delta + \beta s^2 r(\beta + qr). \end{aligned} \tag{5.11}$$

Every summand on the right is nonnegative for $0 \le s \le r \le 1$. This identity follows by expansion and proves the claimed inequality after division by its positive denominator. The formula does not use the value of β beyond $0 < \beta < 1$.

The inaccessible case omitted by this division is $s = \delta = \gamma = 0$. A finite initial cost then requires $r > 0$. Direct conditioning gives $C_0 = 1/r$ and, after the indicated a-failure improvement, $C_1 = f(r)$. If other endpoint choices make one of the costs infinite, they cannot improve a finite optimum. The remaining finite endpoints are covered either by the displayed formulas or by their one-label reduction.

If both costs are at most b, (5.10) implies $b \ge f(1/b)$. The positive solution of $b = f(1/b)$ is B_0, since its reciprocal solves $r^2 + \beta^2 r - \beta^2 = 0$. This proves the lower bound for $r \ge s$. Equality is achieved by (5.7). At a finite optimum one has $0 < r < 1$; the endpoint $r = 0$ cannot succeed in model 0, whereas $r = 1$ gives model 1 cost at least $1 + 1/\beta = 5 > B_0$. In this open range the right-hand side of (5.11) is strictly positive if any of s, δ, γ is positive. Hence equality requires all three to vanish. The two remaining inequalities then force $r = r_0$. Without the a-failure improvement, the model-1 cost at these rows is

$$\frac{\beta + r\alpha}{\beta[\beta(1-r) + r\alpha]}.$$

It is strictly decreasing in α for $0 < r < 1$. Its value equals B_0 only at $\alpha = 1$. This proves the stated rigidity in this orientation.

It remains to exclude $r \le s$. In this orientation, routing every a-failure to label 1 improves model 1 and leaves model 0 unchanged. Make that improvement. The same N_0, D_0 describe model 0, whereas model 1 has

$$\begin{aligned} \widetilde{N}_1 &= \beta + qs + q(1-s)\delta + q(1-r)\gamma, \\ \widetilde{D}_1 &= \beta(1-r)[\beta + qs + q(1-s)(\delta + \gamma)]. \end{aligned}$$

Suppose a proposed bound $b < B_{\mathrm{opp}} < 5$ holds for both models. Model 1 has success probability at most $\beta(1-r)$ at every call, so $b \ge 1/[\beta(1-r)]$. In particular $b \ge 4$ and $br < 1$, the latter because $b < 1 + 1/\beta = 5$. The condition $N_0/D_0 \le b$ now implies $bs > 1$ and

$$\gamma \ge \frac{[s + (1-s)\delta](1-br)}{(1-r)(bs-1)}. \tag{5.12}$$

The model-1 cost increases with γ: it moves a failure from the better first label to the worse second label. Substitute the right-hand side of (5.12) in $\widetilde{N}_1/\widetilde{D}_1$, and call the resulting rational function $F_b(r, s, \delta)$. Its denominator is positive on the range in question, even if the substituted γ exceeds one. The latter substitution only relaxes the original constraint.

For precision, differentiation gives the following signs:

$$\partial_\delta F_b(r,s,\delta) \geq 0,$$
$$\partial_r F_b(r,s,0) > 0,$$
$$\operatorname{sign} \partial_s F_b(0,s,0) = \operatorname{sign}(bs-2).$$

These signs can be verified without estimating a root. Up to positive squared denominators, their respective numerators are

$$(\beta-1)(r-s)(s-1)(br-1)(bs-1), \quad -(bs-1)^2(\beta s-\beta-s), \quad (1-\beta)s(bs-2).$$

Consequently

$$C_1 \geq F_b(r,s,\delta) \geq F_b(0,s,0) \geq F_b(0,2/b,0) = \frac{\beta+4q/b}{\beta(\beta+4q/b-4q/b^2)}.$$

Here $2/b \leq 1$, since $b \geq 4$. The inequality that the final expression be at most b is, at $\beta = 1/4$, $b^2+8b-60 \geq 0$. It contradicts $b < B_{\mathrm{opp}}$. At $b = B_{\mathrm{opp}}$ the choice $r = 0$, $s = \gamma = 2/b$, $\delta = 0$, with every a-failure routed to label 1, attains equality. This proves (5.8) and completes the lemma. □

5.2. **A quantitative consequence of boundary rigidity.** The next estimate concerns the original, unrestricted six-parameter controller. It does not assume that its a-failure updates have already been improved. Write J_0^{rob} for its robust cost in $\mathcal{U}_0$.

Lemma 5.3 (Coercivity near the boundary optimizer)**.** *There are a neighborhood of the optimal controllers in Lemma 5.2 and constants $K, \Delta_0 > 0$ with the following property. If a controller in that neighborhood has $J_0^{\mathrm{rob}} \leq B_0 + \Delta$, where $0 \leq \Delta \leq \Delta_0$, then, with $z = s + \delta$,*

$$z \leq K\Delta, \qquad \gamma \leq K\Delta z, \qquad |r - r_0| \leq K\Delta. \tag{5.13}$$

At $z = 0$ the middle conclusion says $\gamma = 0$.

Proof. Choose a small fixed neighborhood in which r is bounded away from zero and one and s is smaller than $r/2$. The actual model-1 cost is at least its cost after routing all a-failures to label 2. In (5.11), the coefficients of $\gamma, \delta^2, s\delta, s^2$ are bounded below by positive constants on this neighborhood. Also N_0 is bounded above and below by positive constant multiples of $z+\gamma$. The denominator which converts that polynomial to $C_1 - f(D_0/N_0)$ is bounded above by a constant times N_0. Thus, whenever $N_0 > 0$,

$$C_1 - f(1/C_0) \geq c\,\frac{\gamma+z^2}{z+\gamma} \tag{5.14}$$

for some $c > 0$. Here C_0 is the actual baseline model-0 cost and C_1 may be the actual, unimproved model-1 cost. Both are at most $B_0 + \Delta$. Since f is increasing,

$$0 \leq C_1 - f(1/C_0) \leq B_0 + \Delta - f\big(1/(B_0+\Delta)\big) \leq K_1\Delta.$$

The last bound follows from differentiability and $f(1/B_0) = B_0$.

Shrink the neighborhood so that $z \leq 1$. Then $(\gamma+z^2)/(z+\gamma) \geq z$, and it is also at least $\gamma/(z+\gamma)$. These two bounds give $z \leq K_2\Delta$ and $\gamma/(z+\gamma) \leq K_2\Delta$. For $K_2\Delta \leq 1/2$, the latter yields $\gamma \leq 2K_2\Delta z$.

Put $t = 1/C_0$. The same inequalities give $1/(B_0+\Delta) \leq t \leq f^{-1}(B_0+\Delta)$. The derivative $f'(r_0) > 0$, so $|t - r_0| \leq K_3\Delta$. Finally the explicit model-0 formula gives

$$r - t = \frac{(r-s)(1-r)\gamma}{N_0},$$

which is bounded by a constant times $\gamma/(z+\gamma)$. This proves the estimate for r. If $N_0 = 0$, then $z = \gamma = 0$ and $t = r$ directly, so the same conclusions hold. Increasing one constant K proves all of (5.13). □

5.3. **The matching asymptotic.**

Proof of Theorem 5.1. Every model has a successful command for $\varepsilon < 1$. Theorem 3.1 gives feasibility, and Theorem 3.4 gives attainment at two labels. Alternatively, here the parameter cube (5.6) is compact, total expected cost is the supremum of its continuous finite-prefix costs, and a constant mixed row gives a finite competitor. These observations prove attainment directly.

A one-label row with a-probability r has costs $1/[(1-\varepsilon)r]$ and $1/[\beta(1-r)]$. Equating them proves (5.4); this value is at most 6 when $\varepsilon \le 1/2$. The single fixed row $r = 1/3$ has cost at most 6 for every model and every $\varepsilon \in [0, 1/2]$, so the fallback cap can be realized by the same program throughout. Pairing the unchanged model 1 with itself and 0_0 with 0_ε, the scalar nonterminal-map discrepancy is ε. The cost operators coincide. The paired instrument discrepancy used in Theorem 4.3 is therefore exactly ε.

For the upper estimate define the positive constants

$$A = r_0^{-2}, \qquad F = \frac{\beta}{D^2} = f'(r_0), \qquad L = \frac{q r_0^2}{\beta D^2}, \qquad k = \sqrt{\frac{F}{AL}} = \frac{\beta}{\sqrt{q}}, \qquad h = \frac{k^{-1} - Lk}{A + F}. \tag{5.15}$$

For sufficiently small $\varepsilon > 0$, use the controller

$$r = r_0 + h\sqrt{\varepsilon}, \quad s = 0, \quad \alpha = 1, \quad \gamma = 0, \quad \delta = k\sqrt{\varepsilon}, \tag{5.16}$$

with any ξ. These are legitimate probabilities. Both models are proper from every reachable label. Direct substitution in (5.5) gives

$$C_0(\varepsilon) = \frac{1}{(1-\varepsilon)r} + \frac{\varepsilon}{(1-\varepsilon)\delta}, \tag{5.17}$$

$$C_1(\varepsilon) = \frac{\beta + q\delta + r}{\beta[\beta + qr + q(1-r)\delta]}. \tag{5.18}$$

Their expansions are respectively

$$B_0 + (-Ah + k^{-1})\sqrt{\varepsilon} + O(\varepsilon), \qquad B_0 + (Fh + Lk)\sqrt{\varepsilon} + O(\varepsilon).$$

The two leading coefficients are equal by the definition of h, and their common value is

$$\frac{2\sqrt{FAL}}{A+F} = \frac{2\sqrt{q}\, r_0^2}{D^2 + \beta r_0^2} = c_0.$$

This proves the required upper limit, in particular $V_2(\varepsilon) - B_0 = O(\sqrt{\varepsilon})$.

For the converse take an attaining controller for each $\varepsilon > 0$. For any fixed controller the model-0 cost is nondecreasing in ε. Indeed the nonterminal matrix is its matrix at zero plus ε times a nonnegative matrix, so each finite-prefix cost is nondecreasing. Model 1 is unchanged. Consequently $V_2(\varepsilon) \ge B_0$, and every such optimizer has baseline robust cost at most $V_2(\varepsilon)$.

Every limit point of these controllers as $\varepsilon \downarrow 0$ is a baseline optimizer. To see this, use compactness of the parameter cube and lower semicontinuity of the baseline total cost, the latter being the supremum of continuous finite-prefix costs. The upper estimate just proved then bounds a limiting baseline cost by B_0. Lemma 5.2 therefore implies

$$r \longrightarrow r_0, \quad s, \gamma, \delta \longrightarrow 0, \quad \alpha \longrightarrow 1.$$

No convergence of the immaterial coordinate ξ is required. For all sufficiently small parameters the coercivity lemma applies, with $\Delta = V_2(\varepsilon) - B_0 = O(\sqrt{\varepsilon})$. It gives

$$s + \delta = O(\sqrt{\varepsilon}), \qquad \gamma = O(\varepsilon), \qquad r - r_0 = O(\sqrt{\varepsilon}). \tag{5.19}$$

Take any sequence along which the lower limit of the normalized cost gap is realized. Pass to a subsequence so that the bounded scaled coordinates converge:

$$\frac{r - r_0}{\sqrt{\varepsilon}} \longrightarrow H, \qquad \frac{s}{\sqrt{\varepsilon}} \longrightarrow u, \qquad \frac{\delta}{\sqrt{\varepsilon}} \longrightarrow v, \qquad u, v \ge 0.$$

Write $t = u + v$. We first show $t > 0$. In model 0_ε the two success probabilities and off-diagonal nonterminal probabilities are

$$h_1 = (1-\varepsilon)r, \quad h_2 = (1-\varepsilon)s, \qquad t_1 = \varepsilon r\alpha + (1-r)\gamma, \quad t_2 = \varepsilon s\xi + (1-s)\delta. \tag{5.20}$$

Let $\mathcal{D} = h_1h_2 + h_1t_2 + h_2t_1$. For small positive ε, finite cost and $\alpha \longrightarrow 1$ require $s + \delta > 0$; otherwise the positive probability of entering the second label leads to an absorbing failure state. Thus $\mathcal{D} > 0$, and (5.5) gives the exact identity

$$C_0(\varepsilon) - \frac{1}{h_1} = \frac{(h_1 - h_2)t_1}{h_1\mathcal{D}}. \tag{5.21}$$

In the neighborhood under consideration, $\mathcal{D}$ is at most a constant times $s + \delta$, while $t_1 \geq \varepsilon r\alpha$. Hence the right-hand side is at least $c\varepsilon/(s + \delta)$ for a fixed $c > 0$. Since $1/h_1 = B_0 + O(\sqrt{\varepsilon})$ by (5.19), the bounded normalized cost gap excludes $(s + \delta)/\sqrt{\varepsilon} \longrightarrow 0$. Therefore $t > 0$.

Now (5.20) and (5.19) give $\mathcal{D}/\sqrt{\varepsilon} \longrightarrow r_0t$. The nonnegative contribution of γ to t_1 may be dropped in a lower bound, while $\alpha \longrightarrow 1$. After dividing (5.21) by $\sqrt{\varepsilon}$ we obtain

$$\liminf \frac{C_0(\varepsilon) - B_0}{\sqrt{\varepsilon}} \geq -AH + \frac{1}{t}. \tag{5.22}$$

For model 1, route every a-failure to label 2 only for the purpose of a lower bound. The resulting cost N_1/D_1 is a smooth function of r, s, γ, δ in a neighborhood of the baseline optimizer, since D_1 is strictly positive there. Its derivatives with respect to r, s, δ at that point are

$$F, \qquad S = \frac{r_0}{\beta D}, \qquad L, \qquad S - L = \frac{r_0}{D^2} > 0.$$

The γ term is $O(\varepsilon)$ by (5.19). Taylor expansion, valid uniformly on this fixed neighborhood, therefore yields

$$\liminf \frac{C_1(\varepsilon) - B_0}{\sqrt{\varepsilon}} \geq FH + Su + Lv \geq FH + Lt. \tag{5.23}$$

This bound applies to the original arbitrary α, ξ; the routing improvement only reduced its model-1 cost.

Combining (5.22) and (5.23), the lower limit of the robust gap is at least

$$\max\{-AH + t^{-1},\ FH + Lt\} \geq \frac{F/t + ALt}{A + F} \geq \frac{2\sqrt{FAL}}{A + F} = c_0.$$

The first inequality is the convex combination with weights $F/(A+F)$ and $A/(A+F)$; the second is the arithmetic–geometric mean inequality. Since the sequence realizing the lower limit was arbitrary, this proves the lower limit in (5.3). The upper construction gives the matching upper limit. Finally $c_0\sqrt{\varepsilon}/\varepsilon^\alpha \longrightarrow \infty$ for every $\alpha > 1/2$, proving the asserted optimality of the exponent in the general primitive class. □

The limiting controller at zero perturbation is proper from its prescribed start in both models, but its second label is dark and unreachable in model 0_0. At a positive perturbation an a-failure reaches that label with probability of order ε. An escape of order $\sqrt{\varepsilon}$ balances the resulting residence cost against the cost of disturbing the other model's optimal continuation. The proof above quantifies this explanation over all stochastic two-label controllers; it does not restrict the converse to the displayed realizing program. The exact leading constant belongs to this family. The exponent conclusion concerns the full class of Theorem 4.3; it does not contradict the Lipschitz result under the additional success floor in Corollary 4.5.

6. First-order rigidity of all nearly optimal two-label programs

The exact value asymptotic in Theorem 5.1 does not by itself identify every nearly optimal program. We now characterize them to the first nontrivial order. This is an author addition to the fixed scalar family of Section 5, not a classification for arbitrary quantum instruments. The entire six-parameter stochastic controller class and its prescribed initial label remain unchanged.

Keep $\beta, q, r_0, B_0, D, A, F, L, k, h, c_0$ from (5.2) and (5.15). Define also

$$S = \frac{r_0}{\beta D}, \qquad Z = \frac{r_0^2}{D^2}, \qquad T = \frac{r_0^2}{\beta D^2}, \qquad E = \frac{1 - r_0}{r_0}. \tag{6.1}$$

All these constants are positive, and $S - L = r_0/D^2 > 0$. For a controller Θ_ε in (5.6), let J_ε be its worst model cost, with value $+\infty$ if it is improper from the prescribed initial label in either model.

Theorem 6.1 (Necessary and sufficient first-order optimality)**.** *Choose any controller Θ_ε for each sufficiently small $\varepsilon > 0$. No continuity or differentiability of this selection is required. Then*

$$J_\varepsilon - V_2(\varepsilon) = o(\sqrt{\varepsilon}) \tag{6.2}$$

if and only if its coordinates satisfy

$$\begin{aligned} r &= r_0 + h\sqrt{\varepsilon} + o(\sqrt{\varepsilon}), & s &= o(\sqrt{\varepsilon}), \\ \delta &= k\sqrt{\varepsilon} + o(\sqrt{\varepsilon}), & \gamma &= o(\varepsilon), \\ 1 - \alpha &= o(\sqrt{\varepsilon}). \end{aligned} \tag{6.3}$$

The remaining coordinate ξ may be any function into $[0, 1]$. Every exact optimizer satisfies these conclusions. Under either equivalent condition, each model separately has cost $B_0 + c_0\sqrt{\varepsilon} + o(\sqrt{\varepsilon})$.

Proof. Write $\lambda = \sqrt{\varepsilon}$. Assume (6.2). The exact value asymptotic gives $J_\varepsilon = B_0 + c_0\lambda + o(\lambda)$. For a fixed controller, the baseline model-0 failure matrix is entrywise no greater than its perturbed matrix. Termwise comparison of the nonnegative finite-prefix cost series therefore gives $B_0 \le J_0 \le J_\varepsilon$, where J_0 is this controller's baseline robust cost, not the optimum over baseline controllers.

Every limit point in the compact parameter cube is a baseline optimizer: the baseline total cost is the supremum of continuous finite-prefix costs and is lower semicontinuous. The complete equality set in Lemma 5.2 consequently gives

$$r \longrightarrow r_0, \quad s, \gamma, \delta \longrightarrow 0, \quad \alpha \longrightarrow 1,$$

with no restriction on ξ. Applying Lemma 5.3 with $0 \le J_0 - B_0 = O(\lambda)$ gives, uniformly along the selection,

$$r - r_0 = O(\lambda), \qquad s + \delta = O(\lambda), \qquad \gamma = O(\lambda^2). \tag{6.4}$$

The missing update scale. Let C_1 be the actual model-1 cost and let C_1^I denote its cost after setting $\alpha = 1, \xi = 0$ for this comparison only. Near the baseline equality set one has $r > s$. Both row success probabilities are bounded away from zero, so all the following resolvents are proper and smooth, uniformly for $\xi \in [0, 1]$. Put

$$U = \beta(1 - r), \quad W = \beta(1 - s), \quad a = r\alpha + q(1 - r)\gamma, \quad b = s\xi + q(1 - s)\delta, \quad Q = UW + Ub + Wa.$$

The exact cost and its off-diagonal derivatives are

$$C_1 = \frac{W + a + b}{Q}, \qquad \partial_a C_1 = \frac{(U - W)(W + b)}{Q^2}, \qquad \partial_b C_1 = \frac{(W - U)a}{Q^2}. \tag{6.5}$$

In particular, increasing ξ cannot reduce the cost, whereas $\partial_\alpha C_1$ tends uniformly to $-Z < 0$ at the baseline equality set. In a smaller fixed neighborhood, $-\partial_\alpha C_1 \ge Z/2$, including every intermediate α between its actual value and one. Integrating that inequality and then reducing ξ to zero yields

$$C_1 - C_1^I \ge (Z/2)(1 - \alpha).$$

The improved cost C_1^I is a smooth function of r, s, γ, δ. By (6.4), $C_1^I = B_0 + O(\lambda)$. Since $C_1 \le J_\varepsilon = B_0 + O(\lambda)$, we obtain

$$1 - \alpha = O(\lambda). \tag{6.6}$$

Convergence of α alone would not justify its first-order Taylor expansion; this estimate supplies the needed rate.

All scaled subsequences. Take an arbitrary sequence tending to zero. The established bounds allow extraction of a subsequence such that

$$(r - r_0)/\lambda \longrightarrow H, \qquad s/\lambda \longrightarrow u, \qquad \delta/\lambda \longrightarrow v,$$
$$\gamma/\lambda^2 \longrightarrow G, \qquad (1-\alpha)/\lambda \longrightarrow a_*, \qquad \xi \longrightarrow \zeta.$$

Here $u, v, G, a_* \ge 0$ and $\zeta \in [0,1]$. Put $t = u + v$. We must show $t > 0$ before dividing by it.

For model 0_ε, use the four probabilities h_1, h_2, t_1, t_2 in (5.20), and put $\mathcal{D} = h_1h_2 + h_1t_2 + h_2t_1$. Finite cost and $\alpha \to 1$ force $s + \delta > 0$ for small positive ε: otherwise the second label is absorbing and dark, but is entered with probability at least $\varepsilon r\alpha > 0$. Thus $\mathcal{D} > 0$ and the exact identity (5.21) applies. In the fixed neighborhood, $\mathcal{D} \le C(s + \delta)$, $t_1 \ge \varepsilon r\alpha$, and $h_1 - h_2$ is bounded below positively. Consequently

$$C_0 - \frac{1}{h_1} = \frac{(h_1 - h_2)t_1}{h_1\mathcal{D}} \ge \frac{c\varepsilon}{s + \delta}$$

for constants $C, c > 0$ independent of the controller. Also $1/h_1 = B_0 + O(\lambda)$. The bounded normalized robust cost excess therefore rules out $t = 0$.

Now $\mathcal{D}/\lambda \to r_0 t$ and $t_1/\lambda^2 \to r_0 + (1 - r_0)G$. Keeping the leak contribution, rather than dropping it as in a value lower bound, gives the exact limit

$$\frac{C_0 - B_0}{\lambda} \longrightarrow X = -AH + \frac{1 + EG}{t}. \tag{6.7}$$

For model 1, the rational expression in (6.5) has a denominator bounded away from zero in the fixed neighborhood. At the baseline, its derivatives, with ξ held fixed, are

$$\partial_r C_1 = F, \qquad \partial_s C_1 = S + T\xi, \qquad \partial_\delta C_1 = L, \qquad \partial_\alpha C_1 = -Z.$$

These identities follow by direct differentiation of the displayed resolvent; they hold uniformly for $\xi \in [0,1]$. The γ contribution is $O(\lambda^2)$. The bounds (6.4) and (6.6) give a uniform Taylor remainder $o(\lambda)$. Hence

$$\frac{C_1 - B_0}{\lambda} \longrightarrow Y = FH + Lt + (S - L)u + Za_* + T\zeta u. \tag{6.8}$$

Since the robust normalized gap tends to c_0, we have $\max(X, Y) = c_0$. The positive weights $F/(A + F)$ and $A/(A + F)$ give

$$\begin{aligned} c_0 &\ge \frac{FX + AY}{A + F} \\ &= \frac{F/t + ALt}{A + F} + \frac{FEG/t + A(S - L)u + AZa_* + AT\zeta u}{A + F} \\ &\ge \frac{2\sqrt{FAL}}{A + F} = c_0. \end{aligned}$$

Every additional term is nonnegative. Equality in the arithmetic–geometric mean forces $t = k$. Since $E, S - L, Z > 0$, equality also forces $G = u = a_* = 0$, whence $v = k$. Equality of a positive weighted average to its maximum forces $X = Y = c_0$. Solving $-AH + k^{-1} = FH + Lk$ gives $H = h$. Every convergent scaled subsequence has these same limits. Thus the full limits hold and imply (6.3). This argument imposes no limit on ξ.

Sufficiency and realizability. Conversely assume (6.3). For all sufficiently small ε, $\delta > 0$, r is bounded inside $(0, 1)$, and s is close to zero. In model 0_ε the second label can return to the

first through a b-failure with probability $(1-s)\delta > 0$, and the first label has success probability $(1-\varepsilon)r > 0$. Both reachable labels therefore have finite cost. In model 1 both row success probabilities are bounded away from zero. The controller is consequently proper from its prescribed start in both models.

The exact model-0 expansion applies with $H = h, t = k, G = 0$ and gives $(C_0 - B_0)/\lambda \to -Ah + k^{-1} = c_0$. The uniform model-1 expansion has $u = a_* = 0, v = k$ and gives $(C_1 - B_0)/\lambda \to Fh + Lk = c_0$, regardless of ξ. Therefore $J_\varepsilon = B_0 + c_0\lambda + o(\lambda)$; Theorem 5.1 proves (6.2) and the stated componentwise limits. Exact optimizers are a special case because their value difference is zero. This completes both implications. □

The theorem identifies a unique leading repair mechanism in this family. At the square-root scale, activating success at the second label through s, adding baseline leakage through γ at the ε scale, or losing the optimal a-failure update through $1-\alpha$ all incur strictly positive leading penalties. The b-failure return δ has the forced square-root coefficient k, while the initial command row has the forced correction h. This is neither an exact finite-ε optimizer formula nor a universal degeneration classification, and it does not close the original general-memory squared-sign risk or the other independent research questions recorded below.

7. Finite search: arithmetic classification and explicit memory bounds

The arithmetic obstruction extends to every finite number of search locations, with arbitrary positive call costs. The distinction is between logarithms of the failure probabilities that lie in one rational direction and those that occupy several rational directions. The latter case includes families containing both rationally related and irrationally related pairs. We obtain a necessary and sufficient condition for an optimal history law to admit an autonomous stochastic implementation with finitely many counted labels. We also give a primitive-data exponential approximation bound and, for rational data in the saturating case, a computable bound on the sufficient number of labels.

The search ordering, interior least-favourable priors, and finite mixtures of consistent tie orders are classical ingredients [95, 96, 97]. Eventual cycles under a common-power relation are also classical for arbitrary finite n; see Section 6.1 and Appendix C of Clarkson–Lin–Glazebrook [96], with the earlier search-theoretic attribution there to Matula. We give the required arguments below. The common-power sufficiency is therefore a classical search consequence together with counted initialization. The additional resource conclusions concern all-stochastic necessity through commensurability classes, explicit primitive-data bounds, and implementation by counted stochastic memory. A finite mixture of infinite words alone is not such an implementation. This section is a classification for the displayed search family, not an assertion that every instrument in Theorem 3.4 has an arithmetic saturation criterion.

Fix $n \geq 2$, costs $c_i > 0$, and failure probabilities $r_i \in (0,1)$, for $i \in \{1, \dots, n\}$. There are n fixed hidden models and n commands. In model i, command i succeeds with probability $q_i = 1 - r_i$, and every other command fails surely. There is one failure report, and success stops the experiment. Command i costs c_i in every model. Set

$$\tau_i = \frac{c_i}{q_i}, \qquad u_i = -\log r_i > 0, \qquad B = \sum_{i=1}^{n} \tau_i, \qquad H = \sum_{i=1}^{n} \frac{1}{u_i}. \tag{7.1}$$

The hidden model is selected once. The initialization and memory conventions are those of Section 3: the initial label is prescribed and pure, and command and update randomness is permitted only through the fixed controller kernels. Write V_M and $V = V_{\mathrm{hist}}$ for the finite-label and history values in this family.

Lemma 7.1 (A common one-label fallback)**.** *The one-label value is $V_1 = B$. It is attained by the row*

$$p_i = \frac{q_i^{-1}}{\sum_j q_j^{-1}}, \tag{7.2}$$

whose expected cost equals B in every model.

Proof. For a row p, put $x_i = p_i q_i$. A zero x_i gives infinite cost in model i. Otherwise its one-call mean cost is $\sum_i p_i c_i = \sum_i \tau_i x_i$, and its worst model cost is

$$\frac{\sum_i \tau_i x_i}{\min_i x_i} \geq \sum_i \tau_i = B.$$

The displayed row makes all x_i equal and attains the bound. □

7.1. **The search saddle and consistent tie orders.** A planned word $w = (w_0, w_1, \ldots) \in \{1, \ldots, n\}^{\mathbb{N}_0}$ specifies the commands along the hypothetical all-failure path. Let $k_i(t)$ be the number of occurrences of i before position t. Its cost in model i is the nonnegative sum

$$C_i(w) = \sum_{t \geq 0} c_{w_t} r_i^{k_i(t)}. \tag{7.3}$$

Every randomized causal policy induces a probability law on these words; conversely, a word law is realized by its conditional next-letter kernels. Indeed, conditional on a command prefix, its physical failure likelihood is common to all private realizations of that prefix and cancels. Thus these conditional kernels do not use the hidden model. Tonelli's theorem identifies policy costs with the expectations of (7.3). Finite cost in every model requires infinitely many occurrences of every command, almost surely: after the last occurrence of i, its survival factor is a fixed positive number and every later cost is bounded below by $\min_j c_j > 0$.

Lemma 7.2 (The finite-search saddle)**.** *For an interior prior $\pi = (\pi_1, \ldots, \pi_n)$, put $a_i = \pi_i/\tau_i$. A finite-cost word is Bayes optimal if and only if it chooses, at every prefix, a maximal element of*

$$\{a_i r_i^{k_i(t)} : 1 \leq i \leq n\}. \tag{7.4}$$

Every such greedy word has finite cost in every model. If $b_i = -\log a_i$ and

$$R_\pi = \max\{\max_i b_i - \min_i b_i, \max_i u_i\}, \qquad \bar{b} = \frac{\sum_i b_i/u_i}{H},$$

then, uniformly over every choice at ties,

$$r_i^{k_i(t)} \leq e^{b_i - \bar{b} + R_\pi} e^{-t/H}, \qquad \lim_{t\to\infty} \frac{k_i(t)}{t} = \frac{1}{H u_i}. \tag{7.5}$$

The robust history value is attained. There is an interior prior π_* with Bayes value V, and every robust optimum has cost vector $(V, \ldots, V)$ and is supported on greedy words for that prior. One robust optimum is a mixture of at most n greedy words, each of which resolves every tie using one fixed priority permutation of $\{1, \ldots, n\}$.

Proof. Exchanging two adjacent different commands ij and ji after counts k changes only model components i, j. With ij first in the subtraction the risk difference is

$$-c_j q_i r_i^{k_i} e_i + c_i q_j r_j^{k_j} e_j, \tag{7.6}$$

where e_i are the coordinate vectors of $\mathbb{R}^n$. Its Bayes value is $c_i c_j (a_j r_j^{k_j} - a_i r_i^{k_i})$. The greedy rule merges the n decreasing lists $\{a_i r_i^k : k \geq 0\}$ in decreasing numerical order. Given any finite-cost competitor and a desired greedy next letter, move the next occurrence of that letter to the desired position. It exists because every letter occurs infinitely often. Each intervening list item is no larger than the desired item: it was no larger at the starting prefix, and subsequent items in its own list

decrease. The adjacent exchanges therefore do not increase Bayes cost, and a strictly incorrect first choice strictly decreases it. Repeating finitely many times matches any prescribed finite greedy prefix. The cost of that prefix is at most the full competitor cost. Letting its length increase proves optimality of a greedy word once its finiteness has been verified. The strict exchange proves necessity. Because the prior is interior, any infinite component cost gives infinite Bayes cost.

For finiteness set $x_i(t) = b_i + u_i k_i(t)$. A greedy step increments a current minimum by u_i. Its minimum cannot decrease, and its maximum is at most the greater of the old maximum and the old minimum plus $\max_i u_i$. Induction gives

$$\max_i x_i(t) - \min_i x_i(t) \le R_\pi.$$

The weighted mean with weights $1/u_i$ is $\bar{b} + t/H$, since $\sum_i k_i(t) = t$. It lies between the minimum and maximum. Thus $|x_i(t) - \bar{b} - t/H| \le R_\pi$, which proves both assertions in (7.5). The costs are bounded above by a geometric sum times $\max_i c_i$.

We next prove robust attainment and the interior saddle. Let $\mathcal{R} \subset \mathbb{R}^n$ consist of the finite upper bounds on cost vectors attainable by word laws. This nonempty set is convex by mixing and upward closed. It is closed: if finite upper bounds converge, compactness of probability laws on the finite-alphabet product space gives a weakly convergent subsequence. Every finite-prefix cost is continuous and bounded, so the limiting law satisfies all the finite-prefix bounds; monotone convergence gives the full bounds. The same argument applied to a minimizing sequence shows robust attainment, and $V \le B$ by Lemma 7.1.

Separate $\mathcal{R}$ from the open orthant $\{z : z_i < V \text{ for every } i\}$. Upward closure makes the nonzero separating vector nonnegative. Normalize its sum to one, and call it π_*. Since $(V, \dots, V) \in \mathcal{R}$, separation gives $\inf_{z \in \mathcal{R}} \pi_* \cdot z = V$. We claim that π_* has full support. Otherwise let $S = \{i : \pi_{*,i} > 0\}$ be a proper subset. Take an attaining robust word law and delete every command outside S from every word. For a model $i \in S$, the survival factor of each retained call is unchanged, whereas every deleted call had positive cost and strictly positive survival probability. Every original word contains all commands infinitely often almost surely. Consequently the deleted law strictly decreases the π_*-weighted cost, while retaining finite costs and eventual success for each model in S.

The deleted law need not be proper outside S. To obtain admissible competitors, follow its first T calls and then use the common fallback (7.2). The fallback adds at most B times the survival probability at T in each model of S. Those probabilities tend to zero by bounded convergence. The finite-prefix costs increase to the deleted costs. These completed laws are proper in every model and converge in weighted cost to a value strictly below V, a contradiction. Thus π_* is interior. An attaining robust law has costs at most V in every component and weighted cost at least V, so all its component costs equal V. Its word law must be supported on Bayes minimizers, which are precisely the greedy words just characterized.

It remains to prove the finite consistent-tie mixture. At any tied numerical level $z > 0$, each list contributes at most one item. The tied items form a consecutive block; permuting them leaves every later count unchanged. For a tied adjacent pair i, j, equation (7.6) becomes

$$c_i c_j z \left(-\frac{e_i}{\pi_{*,i}} + \frac{e_j}{\pi_{*,j}} \right). \tag{7.7}$$

For any linear functional $\lambda \in \mathbb{R}^n$, its difference is $c_i c_j z(\lambda_j/\pi_{*,j} - \lambda_i/\pi_{*,i})$. It follows that the order minimizing λ in every tied block is the same global priority order: decreasing $\lambda_i/\pi_{*,i}$, with one fixed convention for equal ratios. Sort finitely many tied blocks in that order and then let their number increase. The geometric tail bound (7.5), uniform over tie choices, shows convergence of every component cost. Hence the minimum of every linear functional over the greedy risk vectors is attained by one of the $n!$ consistent priority words.

The greedy word space is closed in the compact product space, and its risk map is continuous by the same uniform tails. Its compact risk image therefore lies in the convex hull of the finitely

many consistent-priority risk vectors: otherwise finite-dimensional separation would contradict the preceding minimum property. The robust risk vector, as a barycenter of greedy risks, lies in that convex hull as well. All these vectors lie on the hyperplane $\pi_* \cdot z = V$. Any convex representation using more than n of them has an affine dependence. Subtract a suitable multiple of that dependence from the weights until one weight becomes zero; the weights remain nonnegative, still sum to one, and give the same vector. Repeating leaves at most n vectors, as asserted. □

7.2. **Rigidity of projected finite-state outputs.**

Lemma 7.3 (A deterministic class suffix from finite stochastic memory)**.** *Let an autonomous controller have finitely many labels, finitely many commands, one failure report, and fixed stochastic command and update kernels. Partition the commands into finitely many classes. If its planned word law projects, after some fixed finite time, onto one prescribed class word almost surely, that class word is ultimately periodic.*

Proof. Let S_t be the nonempty set of labels having positive probability at time t on the hypothetical all-failure path. The next support is the union, over labels in S_t and commands of positive probability in their rows, of the positive supports of the corresponding update rows. It is therefore a fixed deterministic function of S_t. At times when the class output is prescribed, every positive command in every label of S_t belongs to that prescribed class. Since the rows have sum one, this class is determined by S_t as well. There are finitely many nonempty label subsets. The support-set trajectory eventually repeats and thereafter is periodic. Its prescribed class output is consequently ultimately periodic. The argument permits arbitrary stochastic transitions and arbitrary mixing among commands inside the prescribed class. □

Theorem 7.4 (Finite-search saturation for any finite number of locations)**.** *For the family* (7.1)*,*

$$\big(\exists M < \infty :\ V_M = V\big) \quad \Longleftrightarrow \quad \frac{\log r_i}{\log r_j} \in \mathbb{Q} \quad \textit{for every } i, j. \tag{7.8}$$

If the condition fails, every finite M has the strict attained gap $V_M > V$, although $V_M \downarrow V$ as $M \to \infty$. If the condition holds, a robust optimum is realized by a counted pure-start finite controller, obtained from at most n ultimately periodic consistent-priority words.

Proof. Partition the indices into classes by $i \sim j$ if u_i/u_j is rational. Consider a least-favourable interior prior from Lemma 7.2. Two lists from different classes have at most one tied pair. Indeed, two distinct equalities $a_i r_i^k = a_j r_j^l$ would imply $(k-k')u_i = (l-l')u_j$ with nonzero integer differences, contrary to their being in different classes. There are finitely many pairs of lists, hence only finitely many cross-class tied levels. Each positive numerical level has only finitely many list items above it. After a fixed finite position, all remaining ties therefore occur within a single class. Ordering the list items then fixes the same deterministic class suffix for every greedy word, regardless of the within-class choices.

For a class C, (7.5) gives its positive limiting frequency

$$f_C = \frac{H_C}{H}, \qquad H_C = \sum_{i \in C} \frac{1}{u_i}. \tag{7.9}$$

Choose a representative u_C from each class. Then $H_C = A_C/u_C$ for some positive rational A_C. For two distinct classes C, D, the ratio $f_C/f_D = (A_C/A_D)(u_D/u_C)$ is irrational. An ultimately periodic class word has rational limiting frequencies, and therefore rational ratios between any two positive frequencies. Thus the common class suffix cannot be ultimately periodic when there is more than one class.

If a finite controller attained V, its word law would be supported on the greedy words by Lemma 7.2, so its projected class suffix would be the one just described. Lemma 7.3 gives a contradiction. All failure probabilities in the search family are positive, so there is no zero-likelihood

physical history to remove from this argument. The attained-frontier result in Theorem 3.4 makes nonattainment of V equivalent to the strict gap at each finite M. Convergence follows from Corollary 4.4; an explicit rate is also proved below.

Conversely suppose that all u_i/u_j are rational. Write $u_i = h_i\ell$ with positive integers h_i and $\ell > 0$. For one fixed priority word consider the relative scores

$$d_i(t) = x_i(t) - x_n(t), \qquad 1 \le i < n.$$

Each $d_i(t)$ belongs to the fixed coset $(b_i - b_n) + \ell\mathbb{Z}$ and lies in $[-R_{\pi_*}, R_{\pi_*}]$. There are only finitely many such tuples. The tuple determines the next command by minimizing the scores relative to $x_n = 0$ and using the fixed priority at ties. On command $j < n$, only d_j increases, by u_j; on command n, every d_i decreases by u_n. Thus the tuple evolves deterministically in a finite set. The word is ultimately periodic and has a finite deterministic implementation.

The saddle lemma expresses a robust optimum as a mixture of at most n such words. We implement that mixture without a free random initial label. Give the implementation one prescribed initial label and disjoint finite tail controllers for the words. If their weights are λ_j and first commands are a_j, the initial row chooses command a with probability $p_a = \sum_{j:a_j=a} \lambda_j$. For $p_a > 0$, after its failure it selects the corresponding tail j with probability λ_j/p_a among those with $a_j = a$. The update rows for $p_a = 0$ are unreachable and may be assigned arbitrarily. The physical failure likelihood conditional on a is the same for all those branches in every model, so this realizes the desired mixture exactly. The branch identity is stored in the counted tail labels. This proves finite saturation. □

7.3. **Bounds from primitive data.** The preceding proof used an interior prior without estimating how close it can lie to the boundary. The following bound removes this unknown from the memory estimates. Its constants are deliberately explicit; they are not asserted to be sharp.

Lemma 7.5 (An explicit least-favourable prior bound)**.** *Put* $c_- = \min_i c_i$, $c_+ = \max_i c_i$, $r_- = \min_i r_i$, *and define*

$$\begin{aligned} T_0 &= \left\lceil \frac{2B}{c_-} \right\rceil, \qquad g = \frac{c_-}{2} r_-^{T_0}, \\ T_1 &= \left\lceil \frac{2B^2}{c_- g} \right\rceil, \quad K = c_+ T_1 + B, \qquad \delta = \frac{g}{2K}. \end{aligned} \tag{7.10}$$

Every least-favourable prior whose Bayes value is V *satisfies* $\pi_i \ge \delta$ *for every* i.

Proof. Fix an attaining robust word law. In each model its expected number of calls is at most $V/c_- \le B/c_-$, since every call costs at least c_-. Markov's inequality shows that success occurs by call T_0 with probability at least $1/2$. In model i, this implies that its planned word contains command i among the first T_0 positions with probability at least $1/2$.

Delete all occurrences of command i. In any other model j, the survival factor of a retained command is unchanged. On the event just described, the first deleted call costs at least c_- and has model-j survival probability at least $r_-^{T_0}$. The deleted law therefore lowers the cost of every model $j \ne i$ by at least g. Let W_i be the robust history value of the restricted search problem using only the $n-1$ other models and commands. We have

$$W_i \le V - g. \tag{7.11}$$

The same finite-prefix compactness argument used above gives an attaining restricted law, including when $n - 1 = 1$.

Run this law for at most T_1 calls, then switch to the common fallback of value B. For $j \ne i$, the restricted law has expected call count at most $W_i/c_- \le B/c_-$. Its probability of surviving T_1 calls

is at most $B/(c_-T_1)$. The completed policy therefore has cost at most

$$W_i + \frac{B^2}{c_-T_1} \le V - \frac{g}{2} \quad (j \ne i).$$

In model i every restricted call fails, but its total cost is at most $c_+T_1+B = K$. For a least-favourable prior, this competitor must have Bayes cost at least V. Hence

$$V \le \pi_i K + (1-\pi_i)(V - g/2), \qquad \pi_i \ge \frac{g/2}{K - V + g/2} \ge \frac{g}{2K}.$$

The last inequality uses $V \ge c_- > g/2$, which follows from the cost of the compulsory first call and the definition of g. □

Theorem 7.6 (An explicit exponential memory approximation)**.** *With δ from* (7.10)*, put*

$$\begin{aligned} \tau_- = \min_i \tau_i, \qquad \tau_+ = \max_i \tau_i, \qquad C = \frac{\tau_+}{\delta\tau_-}, \\ A = C\max\{C, r_-^{-1}\}, \qquad \kappa = e^{-1/H} \in (0,1). \end{aligned} \tag{7.12}$$

For every $M \ge n$,

$$0 \le V_M - V \le BA\,\kappa^{\lfloor M/n \rfloor}. \tag{7.13}$$

The bound holds whether or not finite saturation occurs.

Proof. Choose the optimal mixture of at most n consistent-priority words from Lemma 7.2. By the prior bound, $a_i = \pi_{*,i}/\tau_i$ lies between δ/τ_+ and $1/\tau_-$. Put $L = \log C$ and $R = \max\{L, \max_i u_i\}$. Then the initial logarithmic-score spread is at most L, and $R_{\pi_*} \le R$. Also $b_i - \bar{b} \le L$ and $e^{R+L} = C\max\{C, r_-^{-1}\} = A$. Equation (7.5) now bounds every model's survival probability after T commands, in each word and hence in the mixture, by $A\kappa^T$.

Retain the first $T \ge 1$ commands of each mixture word and then use the one-label fallback. The tail replacement increases each cost by at most $BA\kappa^T$. A counted implementation uses one initial label, at most $n(T-1)$ disjoint continuation labels, and one common fallback label. The initial branching is implemented after the first command exactly as in the saturation theorem. Since $n \ge 2$, the total is at most $2 + n(T-1) \le nT$. Taking $T = \lfloor M/n \rfloor$ and adding unused labels if necessary proves the claim. □

Corollary 7.7 (A decidable rational criterion and a sufficient label cap)**.** *Suppose that r_i, c_i are positive rational numbers with $r_i < 1$. For each i, let $\nu(r_i)$ be its signed vector of prime exponents, using the finite union of primes occurring in all numerators and denominators. Finite saturation occurs if and only if all these vectors lie on the same positive rational ray. This is decidable by factorization and integer comparisons.*

In the saturating case one can compute $t \in \mathbb{Q} \cap (0,1)$ and positive integers h_i with $r_i = t^{h_i}$. Using the rational constants δ, τ_-, τ_+ above, choose an integer $J \ge 0$ such that

$$\begin{aligned} t^{-J} \ge \frac{\tau_+}{\delta\tau_-}, \qquad N_0 = \max\{J, \max_i h_i\}, \\ N = (2N_0+1)^{n-1}, \qquad M_0 = 1 + nN. \end{aligned}$$

Then $V_{M_0} = V$. In particular, the history value in this saturating rational family is a real algebraic number and is computable by the finite quantified description of Theorem 3.4 at the explicitly bounded budget M_0.

Proof. For rational numbers, a rational relation between their logarithms is equivalent to a positive integer power identity. Unique prime factorization makes this equivalent to positive rational collinearity of their signed exponent vectors. This proves the criterion using Theorem 7.4. To construct the common base, divide $\nu(r_1)$ by the greatest common divisor of its nonzero coordinates in absolute value, giving a primitive integer vector w. Each other exponent vector is $h_i w$ with

$h_i > 0$. Since the coordinates of w have greatest common divisor one, an integer linear combination of them is one; applying it to $h_i w$ shows that h_i is an integer. Then $t = \prod_p p^{w_p}$ is rational, satisfies $0 < t < 1$, and has $r_i = t^{h_i}$. A common divisor of all h_i may be absorbed in t if desired.

All constants in (7.10) are rational and effectively obtained by finite arithmetic and ceilings. The inequality defining J is found by increasing J and making exact rational comparisons. Write $\ell = -\log t$. It implies $L = \log C \le J\ell$, while $u_i = h_i \ell$, so $R = \max\{L, \max_i u_i\} \le N_0 \ell$. Each relative-score coordinate in the saturation proof belongs to one fixed coset of $\ell\mathbb{Z}$ inside $[-N_0\ell, N_0\ell]$. It has at most $2N_0 + 1$ possible values, independently of the unknown offset of that coset. Each consistent priority word therefore uses at most N relative-score states. The counted pure-start mixture uses at most one initial label and n copies of those state sets, giving $M_0 = 1 + nN$.

The family has rational scalar instrument and cost data. Its finite-budget optimum is attained, finite, algebraic, and specified by the real-algebraic quantified formula of Theorem 3.4. Since the proven cap gives $V = V_{M_0}$, that finite procedure computes the history value here. The procedure is an effectivity result; no practical complexity bound for quantifier elimination is asserted. □

The two-location normalized bounds in Section S1 retain their sharper constants. The present results add arbitrary finite model number, arbitrary positive deterministic command costs, mixed commensurability classes, a primitive-data convergence rate, and an effective sufficient memory cap in the rational saturating case. They do not infer a positive numerical lower bound on the gap $V_M - V$ from irrationality alone, nor a comparable arithmetic classification for general noncommuting instruments.

8. An exact hidden-model memory frontier

The model family in this example consists of two scalar instruments. The model $\theta \in \{0, 1\}$ stays fixed during the run and is not observed. There are two commands, a and b, one failure report, unit call cost, and compulsory stopping at success. In model 0, command a succeeds with probability one and b fails with probability one; in model 1 the roles are reversed. The controller knows its sampled command, and may use that command together with the failure report in its update. It is autonomous, has a prescribed pure initial live label, and has no uncounted clock or persistent random seed.

Proposition 8.1 (Golden-ratio memory cost)**.** *Writing $\varphi = (1 + \sqrt{5})/2$, the worst-model expected total cost obeys*

$$V_1 = 2, \qquad V_2 = \varphi, \qquad V_M = \frac{3}{2} \quad (M \ge 3).$$

The last value is also the optimum among unrestricted history-dependent controllers. For the two-label optimum, one label is unreachable and improper in one model, although the prescribed-start controller is proper in both models. Thus model-dependent reachable supports cannot in general be replaced by a full-space resolvent or full-space Lyapunov inequality.

Proof. At a live label let r denote the probability of command a. A single label has expected costs $1/r$ and $1/(1 - r)$, interpreted as infinite at the corresponding endpoints. Their maximum is minimized at $r = 1/2$ and equals 2.

For two labels let the command probabilities be r, s, with initial row r. At every later call the success probability in model 0 is at most $\max(r, s)$ and in model 1 at most $1 - \min(r, s)$. A process whose conditional success probability is at most $h > 0$ has expected number of remaining calls at least $1/h$: its survival probabilities are at least $(1 - h)^n$, by induction. If $h = 0$, that expectation is infinite. Conditioning on the first call therefore gives

$$C_0 \ge 1 + \frac{1 - r}{\max(r, s)}, \qquad C_1 \ge 1 + \frac{r}{1 - \min(r, s)},$$

whenever the displayed terms have positive numerator; endpoint cases with zero numerator have the evident zero continuation contribution. If $r \geq s$, these bounds imply

$$\max(C_0, C_1) \geq \max\{1/r, 1+r\} \geq \varphi.$$

If $r \leq s$, they imply

$$\max(C_0, C_1) \geq \max\{2-r, 1/(1-r)\} \geq \varphi.$$

The boundary cases satisfy the same lower bound by direct inspection. Both one-variable minima follow by equating their decreasing and increasing entries.

For attainment take $r = 1/\varphi = \varphi - 1$ and $s = 0$. From label 1, a failed a command writes label 2, whose command is always b, while a failed b command retains label 1. At label 2, every failure retains label 2. In model 0, only b can fail, so label 2 is never reached and the expected cost is $1/r = \varphi$. In model 1, the first call either succeeds on b or fails on a and is followed by one certain success, giving cost $1 + r = \varphi$. This computes the actual cost without assuming properness of the unused model-0 label.

For an arbitrary history-dependent controller, let r be its first-call probability of a. A failed first call needs at least one additional unit-cost call before success. The two model costs are therefore at least $2 - r$ and $1 + r$, whose maximum is at least $3/2$. Three counted labels attain the bound: use an initial 1/2–1/2 row and two continuing rows, one always a and one always b. A failed a command writes the pure-b label and a failed b command writes the pure-a label. The next call succeeds surely. Duplicating labels gives the assertion for every $M \geq 3$.

Finally, under the two-label optimizer the scalar failure operator in model 0 leaves label 2 invariant, since its b command always fails. A full-space cost supersolution would require $W_{0,2} \geq 1 + W_{0,2}$, which is impossible. The valid reachable projections are $P_0 = (1, 0)$ and $P_1 = (1, 1)$; the corresponding cost certificates are $W_0 = (\varphi, 0)$ and $W_1 = (\varphi, 1)$. They obey the support-restricted Bellman equalities and the same initial bound $W_{\theta,1} = \varphi$. □

The pure initial label matters. If an uncounted randomized initial choice of labels were free, two pure command rows would already attain $3/2$ by an equal initial mixture and switching after the first failure. That is a different memory convention. Likewise a continuously sampled persistent seed or externally supplied time index would change the controller class.

9. Primitive optimal control for finite reset instruments

An exact application should not require that a balancing group has already been built into the experiment. We give a class in which the best continuation and a least-favorable preparation can instead be derived from the primitive measurement instrument. The class includes any finite family of rank-one failure projections in any finite Hilbert dimension, as well as mixed post-failure states. It does not subsume an arbitrary-rank failure projection, whose posterior still depends on its input. The complete arbitrary-rank two-detector theorem is retained in Section S38 of the supplement.

Let $\mathcal{H}$ be finite-dimensional. There is a nonempty finite command set $\mathcal{A}$, nonterminal reports $i = 1, \ldots, s$, and a terminal success. Assume $s \geq 1$. A nonterminal branch has the physically specified map

$$\mathcal{I}_{a,i}(\rho) = \operatorname{tr}(E_{a,i}\rho)\sigma_i, \qquad E_{a,i} \succeq 0, \quad \sum_i E_{a,i} \preccurlyeq I, \tag{9.1}$$

where σ_i is a known density matrix and the report i is observed. The aggregate success effect is $B_a = I - \sum_i E_{a,i}$. The instrument itself prepares the state σ_i on failure; no additional freely available reset is assumed. One call has a nonnegative cost with conditional mean $\operatorname{tr}(G_a\rho)$, where

$$0 < c_* I \preccurlyeq G_a \preccurlyeq c^* I < \infty.$$

Success requires immediate commitment and failure forbids it. The decoder has one fixed terminal value. All commands are legal. The initial state is arbitrary and unknown to the observer; the

known post-report state is part of the physical model, not an unknown-state oracle. Commands and reports may be relabelled without changing these data.

Set

$$Q_{iaj} = \operatorname{tr}(E_{a,j}\sigma_i), \qquad c_{ia} = \operatorname{tr}(G_a\sigma_i), \qquad q_{ia}^{\mathrm{suc}} = 1 - \sum_j Q_{iaj}.$$

A nonempty set $S \subset \{1, \dots, s\}$ is a *closed failure set* when $\sum_{j\in S} Q_{iaj} = 1$ for every $i \in S$ and every command a.

Theorem 9.1 (Finite reset classification, optimum and preparation saddle)**.** *There exists one common visible-history policy which is proper at finite uniform expected cost for every initial density matrix if and only if there is no closed failure set. If there is none, the finite Bellman system*

$$v_i = \min_{a\in\mathcal{A}} \left(c_{ia} + \sum_j Q_{iaj} v_j \right), \qquad 1 \le i \le s, \tag{9.2}$$

has a unique finite nonnegative solution. It is the optimal remaining cost from each reported state σ_i. A deterministic minimizing command at every reported state attains it and is proper.

Define Hermitian start-cost operators

$$L_a = G_a + \sum_j v_j E_{a,j}.$$

The unknown-initial-state value is exactly

$$V_* = \min_{p\in\Delta(\mathcal{A})} \lambda_{\max} \left(\sum_a p_a L_a \right) \tag{9.3}$$

$$= \max_{\rho\succeq 0,\ \operatorname{tr}\rho=1} \min_a \operatorname{tr}(\rho L_a). \tag{9.4}$$

Both extrema are attained. A minimizing initial command distribution followed by the deterministic Bellman continuation is one robustly minimax policy whose uniform bound holds for every initial state; its initial row need not minimize cost separately at each known state. A maximizing density matrix is a primitive least-favorable preparation against every visible-history policy. The latter statement is for unrestricted history; it is not automatically a saddle at every smaller memory budget.

If g distinct command values occur in a chosen deterministic Bellman selector, then $g + 1 \le \min(s, |\mathcal{A}|) + 1$ live labels suffice with a prescribed initial label. If a minimizing initial distribution is a pure command already used by that selector, g suffice. Event stopping permits reuse of a live label for the constant terminal value; state stopping adds one disjoint terminal label.

Proof. If a closed failure set exists, start in any σ_i with $i \in S$. Every command fails with probability one and reports another state in S. Induction makes success impossible under every policy; the positive cost floor makes its expected cost infinite.

Suppose there is no such set. Choose every command independently with strictly positive probability. From each failure state there is a positive-probability directed path to success: otherwise the set of failure states reachable from that state would be nonempty and closed under every command, contradicting the assumption. A shortest such path visits no failure state twice and has at most s edges. Finiteness gives a positive lower bound δ for the probability of following one such path in the next s calls, uniformly in its starting failure state. Consequently survival after ns calls is at most $(1-\delta)^n$. From an arbitrary initial state the first call either succeeds or enters one of the reported states. The expected number of calls is therefore uniformly at most $1 + s/\delta$, and the expected cost at most $c^*(1 + s/\delta)$. This proves existence using even a one-live-label randomized command row. Its role is feasibility, not optimality.

For explicit primitive constants, if the row chooses each command with probability at least $\eta > 0$ and $\epsilon > 0$ is the smallest positive number among the finite collection $Q_{iaj}, q_{ia}^{\mathrm{suc}}$, then one may take $\delta = (\eta\epsilon)^s$. Factors for a shorter path can be bounded below by this same number. Zero probabilities are not divided out or assigned a false positive floor.

Start value iteration at $v^{(0)} = 0$. The finite-horizon dynamic programming recursion gives $v_i^{(n+1)} = \min_a(c_{ia} + \sum_j Q_{iaj} v_j^{(n)})$. These values increase and are bounded by the proper randomized policy's finite cost. Their limit v is finite and solves (9.2), since a finite minimum commutes with this coordinatewise limit. Select a minimizing command $\alpha(i)$ in each row. Iterating its equality bounds every partial expected cost by v_i. The positive floor then bounds the sum of survival probabilities, so the selected policy is proper and has finite cost. The bounded terminal remainder $\mathbb{E}[v_{I_n}\mathbf{1}_{\{\tau>n\}}]$ vanishes. Passing to the limit in the iterated equality identifies its cost as v_i.

For any other proper policy, Bellman's inequality for each chosen command telescopes to v_i no greater than its expected cost plus the same bounded survival remainder. Letting the horizon grow proves its cost at least v_i. An infinite-cost policy cannot improve this bound. The same argument for any finite nonnegative Bellman solution w shows that its greedy policy realizes w and that w is below the cost of every proper policy. Comparing to v in both directions proves uniqueness.

Before the first call the observer has no report, so its first command has one model-independent distribution p. After a failure j, the physical state is exactly σ_j, irrespective of the original state or previous private coins. The already proved continuation lower therefore makes its initial cost at least $\mathrm{tr}[\rho \sum_a p_a L_a]$. The Bellman continuation attains this bound for every ρ. Taking the worst density matrix gives the largest eigenvalue, proving (9.3). Compactness of the probability simplex gives its minimum.

For completeness the dual equality follows directly from finite-dimensional separation. Let $\mathcal{K} = \mathrm{conv}\{L_a\}$ and let $v_* = \min_{L\in\mathcal{K}} \lambda_{\max}(L)$. The compact convex set $\mathcal{K}$ is disjoint from the open convex set $\{H : H \prec v_* I\}$. A separating real linear functional on Hermitian matrices is pairing with a nonzero Hermitian R, with $\inf_{L\in\mathcal{K}} \mathrm{tr}(RL) \geq \sup_{H\prec v_* I} \mathrm{tr}(RH)$. Finiteness of the right side forces $R \succeq 0$, since otherwise moving H arbitrarily far in a negative semidefinite direction makes that pairing unbounded above. Normalize $\mathrm{tr}\, R = 1$; the right supremum is then v_*. This gives a density matrix with $\min_a \mathrm{tr}(RL_a) \geq v_*$. Conversely, pairing any density matrix with a minimizing L bounds that minimum by v_*. This proves equality and dual attainment. The continuation lower makes the maximizing state a witness against every policy, not merely against first-call mixtures.

To implement the policy, use one label for its initial distribution and a pure command label for each distinct value of α. On failure report j, retain the label for $\alpha(j)$. The next row depends only on that retained label. Success commits immediately. If the initial row is already one of these pure rows, use that existing label as the start. The event/state packing lemma gives the terminal counts. No running clock, earlier report or persistent seed is available in addition to the labels. □

Proposition 9.2 (Equality, rigidity and a memory lower certificate)**.** *Let ρ_* be an attaining preparation in* (9.4)*, and put*

$$d_{ia} = c_{ia} + \sum_j Q_{iaj} v_j - v_i \geq 0, \qquad \mathcal{A}_i^* = \{a : d_{ia} = 0\}.$$

For every finite-cost policy with first-command distribution p, its cost satisfies the exact residual identity

$$\mathbb{E}_{\rho_*} C = \mathrm{tr}\left(\rho_* \sum_a p_a L_a\right) + \mathbb{E}_{\rho_*} \sum_{n\geq 1:\, \tau>n} d_{I_n, A_{n+1}}. \tag{9.5}$$

Consequently an unrestricted optimum uses only commands in $\mathcal{A}_i^*$ at every positive-probability continuation from reported state i, under this preparation.

Suppose a set I of reported states is visited with positive probability by every robust optimum evaluated at ρ_*. Let $\chi(I)$ be the least number of subsets covering I whose respective intersections of

optimal command sets are nonempty. Every fixed-row autonomous optimum needs at least $\chi(I)$ live labels. If every minimizing initial row has support not contained in $\mathcal{A}_i^*$ for every $i \in I$, it needs at least $\chi(I)+1$. These are checkable lower certificates once the mandatory set and the initial-row condition have been proved, not assertions that every experiment has the same minimal memory.

Proof. After each failed call, subtract v_{I_n} from the cost of the next call plus the conditional next potential. The conditional difference is exactly $d_{I_n,A_{n+1}}$. Sum to a deterministic horizon, including the first-call cost and potential. The remaining potential has absolute value at most $\max_i v_i$ times the survival indicator. Finite expected cost and the floor make survival tend to zero. Monotone convergence for the nonnegative residual sum yields (9.5). Its first term is at least V_* by the dual witness. A robust optimal policy's cost at ρ_* is at most V_*, so every visited positive-probability residual vanishes.

A retained label serving states in a subset J has one command row. Its support must lie in every $\mathcal{A}_i^*$, $i \in J$, because each such visit has positive probability and any action of positive row weight with positive residual would contradict equality. Thus $\bigcap_{i\in J}\mathcal{A}_i^*$ is nonempty. The labels used by mandatory states cover I, proving the first lower. For an optimum, its first row is a minimizing row in (9.3): its worst-state cost dominates the first-call lower, whose minimum is V_*. Under the additional support condition this initial row can serve none of the mandatory states, so its label is additional to their covering labels. This proves the sharpened count. □

Corollary 9.3 (Arbitrary finite rank-one detector families)**.** *Let the failure projection of detector a be $P_a = |u_a\rangle\langle u_a|$ on any finite-dimensional Hilbert space, with $\|u_a\| = 1$, and let its deterministic call cost be $k_a > 0$. There is no intervening evolution or additional control. If all rays $\mathbb{C}u_a$ coincide, that ray is dark and the all-state value is infinite. Otherwise the value is finite and is given by Theorem 9.1, with*

$$v_i = \min_a\{k_a + |\langle u_i, u_a\rangle|^2 v_a\}, \qquad L_a = k_a I + v_a P_a.$$

Thus the continuation is a finite scalar system, while the unknown-initial-state optimization retains the full spectral geometry of the physical projections. There is no fixed prior or restriction to two detectors or two physical dimensions.

Proof. A rank-one failure satisfies $P_a\rho P_a = \mathrm{tr}(P_a\rho)P_a$, so it is a reset branch with report a and state P_a. If all rays agree, their common ray never succeeds. Otherwise choose two distinct rays and alternate their detectors. Their squared overlap is less than one. After the first failure every switch has this same strictly positive success probability, giving finite uniform cost. The classification theorem applies. Substitution of the reset transition probabilities gives the displayed formulas. The general equality and memory certificates retain the actual observed failure label and therefore apply to this family as well. □

The reduction to a finite absorbing decision problem, Bellman verification, convex separation and spectral duality are classical methods; quantum decision-process frameworks already provide much broader control languages [12, 13, 14]. The contribution claimed here is the explicit primitive reset-instrument criterion, the common all-initial-state spectral value, its realization with charged rows, and the residual-based memory certificate within this experiment. It is distinct from optimizing the probing interval for a fixed detector with unitary evolution [19, 20]. It also does not replace the preserved two-stage reset communication and rank-constrained discrimination results by a first-detection theorem.

10. The complete finite-memory frontier for reset control

The optimal-memory threshold and the cost at every smaller memory are different invariants. We now determine the latter by an attained finite variational principle. The result applies to the full reset class of Section 9; strict positivity of the failure effects is not required. In particular it

includes reports which a least-favorable preparation cannot reach. The key reduction preserves the controller's existing command rows while removing all dependence of the post-report update on the preceding label and command.

Keep the primitive data $E_{a,j}, G_a, \sigma_i$, the positive cost floor, and the absence of a closed failure set from Theorem 9.1. For a command distribution $p \in \Delta(\mathcal{A})$ define

$$c_i(p) = \sum_a p_a c_{ia}, \qquad Q_{ij}(p) = \sum_a p_a Q_{iaj}, \qquad G(p) = \sum_a p_a G_a, \qquad E_j(p) = \sum_a p_a E_{a,j}.$$

All these quantities concern the actual reported reset state. There is no additional measurement, cost observation, free clock, or access to the unknown initial density matrix.

Let V_M be the infimum of the worst-initial-state expected cost over autonomous controllers with at most $M \geq 1$ live labels, a prescribed pure initial label, and mandatory stopping at success. The command row at a label can be mixed. Updates after a failure may initially depend stochastically on the old label, the sampled command, and its report. Controllers with infinite worst-state cost do not affect the infimum.

10.1. **A memory-preserving deterministic update theorem.**

Theorem 10.1 (Dominance with the same command rows)**.** *Suppose an M-label controller has finite worst-initial-state cost C, and its command rows are $p^1, \dots, p^M$, with pure initial label 1. There is another controller with these same rows and the same initial label whose update after report i is one deterministic label $k(i)$, independent of the preceding label and command. Its expected cost is no greater than the original cost at every initial density matrix.*

More precisely, the finite system

$$x_i = \min_{1 \leq k \leq M} \left\{ c_i(p^k) + \sum_j Q_{ij}(p^k) x_j \right\} \tag{10.1}$$

has a unique finite nonnegative solution, with $0 \leq x_i \leq C$. Choose $k(i)$ attaining the minimum. The resulting continuation is proper from every reset state, and its full cost from an initial ρ is exactly

$$\operatorname{tr}(\rho H(p^1, x)), \qquad H(p, x) = G(p) + \sum_j x_j E_j(p). \tag{10.2}$$

This is a pointwise cost comparison, without a convexification of the hard memory constraint.

Proof. Regard the finite list of rows as the available choices at each reset state. A choice k has mean cost $c_i(p^k)$ and substochastic reset transition row $Q_{ij}(p^k)$. Any extra information about the sampled command or earlier private coins does not alter the physical state σ_j after report j.

The original controller can be started with label 1 and physical state σ_i, for every i: the robustness quantifier includes *all* initial density matrices. This gives, for the same list of rows, a proper history-dependent policy of cost at most C from each reset state. This comparison restart is used only to prove finiteness; it is not an additional operation supplied to the eventual controller. In particular it remains valid when some reset reports are inaccessible from a least-favorable preparation.

Starting the finite-horizon recursion for the displayed row choices at zero gives increasing vectors bounded above by $C\mathbf{1}$. Their limit x is finite and satisfies (10.1). For a minimizing selector $k(i)$, iterate its equality through a finite horizon. The accumulated expected cost is bounded by x_i. The positive cost floor therefore bounds the sum of survival probabilities, so the selector is proper from every state. The bounded remaining potential tends to zero; its expected total cost is x_i.

Conversely Bellman's inequality holds conditionally at each call of any policy using these rows, even if it retains the sampled command and earlier coins. Telescope it to a finite horizon. For a finite-cost policy, the survival remainder vanishes by the same positive floor, proving that its cost from reset state i is at least x_i. Infinite-cost policies cannot improve the bound. The same verification applied to any finite nonnegative solution gives uniqueness.

After the original first call, each reached continuation from report j uses the same list of rows and hence costs at least x_j. Conditioning on that call proves that its cost at every ρ is at least (10.2). The new controller starts with row p^1 and, after report j, writes precisely the existing label $k(j)$. Thereafter it implements the minimizing selector and attains this expression. The actual device thus uses at most the same M labels, with no uncounted row-selection memory. □

The all-initial-state quantifier is essential to the bounded comparison in this proof. Properness for just one supplied initial state would not by itself imply a finite continuation value at every reset state. Nor does the theorem assert that mixed command rows can always be replaced by pure commands: only their post-report selection is made deterministic.

10.2. **A compact exact variational principle.** Choose once and for all a full-support row $\bar{p}$, for example the uniform row on $\mathcal{A}$. The proof of Theorem 9.1 makes $\bar{Q} = (Q_{ij}(\bar{p}))$ transient. Consequently

$$\bar{x} = (I - \bar{Q})^{-1}(c_i(\bar{p}))_{i=1}^{s}, \qquad C_0 = \lambda_{\max} H(\bar{p}, \bar{x}) < \infty. \tag{10.3}$$

The constant one-row controller has worst-state cost C_0, and $\bar{x}_i = \mathrm{tr}(\sigma_i H(\bar{p}, \bar{x})) \le C_0$. Thus this is a primitive, explicit compactness bound, not an assumed upper bound on an unknown optimal trajectory.

For $k : \{1, \dots, s\} \to \{1, \dots, M\}$ let $\mathcal{F}_{M,k}$ consist of the rows $p^1, \dots, p^M \in \Delta(\mathcal{A})$ and $x \in [0, C_0]^s$ satisfying

$$x_i \ge c_i(p^{k(i)}) + \sum_j Q_{ij}(p^{k(i)}) x_j \qquad (1 \le i \le s). \tag{10.4}$$

There are M^s possible selectors and each corresponding feasible set is compact; some of them may be empty.

Theorem 10.2 (Exact attained frontier at every memory)**.** *For every $M \ge 1$,*

$$V_M = \min_k \min_{(p^1, \dots, p^M, x) \in \mathcal{F}_{M,k}} \lambda_{\max} H(p^1, x). \tag{10.5}$$

The minimum is finite and attained, and an attaining controller has deterministic report-dependent updates as in Theorem 10.1. For $0 \le b \le C_0$, there is an M-label controller of worst-state cost at most b if and only if one of the finite systems

$$(p^1, \dots, p^M, x) \in \mathcal{F}_{M,k}, \qquad H(p^1, x) \preccurlyeq bI \tag{10.6}$$

is feasible.

In particular V_M is nonincreasing in M and equals the unrestricted value V_* once $M \ge \min(s, |\mathcal{A}|) + 1$. The exact optimal label count, without a strict-exposure hypothesis, is

$$m_{\mathrm{opt}} = \min\{M : (10.6) \text{ is feasible at } b = V_*\}. \tag{10.7}$$

Every $M < m_{\mathrm{opt}}$ has the strict gap $V_M > V_*$. For algebraic primitive data, V_M, m_{opt}, and an attaining controller admit finite exact real-algebraic procedures. No polynomial complexity or general minimax saddle at fixed M is asserted.

Proof. A feasible supersolution bounds every finite-prefix continuation cost under the displayed selector by x_i. The positive cost floor therefore makes that selector proper from every reset state. Its true continuation vector is at most x, and its full expected cost operator is at most $H(p^1, x)$, because every $E_j(p^1)$ is positive semidefinite. Thus every feasible point is an actual controller upper certificate.

Conversely take any controller of worst-state cost $b \le C_0$. Theorem 10.1 supplies its row-list Bellman vector $x \le b\mathbf{1}$, a minimizing selector, and a cost operator bounded by bI. These are feasible in (10.6). The bound $x \le b\mathbf{1}$ uses the comparison starting at each σ_i with label 1; it is not inferred from positive exposure of a particular dual state.

All rows equal to $\bar{p}$, with $x = \bar{x}$, give a feasible point of value C_0, for every M. A finite union of the compact $\mathcal{F}_{M,k}$ is therefore nonempty and the continuous objective attains its minimum. If the controller infimum is strictly below C_0, apply the converse to an approximating sequence below C_0; if it equals C_0, use the displayed one-row certificate. Together with the first implication, this identifies the minimum with the controller infimum. At a minimizing feasible point the constructed controller cannot have strictly smaller worst-state cost, since that would contradict the equality of the infima. This proves actual attainment as well as the budget equivalence.

Allowing duplicate rows shows monotonicity. The unrestricted implementation in Theorem 9.1 proves the stated finite saturation bound. At every smaller M, equality with V_* is equivalent to an attaining optimal controller by the result already proved, giving (10.7) and the strict-gap assertion. This supplies a general variational classification even where the simpler transversal formula of Theorem 11.1 is inapplicable.

For algebraic data, $\bar{x}$ and C_0 are algebraic. For each selector, the row constraints and (10.4) are polynomial inequalities with algebraic coefficients. On separating real and imaginary parts, the Hermitian inequality in (10.6) is equivalent to nonnegativity of its principal minors. Exact real-algebraic elimination determines its feasible budget set and least endpoint; compactness ensures that endpoint is included and provides algebraic attaining coordinates. Finally enumerate the finite range in (10.7). This is an exact computability conclusion from the proved finite system, not a claim that a numerical optimization certifies a tie or that an elimination computation has been performed for arbitrary inputs. □

With the inherited single terminal decoder, event stopping has the same total-label frontier V_M. A state-stopped device with $M+1$ total labels has the live-label frontier V_M, since its terminal label is disjoint. The prescribed pure start is already charged in (10.5): p^1 is the command distribution at a counted initial label, not a freely mixed choice of hidden initial labels.

Proposition 10.3 (Continuity of the full frontier)**.** *Fix the Hilbert dimension and the finite command and report alphabets. For every fixed M, V_M is continuous in the primitive effects, costs, and reset states on the set of valid reset instruments with a positive cost floor and no closed failure set. Thus if $V_M > V_*$ at one instrument, the same strict inequality holds throughout some neighborhood within this class. This assertion concerns values and strict gaps; the exact optimal-memory count need not be continuous.*

Proof. Near a fixed instrument, the same full-support row $\bar{p}$ has transient $\bar{Q}$: its spectral radius is strictly less than one, an open condition. The resolvent in (10.3) and its spectral value C_0 are continuous, so the compactness bounds are locally uniform.

An attaining controller can be chosen with deterministic report updates and with its true continuation vector finite from every reset state. Its substochastic reset matrix Q^k, whose ith row is $Q_{ij}(p^{k(i)})$, is transient. Indeed its finite expected cost and the positive cost floor bound expected duration from every state, so the entries of $\sum_{n\geq 0}(Q^k)^n$ are finite. Keep its rows and selector fixed under a small perturbation. The same matrix remains transient, and both $(I - Q^k)^{-1}(c_i(p^{k(i)}))_i$ and the resulting initial cost operator vary continuously. This proves upper semicontinuity of V_M.

For the reverse inequality, take a convergent sequence of primitive data and attaining points in (10.5), along a subsequence realizing the lower limit of the values. The locally uniform cap bounds the continuation vectors, the rows lie in compact simplexes, and a further subsequence has a constant selector. Pass to a limit in the supersolution inequalities and in the initial matrix upper bound at their attaining values. The limit is an actual controller certificate by the first part of the preceding proof. Its value is no greater than the lower limit, proving lower semicontinuity. This argument does not need the limiting continuation vector to lie below one particular choice of cap; its finite supersolution already constructs a proper controller.

The unrestricted value equals $V_{\min(s,|\mathcal{A}|)+1}$, so it is continuous as well. Continuity of their difference proves the strict-gap statement. □

The domination step is specific to observed resetting: after each report the physical state no longer depends on the earlier record. Bellman verification, finite-state-controller optimization, spectral minimax at unrestricted memory, and real-algebraic elimination are classical ingredients [81, 82, 83, 86, 15]. The present result combines them in an exact counted-memory normal form and compact necessary-and-sufficient budget certificate. It does not prove such a reduction for a general partially observed process or for the original squared-sign risk frontier.

10.3. **Closed frontiers for arbitrary action-cover instances.** The preceding variational formula can be solved explicitly for the rational family in Theorem 11.3. Let $A_1, \dots, A_s$ be nonempty proper subsets of $[q] = \{1, \dots, q\}$, $q \geq 2$. On the basis $e_1, \dots, e_s, f_1, \dots, f_q$, keep

$$\sigma_i = |e_i\rangle\langle e_i|, \quad E_{a,i} = \frac{r}{s} I, \quad G_a(e_i) = 1 + \mathbf{1}_{\{a \notin A_i\}}, \quad G_a(f_j) = 3 + \mathbf{1}_{\{a=j\}}, \qquad 0 \leq r < 1.$$

Here $G_a(e_i)$ and $G_a(f_j)$ denote the corresponding diagonal entries. The success effect is $(1-r)I$. This also defines the degenerate endpoint $r = 0$, where every first call succeeds. For $H \subseteq [q]$ put

$$U(H) = \{i : H \cap A_i = \varnothing\}, \qquad b_a(H) = |\{i \in U(H) : a \notin A_i\}|, \qquad \lambda = \frac{r}{s(1-r)}.$$

Let $b_{(1)}(H) \leq \cdots \leq b_{(q)}(H)$ be these q nonnegative integers in increasing order, with repetitions.

Theorem 10.4 (The complete action-cover cost frontier)**.** *For every $M \geq 1$, the exact worst-state cost at M live labels is*

$$V_M = 3 + \frac{r}{1-r} + \min_{\substack{H \subseteq [q] \\ |H| \leq M-1}} \min_{1 \leq k \leq q} \frac{1 + \lambda \sum_{j=1}^{k} b_{(j)}(H)}{k}. \tag{10.8}$$

An optimum uses pure continuing command rows indexed by H and an initial row uniform on k commands with the k smallest values of $b_a(H)$. The initial row and these continuing rows suffice for deterministic report-dependent routing. Both minima are finite enumerations, including all tied and degenerate cases.

For a fixed set system, after subtracting $3 + r/(1-r)$ the frontier is a continuous, nondecreasing, piecewise affine concave function of $\lambda \geq 0$. At $r = 0$ all memory values equal $3 + 1/q$ and one label is optimal; for $r > 0$ the unrestricted-memory threshold is the transversal number plus one, as in Theorem 11.3.

Proof. Fix any list of rows and put

$$h_i = \min_{1 \leq k \leq M} p^k(A_i^c), \qquad \overline{h} = \frac{1}{s} \sum_i h_i.$$

The reset probabilities are independent of the command, so (10.1) has the explicit solution

$$x_i = 1 + h_i + \frac{r(1+\overline{h})}{1-r}, \qquad \frac{1}{s} \sum_i x_i = \frac{1+\overline{h}}{1-r}.$$

Its first-cost eigenvalues on f_j are $3 + p_j^1 + r(1+\overline{h})/(1-r)$; those on e_i are at most $2 + r(1+\overline{h})/(1-r)$. Hence the former strictly dominate, and the exact cost for these rows after optimal routing is

$$3 + \frac{r}{1-r} + \max_a p_a^1 + \lambda \sum_i \min_k p^k(A_i^c). \tag{10.9}$$

This formula applies to all row choices; success has probability $1 - r > 0$ on every call, so no properness exception is present.

For each row p^k with $k \geq 2$, with the other rows fixed, the last sum in (10.9) is a continuous concave function of p^k: each summand is the minimum of one linear function and constants. Its minimum over a simplex has a vertex minimizer. Start from a global minimum, which exists by compactness, and replace these rows successively by such vertices. No replacement can increase the objective; global minimality then keeps it unchanged. Thus all continuing rows may be pure.

Let H be their distinct actions. When $i \notin U(H)$ some continuing row has zero failure penalty and $h_i = 0$. When $i \in U(H)$ all continuing rows have penalty one, giving $h_i = p^1(A_i^c)$, also when H is empty. It follows that

$$\sum_i h_i = \sum_a b_a(H) p_a^1.$$

The remaining initial-row minimization is therefore

$$\min_{p \in \Delta([q])} \left\{ \max_a p_a + \lambda \sum_a b_a(H) p_a \right\}. \tag{10.10}$$

For completeness introduce $t \in [1/q, 1]$ and impose $p_a \leq t$. For fixed t, the linear term is minimized by filling the smallest $b_a(H)$ coordinates first, each up to t, with at most one partially filled coordinate. An exchange of mass from a larger coefficient to a smaller unsaturated one proves this assertion, including ties. On each interval $[1/(k+1), 1/k]$ the minimized expression $t + \lambda \sum_a b_a(H) p_a$ is affine in t. Its minimum has an endpoint minimizer; the endpoints are exactly $t = 1/k$, $1 \leq k \leq q$. At such an endpoint a minimizing p is uniform on k smallest coefficients, and its value is $(1 + \lambda \sum_{j=1}^k b_{(j)}(H))/k$. The epigraph problem in t, p has the same optimum as (10.10), since one may always reduce t to $\max p_a$. This proves (10.8) and the attaining construction.

For fixed H, k the displayed value after subtracting the common term is affine with nonnegative slope in λ. The finite minimum is therefore continuous, piecewise affine, concave and nondecreasing. At $r = 0$ its minimum is $1/q$, attained by a single uniform row. For $r > 0$, an optimal initial row at the unrestricted value must be uniform on all q actions; because each A_i is proper it has positive penalty at every uncovered state. Thus zero excess cost requires H to meet every A_i, and a smallest transversal is sufficient. Equivalently this is the already proved positive-effect threshold. □

Corollary 10.5 (Arbitrarily long strict memory hierarchies)**.** *For every $n \geq 2$ there is a rational reset experiment in dimension $2n$, with n commands, n failure reports, success probability $1/2$ at every call and all costs in $[1, 4]$, for which*

$$V_M = 4 + \frac{1}{n} + \frac{(n-1)\max\{n - M + 1, 0\}}{n^2} \qquad (M \geq 1). \tag{10.11}$$

In particular $V_1 > V_2 > \cdots > V_{n+1} = V_*$, and each of the first n memory increments saves exactly $(n-1)/n^2$ in worst-state expected cost. All failure effects are positive definite. In this family the retained first-residual certificate satisfies

$$D_M = \frac{V_M - V_*}{2}. \tag{10.12}$$

Thus it is a strict lower certificate below the optimum threshold, whereas (10.11) *is the full repeated-experiment cost frontier.*

The n strict inequalities persist in an open neighborhood within the class of valid instruments. They also persist under explicit arbitrarily small perturbations for which a failure effect and a cost operator do not commute. These perturbations retain positive failure effects, dimension $2n$, and mean cost operators $I \preccurlyeq G_a \preccurlyeq 4I$; the exact formula (10.11) *refers to the unperturbed instrument.*

Proof. Use the preceding construction with $q = s = n$, $A_i = \{i\}$ and $r = 1/2$, so $\lambda = 1/n$. For a continuing action set H, write $u = n - |H|$ for the number of uncovered singleton sets. Then

$b_a(H) = u - 1$ on the u uncovered actions and $b_a(H) = u$ on the remaining actions. If $u > 0$, the candidate initial-row value at a support size k is

$$\begin{cases} k^{-1} + (u-1)/n, & 1 \le k \le u, \\ u/n + (1 - u/n)/k, & u \le k \le n. \end{cases}$$

The first expression is minimized at $k = u$; the second is nonincreasing in k and agrees with the first at u. Consequently a minimum always occurs at $k = n$, with value $1/n + u(n-1)/n^2$. For $u = 0$ the same value is $1/n$. Taking the largest allowable H makes $u = \max\{n - M + 1, 0\}$ and proves (10.11). The construction uses a uniform initial command row and one pure row for each action of H; reports in H select their pure row and the remaining reports select the initial row. The stated data and positivity properties follow directly from the displayed primitive construction.

Finally, the Bellman residuals for this family are $d_{ia} = \mathbf{1}_{\{a \ne i\}}$. The matrix in (11.4) retains only the first post-failure penalty, so its minimized eigenvalue is obtained from (10.9) by replacing $\lambda = 1/n$ by $r/s = 1/(2n)$. The same row-purification argument applies. For $k \le u$ the candidate is $1/k + (u-1)/(2n)$; for $k \ge u$ it is $u/(2n) + (1 - u/(2n))/k$. Again $k = n$ is a minimizer, and subtracting $V_* = 4 + 1/n$ gives $D_M = u(n-1)/(2n^2)$, proving (10.12). This equality is specific to the present family and is not a universal identity for D_M.

Continuity from Proposition 10.3, applied to the finitely many adjacent gaps, proves the neighborhood assertion. For an explicit perturbation put $X = |e_1\rangle\langle f_1| + |f_1\rangle\langle e_1|$ and replace only $E_{1,1}$ by $E_{1,1} + \epsilon X$, where

$$0 < \epsilon < \frac{n-1}{8n^2}.$$

The failure effect remains positive definite, and the corresponding success effect $\frac{1}{2}I - \epsilon X$ remains positive definite. All other primitive data are unchanged. Since $G_1(e_1) = 1$ and $G_1(f_1) = 4$, this modified failure effect does not commute with G_1.

These cost and report effects have a joint physical implementation. Charge one on every failure. Split the success effect B_a into

$$F_{a,9} = (G_a - I)/8, \qquad F_{a,1} = B_a - F_{a,9},$$

charging nine and one, respectively, and retaining only the common success report at the control port. Both effects are positive: $0 \preccurlyeq F_{a,9} \preccurlyeq 3I/8$ and $B_a \succeq (1/2 - \epsilon)I$, with $\epsilon < 1/32$. Their sum is B_a, and the aggregate mean cost is $I + 8F_{a,9} = G_a$. Each failure branch uses its stated measure-and-prepare reset map, and each success subbranch can use any terminal prepared state. Thus the perturbation is an actual instrument with positive, bounded paid costs and no additional observation, rather than an unsupported independent assignment of noncommuting cost and report operators.

The perturbation has zero expectation in every σ_i, so it changes none of the reset transition probabilities or continuation Bellman vectors for any chosen list of command rows. Each such vector satisfies $x_i \le 4$, since reset costs are at most two and each subsequent call succeeds with probability $1/2$. Its initial cost operator changes only by $\epsilon p_1^1 x_1 X$, of norm at most 4ϵ. Hence the optimized values satisfy $|V_M(\epsilon) - V_M(0)| \le 4\epsilon$ for every M. Each adjacent gap is therefore at least $(n-1)/n^2 - 8\epsilon > 0$. The common unrestricted upper memory bound still gives $V_{n+1}(\epsilon) = V_*(\epsilon)$. This proves the claimed noncommuting realization without extending the unperturbed closed formula to perturbed data. □

The family shows that the full frontier carries information absent from its saturation threshold: each intermediate memory can have a different, attained optimal value under a fixed success probability and uniformly bounded primitive costs. Formula (10.8) retains the combinatorial difficulty of the arbitrary incidence table, while giving a complete finite answer rather than an efficient algorithm claim. The reset assumption and the declared observation port remain essential;

none of these statements completes the separate general stopping, phase-type timing, or squared-sign minimax-risk problems.

11. Exact optimal memory for reset experiments with tied actions

The realization of a prescribed response and the implementation of an optimal feedback rule are different problems. We now give an exact memory classification for a class of controlled experiments in which optimal actions may be tied. The class allows any finite command set, any finite number of reset reports, arbitrary finite Hilbert dimension, state-dependent call costs, and an unknown initial density matrix. The additional hypothesis is strict positivity of each failure effect. It supplies the reachability needed for a sharp converse; it is not a uniqueness assumption on an optimal decision.

Use the reset instrument of Section 9:

$$\mathcal{I}_{a,i}(\rho) = \operatorname{tr}(E_{a,i}\rho)\sigma_i, \qquad \sum_{i=1}^{s} E_{a,i} \preccurlyeq I, \qquad c_* I \preccurlyeq G_a \preccurlyeq c^* I.$$

Here $a \in \mathcal{A}$, $1 \le i \le s$, $c_* > 0$, and success requires immediate commitment. There is no additional state measurement or free reset. Throughout this section assume

$$E_{a,i} \succ 0 \quad (a \in \mathcal{A},\ 1 \le i \le s), \qquad \text{there is no closed failure set.} \tag{11.1}$$

Theorem 9.1 supplies the finite Bellman vector v, the start operators $L_a = G_a + \sum_i v_i E_{a,i}$, and the value

$$V_* = \min_{p \in \Delta(\mathcal{A})} \lambda_{\max}\Big(\sum_a p_a L_a\Big).$$

Define the Bellman residuals and optimal action sets by

$$d_{ia} = \operatorname{tr}(G_a\sigma_i) + \sum_j \operatorname{tr}(E_{a,j}\sigma_i)v_j - v_i, \qquad \mathcal{A}_i^* = \{a : d_{ia} = 0\}.$$

Every d_{ia} is nonnegative and every $\mathcal{A}_i^*$ is nonempty. Put

$$\mathcal{P}_* = \left\{ p \in \Delta(\mathcal{A}) : \lambda_{\max}\Big(\sum_a p_a L_a\Big) = V_* \right\}, \qquad I(p) = \{i : \operatorname{supp} p \subseteq \mathcal{A}_i^*\}.$$

For $J \subseteq \{1, \dots, s\}$ let

$$t(J) = \min\{|H| : H \subseteq \mathcal{A},\ H \cap \mathcal{A}_i^* \ne \varnothing\ (i \in J)\}, \qquad t(\varnothing) = 0. \tag{11.2}$$

This is an ordinary transversal number of the displayed finite set system. It counts distinct continuing command rows, rather than physical states or the support size of a least-favorable preparation.

Theorem 11.1 (Exact optimal-memory formula with ties)**.** *Under* (11.1)*, the least number of live labels of a fixed-program, prescribed-pure-start autonomous controller attaining the all-initial-state optimum V_* is*

$$m_* = 1 + \min_{p \in \mathcal{P}_*} t\big(\{1, \dots, s\} \setminus I(p)\big). \tag{11.3}$$

Every minimum is realized by one optimal initial command row and pure continuing command rows. The post-report transition may be chosen as a deterministic function of the current reset report alone. Arbitrary stochastic transitions, dependence on the previous label and command, and mixed continuing rows cannot lower (11.3)*.*

Event stopping has the same total-label minimum. State stopping has total-label minimum $m_* + 1$. For algebraic primitive data the value, optimal label count, and one realizing controller can be determined by a finite exact procedure; polynomial running time is not asserted.

Proof. Choose an attaining dual density matrix ρ_* from Theorem 9.1. Consider any robust optimal controller, write its fixed command rows as $p^1, \dots, p^M$, and index its pure initial label by 1. Its cost from every ρ is at least $\operatorname{tr}(\rho \sum_a p_a^1 L_a)$. Consequently $\lambda_{\max}(\sum_a p_a^1 L_a) \le V_*$, and the reverse inequality holds by the definition of V_*. Thus $p^1 \in \mathcal{P}_*$.

The Bellman residual identity of Proposition 9.2 holds at ρ_*. Its initial term is at least V_*, whereas the whole cost is at most V_*. Every nonnegative residual on a reached continuation therefore vanishes. For every reset report i, its probability on the first call is

$$\sum_a p_a^1 \operatorname{tr}(E_{a,i}\rho_*) > 0.$$

This strict inequality follows from positive definiteness and holds even if ρ_* is singular or p^1 has proper support. After this report at least one label k is reached with positive probability. The next command has row p^k, so $\sum_a p_a^k d_{ia} = 0$. Nonnegativity implies $\operatorname{supp} p^k \subseteq \mathcal{A}_i^*$.

If $i \notin I(p^1)$, that label cannot be 1. For each other label which serves one or more such states, choose any action in the support of its row. It belongs to every optimal action set served by that row. These at most $M-1$ chosen actions meet all $\mathcal{A}_i^*$ with $i \notin I(p^1)$. This proves the lower bound in (11.3). It uses only first-failure events; no reachability conclusion is inferred from an unrelated path law.

Conversely, fix $p \in \mathcal{P}_*$ and a smallest transversal H of the remaining states. Use one initial label with command row p and one pure command label for each $a \in H$. After a failure report i, return to the initial label when $i \in I(p)$; otherwise select a fixed $a \in H \cap \mathcal{A}_i^*$. Every continuing row is then Bellman optimal for its actual reset state. Bellman's equality bounds every finite-prefix cost from σ_i by v_i. The positive cost floor implies finite expected duration, and the bounded remaining potential vanishes. Hence the continuation cost is exactly v_i. From an arbitrary initial state the full cost is $\operatorname{tr}(\rho \sum_a p_a L_a) \le V_*$. This is a proper robust optimum. There are finitely many supports of p, so the integer minimum in (11.3) is attained. Reusing a live label for the single terminal decoder is permitted exactly in the event-stopped interface, giving the claimed terminal counts.

For the effective assertion enumerate deterministic command selectors on the s reset states. A proper selector has cost $(I - Q_\alpha)^{-1} c_\alpha$; properness and this vector can be checked by finite algebraic operations and spectral or graph tests. Some proper selector is simultaneously optimal by Theorem 9.1, so the enumeration determines v. The spectral minimization defining V_* is a compact semialgebraic optimization. For each nonempty support $T \subseteq \mathcal{A}$, the assertion that some $p \in \mathcal{P}_*$ has exactly that support is a first-order real-algebraic condition. For example, impose $p_a > 0$ on T, $p_a = 0$ off T, $\sum_a p_a = 1$, and $\sum_a p_a L_a \preccurlyeq V_* I$; positivity of a Hermitian matrix is expressed by its principal minors after separating real and imaginary parts. Exact elimination decides this condition and supplies a sample when it is true. Finite transversal enumeration then gives the minimum and the stated implementation. The procedure does not assume that a numerical optimizer has certified a tie. □

The positive-effect hypothesis can be weakened to the existence of one attaining ρ_* for which $\operatorname{tr}(E_{a,i}\rho_*) > 0$ for all a, i; the same proof applies. The stated version is a direct primitive condition, without an unknown optimal-trajectory assumption. When some reports can be inaccessible, the formula for all s reset states need not be necessary: an unreachable state may carry an arbitrary optimal action set. In that case the lower certificates of Proposition 9.2 retain their original scope.

11.1. **A finite gap certificate for every insufficient memory.** For $M \ge 1$ and rows $p^1, \dots, p^M \in \Delta(\mathcal{A})$ put

$$r_i(p^1, \dots, p^M) = \min_{1 \le k \le M} \sum_a p_a^k d_{ia}, \qquad K_M(p^1, \dots, p^M) = \sum_a p_a^1 L_a + \sum_{a,i} p_a^1 r_i(p^1, \dots, p^M) E_{a,i},$$

and define the finite variational quantity

$$D_M = \min_{(p^1,\dots,p^M)\in\Delta(\mathcal{A})^M} \lambda_{\max}(K_M(p^1,\dots,p^M)) - V_*. \tag{11.4}$$

Theorem 11.2 (Attained certificate of a strict optimality gap)**.** *Under* (11.1)*, every robustly proper controller with at most M live labels has worst-initial-state expected cost at least $V_* + D_M$. The minimum defining D_M is attained, and*

$$D_M > 0 \quad\Longleftrightarrow\quad M < m_*.$$

For algebraic primitive data, D_M is computable by exact real algebraic optimization. It is a lower certificate, not a claimed formula for the entire memory-constrained cost frontier.

Proof. For the fixed rows of a controller, condition on the first command and its failure report i. Whichever label the controller selects, the expected Bellman residual on its next call is at least r_i. All later residuals are nonnegative. The residual identity, now at any initial ρ, gives cost at least $\operatorname{tr}(\rho K_M)$, and maximization over density matrices gives $\lambda_{\max}(K_M)$. A controller of infinite worst-state cost satisfies the lower bound trivially.

The function in (11.4) is continuous on a compact product of simplexes. Moreover $K_M \succeq \sum_a p^1_a L_a$, so $D_M \geq 0$. If $D_M = 0$, take minimizing rows. Then $p^1 \in \mathcal{P}_*$. Evaluate K_M at an attaining dual ρ_*. Its first term is at least V_* and each coefficient $\sum_a p^1_a \operatorname{tr}(E_{a,i}\rho_*)$ is strictly positive. Since its value is at most $\lambda_{\max}(K_M) = V_*$, every r_i is zero. Thus every reset state has at least one row supported on its optimal action set. The construction in Theorem 11.1 implements an optimal controller with these rows, hence $M \geq m_*$. Conversely an optimal controller supplied there has $r_i = 0$ for all i, so $D_M = 0$ whenever $M \geq m_*$. Compactness makes the remaining cases strictly positive. Finite minima, products and Hermitian eigenvalue inequalities are semialgebraic, proving the last assertion. □

11.2. **Every finite optimal-action cover has a physical realization.** The preceding set system is not merely an artifact of a bound. Arbitrary finite covering problems occur in reset instruments with commuting cost and effect operators with strictly positive failure effects and constant success probability.

Theorem 11.3 (Universality of the optimal-memory covering problem)**.** *Let $q \geq 2$ and let $A_1, \dots, A_s$ be nonempty proper subsets of $\{1, \dots, q\}$. Let t be their transversal number. For every fixed rational $0 < r < 1$ there is an explicit rational reset instrument with commuting cost and effect operators in dimension $s + q$, with q commands, s failure reports, success probability $1 - r$ at every call, and costs in $[1, 4]$, such that*

$$V_* = 3 + \frac{r}{1-r} + \frac{1}{q}, \qquad m_* = t + 1.$$

Its unique optimal initial command row is uniform. The minimum is over all stochastic autonomous controllers with a prescribed pure initial label, and includes arbitrary randomized post-report updates. Thus determining optimal memory is NP-hard even on this class, while the unrestricted value and an optimal command distribution are explicit.

Proof. Take an orthonormal basis $e_1, \dots, e_s, f_1, \dots, f_q$ and set

$$\sigma_i = |e_i\rangle\langle e_i|, \qquad E_{a,i} = \frac{r}{s} I.$$

The failure map is $\mathcal{I}_{a,i}(\rho) = (r/s)\operatorname{tr}(\rho)\sigma_i$. For instance its Kraus operators are $\sqrt{r/s}\,|e_i\rangle\langle b|$ for all displayed basis vectors b. Add any success measure-and-prepare branch with effect $(1 - r)I$. Their sum is trace preserving, so these are actual quantum instruments. They commute at the level of the relevant effects and all post-failure states are classical basis states.

Specify the positive diagonal cost operator G_a by

$$\langle e_i, G_a e_i\rangle = 1 + \mathbf{1}_{\{a\notin A_i\}}, \qquad \langle f_j, G_a f_j\rangle = 3 + \mathbf{1}_{\{a=j\}}.$$

These means are physically realizable without extra controller information. In each call measure the displayed diagonal basis, charge its indicated cost, then use fresh coins to choose success or one of the reset reports and prepare the corresponding state. The intermediate basis outcome and realized cost are not an additional observation at the control port. Averaging them gives exactly the declared instrument and mean cost; no coherence-dependent hidden assumption is needed.

All commands have $Q_{iaj} = r/s$. Since every A_i is nonempty, $v_i = (1-r)^{-1}$ solves the Bellman equations and is their finite solution. Its optimal action set at i is exactly A_i. The start operator is $L_a = G_a + r(1-r)^{-1}I$. For any initial command row p, its eigenvalues on the e_i coordinates are

$$\frac{1}{1-r} + p(A_i^c) \le 2 + \frac{r}{1-r},$$

whereas its eigenvalues on the f_j coordinates are $3 + r(1-r)^{-1} + p_j$. The latter strictly dominate, hence

$$\lambda_{\max}\Big(\sum_a p_a L_a\Big) = 3 + \frac{r}{1-r} + \max_j p_j.$$

Its unique minimum is the uniform row, with the displayed value. Its full support lies in none of the proper sets A_i, so Theorem 11.1 gives $m_* = 1 + t$ and constructs an attaining controller.

The matrices have dimension and rational encoding length polynomial in the incidence table of the set system; one may fix $r = 1/2$. Minimum transversal is the classical covering problem [84]. Restriction to proper nonempty subsets does not remove its hardness: adjoining an unused action makes every nonempty set proper without changing its transversal number. An oracle for the optimal label number in the displayed physical family would therefore return the transversal number by subtracting one. This proves the complexity assertion by an explicit reduction, not by a general complexity claim about POMDPs. □

Example 11.4 (Why positive reachability cannot be dropped)**.** On $\mathbb{C}^2$ take one reset state $\sigma = |0\rangle\langle 0|$, two commands a, b, and

$$E_a = E_b = \tfrac{1}{2}|0\rangle\langle 0|, \qquad G_a = \mathrm{diag}(1,6), \qquad G_b = \mathrm{diag}(2,5).$$

There is no closed failure set. The Bellman value is $v = 2$ and its unique optimal continuation command is a. The start operators are $L_a = \mathrm{diag}(2,6)$ and $L_b = \mathrm{diag}(3,5)$, so $V_* = 5$ and the unique minimizing initial row is the pure command b. Applying (11.3) without positive reachability would give two labels. Nevertheless the constant b controller has cost operator $\mathrm{diag}(4,5)$: from σ it pays 2 per call with failure probability $1/2$, and from $|1\rangle$ it pays 5 and succeeds immediately. It is a one-label robust optimum. The least-favorable preparation $|1\rangle\langle 1|$ never sees the reset report. Thus a locally nonoptimal continuation can be hidden below the robust optimum when the exposing preparation cannot reach it. This verifies the need for the reachability hypothesis by exact primitive data.

For example, the three optimal sets $\{1,2\}$, $\{2,3\}$ and $\{1,3\}$ give $m_* = 3$, although every report state has two optimal commands. Pairwise uniqueness of Bayes actions plays no role. More generally the count can grow with an arbitrary finite covering instance while the success probability remains fixed and every failure effect remains strictly positive.

Bellman verification and shortest-path optimization are classical [81]; search over finite stochastic controllers also has an extensive history [82, 83]. The assertions proved here concern the exact counted-row invariant, its initializer interaction, the strict finite gap certificate and its realization inside the stated reset experiment. They do not assert a general classification of partially observed control or replace the distinct squared-sign minimax-risk problem.

12. Reversible response groups and common controlled memory

The fixed-phase results concern one transition matrix. Several commands require one family of matrices on the same retained labels, and checking every command in isolation is insufficient. We give an intrinsic finite-realization criterion for responses whose minimal linear dynamics is reversible and compact, including any finite family of unitary commands on a prepared finite-dimensional quantum system. We then obtain a matching arithmetic memory law for a pair of noncommuting involutions. This is a theorem about exact controlled response realization, not an assertion that a simulator's private label is an observed sufficient statistic for arbitrary feedback optimization.

Positive realization through invariant polyhedral sets is classical [48, 47, 46]. The argument below uses a section of the realizing simplex before passing to the observable quotient; auxiliary hidden-state responses need not belong to the minimal response space. Compact reversibility then turns containment into equality. The resulting finite-group obstruction concerns all command words, rather than a single spectral restriction.

12.1. **The minimal response space.** Let A and C be nonempty finite alphabets of commands and terminal outcomes. A supplied response is a function $b : A^* \to \Delta(C)$. Its residual after a word w is $b^w(z) = b(wz)$, including the empty word $\varnothing$. Put

$$V_b = \operatorname{span}_{\mathbb{R}}\{b^w : w \in A^*\}, \qquad (L_a f)(z) = f(az).$$

Assume V_b is finite dimensional. It is invariant under each L_a, since $L_a b^w = b^{wa}$. Call the response *compactly reversible* if the restrictions $L_a|_{V_b}$ are invertible and the closure G_b of their generated group is a compact subgroup of $\mathrm{GL}(V_b)$. This concerns the observable quotient; it does not require the full physical command group to be finite or fully observable.

An exact stochastic realization with M live labels has row-stochastic matrices Q_a, one row-stochastic terminal decoder matrix D, and the prescribed initial row e_1^{T}, such that

$$b(a_1 \cdots a_n) = e_1^{\mathsf{T}} Q_{a_1} \cdots Q_{a_n} D. \tag{12.1}$$

Randomized terminal readout is allowed even when all Q_a are deterministic. The target input word is not encoded into an uncounted clock or persistent seed.

Theorem 12.1 (Finite realization of compactly reversible responses)**.** *For a compactly reversible response, the following are equivalent:*

(i) *An exact finite stochastic realization exists.*
(ii) *The residual set $\{b^w : w \in A^*\}$ is finite.*
(iii) *The observable group G_b is finite.*
(iv) *An exact realization with deterministic transitions and a possibly randomized decoder exists.*

The least number of deterministic-transition labels is the residual cardinality. This is not in general the least stochastic label count. If an M-label stochastic realization exists, then

$$|G_b| \le (2^M - 1)!.$$

This last bound is only a finite combinatorial certificate, not a sharp memory law.

Proof. Suppose (12.1) holds, and write $r_w = e_1^{\mathsf{T}} Q_w$. Let $R = \operatorname{span}_{\mathbb{R}}\{r_w\} \subseteq \mathbb{R}^M$. The assignment $\pi(r_w) = b^w$ extends to a well-defined surjective linear map $\pi : R \to V_b$: a relation $\sum_w c_w r_w = 0$ implies, after right multiplication by $Q_z D$, the identical relation among $b^w(z)$ for every z. Moreover

$$\pi(rQ_a) = L_a \pi(r) \qquad (r \in R).$$

The simplex section $K = R \cap \Delta_M$ is a compact polytope, invariant under right multiplication by every Q_a. Its image $P = \pi(K)$ is a compact polytope, contains every residual, and spans V_b. It satisfies $L_a P \subseteq P$. It is important that we intersect the simplex with R first: projecting arbitrary hidden-state responses directly into V_b would not be justified.

For an element T of a compact matrix group, positive powers have a subsequence tending to the identity with exponents tending to infinity. One elementary proof takes a convergent subsequence of powers with successively growing index gaps; quotients of two close powers approach the identity. A finite-order element is covered by its period multiples. Thus positive powers also approach T^{-1}. Applying this to L_a and using closedness of P gives $L_a^{-1}P \subseteq P$, and hence $L_aP = P$. All elements of G_b preserve P. They permute its finitely many vertices, and this permutation action is faithful because those vertices span V_b. Consequently G_b is finite. This proves (i)⇒(iii).

For the displayed bound, a vertex of K is determined by its positive support. If two distinct vertices had the same support, their difference could be added and subtracted in sufficiently small multiples while preserving R, unit mass and nonnegativity, contradicting extremality. Thus K has at most $2^M - 1$ vertices. A linear image of a polytope has no more vertices than the original polytope, so the faithful vertex action proves the bound.

If G_b is finite, the residual orbit is finite. Conversely, if the residual set is finite, each invertible L_a maps it injectively into itself, hence permutes it. Since it spans V_b, the generated group and its closure are finite. This proves (ii)⇔(iii). Under (ii), use the residuals as labels, send b^w to b^{wa} on input a, and use decoder $b^w(\varnothing)$. This realizes b with deterministic transitions. If any such deterministic-transition machine sends u and v to the same label, every suffix yields the same decoder law, so $b^u = b^v$. This proves its minimal count. Finally (iv)⇒(i) is immediate. No assertion that stochastic transitions can always be determinized without increasing size was used. □

Corollary 12.2 (Prepared unitary commands)**.** *Let U_a be finitely many unitary matrices on $\mathbb{C}^d$, let ρ be a prepared density matrix, and let $\{E_r : r \in C\}$ be a terminal positive-operator-valued measurement. For*

$$b(w, r) = \operatorname{tr}(E_r \Phi_w(\rho)), \qquad \Phi_a(X) = U_a X U_a^*,$$

exact finite stochastic realization for all command words is possible if and only if its observable response group is finite. The group is computed after quotienting only those directions invisible to every future commanded terminal test. Its dimension is at most d^2.

Proof. Let $W = \operatorname{span}_{\mathbb{R}}\{\Phi_w(\rho)\}$ in the Hermitian matrices. Each conjugation is a Hilbert–Schmidt isometry. Its forward invariant subspace W is also backward invariant, by finite-dimensional invertibility. Let

$$N = \{X \in W : \operatorname{tr}(E_r \Phi_z X) = 0 \text{ for all } z, r\}.$$

It is invariant under every Φ_a and hence under its inverse. The orthogonal complement of N in W is invariant as well. Therefore the induced transformations on W/N are orthogonal for the quotient inner product. The map $X \mapsto (z, r \mapsto \operatorname{tr}(E_r \Phi_z X))$ identifies W/N with V_b and intertwines them with L_a. Their generated group has compact closure in the orthogonal group. Apply Theorem 12.1. A constant measurement can make this induced group trivial even when the physical unitary group is infinite, so finiteness of the full physical group is not a necessary condition. □

12.2. **A matching memory law for two noncommuting commands.** Fix $x \in \mathbb{R}$ with $\sin(2\pi x) \neq 0$. On a protected qubit take

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} \cos(\pi x) & \sin(\pi x) \\ \sin(\pi x) & -\cos(\pi x) \end{pmatrix}, \quad \rho = E = |0\rangle\langle 0|.$$

Both commands are involutions. For known $0 \leq \alpha \leq 1$, their channels are

$$\Phi_{a,\alpha}(X) = \alpha U_a X U_a^* + (1 - \alpha) \operatorname{tr}(X) I/2.$$

Let $M_x^{\mathrm{ctrl}}(\alpha)$ be the least stochastic live memory of the whole binary response $w \mapsto \operatorname{tr}(E \Phi_{w,\alpha} \rho)$, using the same matrices and decoder for every word. In the real Bloch plane, with coordinates (z, x), these commands are αS_0 and αS_1, where S_0 reflects the angle t to $-t$ and S_1 reflects it to $2\pi x - t$. Their product is $\alpha^2 R_{2\pi x}$.

Theorem 12.3 (All-word arithmetic law for an involutive command pair)**.** *For $0 < \epsilon < 1$ and $\alpha = 1 - \epsilon$, the arithmetic envelope of Section S19 gives*

$$\tfrac{1}{4}\mathfrak{A}_x(\epsilon) \le M_x^{\mathrm{ctrl}}(1-\epsilon) \le (1+4\pi^2)\mathfrak{A}_x(\epsilon). \tag{12.2}$$

The upper is one explicit finite stochastic machine valid for every mixed command word, not separately chosen unary machines. The lower ranges over arbitrary common stochastic realizations.

For irrational x, the lower and upper logarithmic growth exponents as $\epsilon \downarrow 0$ are respectively

$$\frac{1}{\mu(x)+1}, \qquad \frac{\mu(x)-1}{2\mu(x)-1},$$

with values $0, 1/2$ at $\mu = \infty$. A uniform cubic-root law holds precisely for badly approximable x; otherwise the normalized count has lower limit zero and upper limit infinity.

At $\alpha = 1$, the exact count is the reduced denominator q if x is rational, and is infinite otherwise. At $\alpha = 0$, it is two. Every strictly damped pair has a finite exact common realization. These statements concern controlled response memory, not a general multi-command boundary-optimal policy theorem.

Proof. Any common realization restricted to the words $(01)^n$ realizes

$$\tfrac{1}{2}[1 + \alpha^{2n}\cos(2\pi n x)]$$

with transition matrix Q_0Q_1. Its size is therefore at least $m_x(1-\alpha^2) \ge \mathfrak{A}_x(\epsilon(2-\epsilon))/2$ by Theorem S19.1. For $1 \le c \le 2$, the defining maximum of the envelope obeys $\mathfrak{A}_x(c\epsilon) \ge \mathfrak{A}_x(\epsilon)/c$ whenever both arguments are in its domain. This follows for each denominator before taking its infimum. Taking $c = 2 - \epsilon$ proves the lower in (12.2).

For the upper, the rational multiple-polygon construction in the proof of Theorem S19.1 gives a regular l-gon P on the unit circle, with vertex $v_0 = (1, 0)$, such that

$$l \le (1+4\pi^2)\mathfrak{A}_x(\epsilon), \qquad \alpha R_{2\pi x} v_j \in P \quad \text{for every vertex } v_j.$$

The first reflection S_0 permutes this regular polygon and zero belongs to it. Consequently $\alpha S_0 P \subseteq P$, while $S_1 = R_{2\pi x}S_0$ gives $\alpha S_1 P \subseteq P$. Choose barycentric rows $Q_a(j,k)$ of $\alpha S_a v_j$ in the same vertices. Let $d_j = (1 + (v_j)_1)/2$. Starting at vertex zero, induction on the actual command word shows that the expected retained vertex equals the protected Bloch vector. Pairing with the first coordinate proves every response identity. The start is one of the counted vertices; there is no extra initialization mixture.

The two constants in (12.2) transfer both exponents and both uniform-law statements from Theorems S19.3 and S19.4. At zero damping, a rational phase of reduced denominator $q \ge 3$ has a common deterministic q-gon realization. The word-pair transition has a primitive qth-root mode and needs at least q states by the unit-circle spectral obstruction. For irrational x that same obstruction rules out every finite size. At $\alpha = 0$ the response is one at the empty word and $1/2$ after every nonempty word, realized by a first label followed by an absorbing fair label; one label cannot change its decoder. This proves all endpoints. □

A fully algebraic example uses $U_1 = 5^{-1/2}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$. Then $e^{2\pi i x} = (3+4i)/5$ is not a root of unity, since its sum with its conjugate is the rational noninteger $6/5$. Each undamped command in isolation has period at most two; their common all-word response has no finite stochastic realization. The arithmetic exponent of this particular angle is not determined here. The example isolates the failure of commandwise testing without inferring a Diophantine class from finitely many coefficients.

If an independent herald with fixed probability p is placed before each command operation, and only failures apply the supplied command, every common realization above preserves the terminal response under every external command schedule and the first-herald cost $1/p$. The commands are actual inputs to that same retained-state transducer. This does not make its private label observable to the caller, allow uncharged midstream quantum measurements, or prove optimality for a controller

free to choose an easier command sequence. The unary optimal squared-task interpretation below continues to use its separately proved physical embedding.

13. Convex specifications and optimal uniform approximation

The exact normal form has a robust extension. The target need not be an exactly specified probability table, and the stochastic controller need not satisfy the group relations, even on its observable states. What is required is a constraint on every word representing a given group element. The quantifier over all words is essential.

Fix a finite command alphabet A, a finite group G, and a surjective homomorphism $\gamma : A^* \to G$. For every $g \in G$ let K_g be a closed convex subset of the finite probability simplex $\Delta(C)$. Empty sets are allowed. An M-label pure-start controller consists of a prescribed point row e_o, stochastic command matrices Q_a, and a row-stochastic decoder D_0. It satisfies the specification if

$$e_o Q_w D_0 \in K_{\gamma(w)} \qquad (w \in A^*). \tag{13.1}$$

For a mixed start, replace e_o by an arbitrary probability row and do not charge an additional label for preparing that row.

Theorem 13.1 (Finite-group convex-specification normal form)**.** *The specification* (13.1) *has a pure-start realization on at most M labels if and only if there are a finite G-set X, a probability row λ on X, and a row-stochastic decoder D such that*

$$\lambda P_g D \in K_g \quad (g \in G), \qquad |X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \le M. \tag{13.2}$$

For mixed starts the corresponding criterion is $|X| \le M$. The action on X is permitted to have a kernel. In particular no faithfulness hypothesis is imposed on an approximating action.

Proof. We give the projection argument explicitly because its use of convexity replaces the exact observable intertwining in Theorem S2.1. Let $\mathcal{L} = \gamma^{-1}(e)$. Select finitely many identity words $\ell_1, \dots, \ell_s$ whose positive matrix entries cover every directed edge occurring in any Q_ℓ, $\ell \in \mathcal{L}$, and put

$$T = (I + Q_{\ell_1} + \cdots + Q_{\ell_s})/(s+1).$$

At most M^2 words suffice. Each closed communicating class C_i of T is closed under every identity-word matrix. The positive diagonal and Lemma S17.1 give a stochastic idempotent $E = \lim_n T^n$. Its image simplex has disjointly supported vertices $\mu_1, \dots, \mu_k$, the stationary rows of the closed classes. Every identity word preserves those classes, and consequently

$$EQ_\ell E = E \qquad (\ell \in \mathcal{L}). \tag{13.3}$$

For a word u write $F_u = EQ_uE$ on this image simplex. Choose an inverse word $\bar{u}$ with $\gamma(\bar{u}) = \gamma(u)^{-1}$. Since T^n is a convex combination of identity-word matrices, (13.3) applied to $u\ell\bar{u}$ and $\bar{u}\ell u$ gives

$$F_u F_{\bar{u}} = E = F_{\bar{u}} F_u.$$

Thus F_u is an affine automorphism of the image simplex and permutes its vertices. If $\gamma(u) = \gamma(v)$, multiplication by the common inverse $F_{\bar{u}}$ shows $F_u = F_v$. Replacing the middle E by T^n in EQ_uEQ_vE also gives $F_uF_v = F_{uv}$, since all words $u\ell v$ have the same group image. This defines a G-action P_g on $X = \{1, \dots, k\}$.

Write $e_oE = \sum_i \lambda_i \mu_i$ and put $D(i, \cdot) = \mu_i D_0$. For any representative u of g,

$$\lambda P_g D = e_o EQ_u E D_0 = \lim_{n\to\infty} e_o T^n Q_u T^n D_0.$$

The last expression before passage to the limit is a convex combination of $e_o Q_{\ell u \ell'} D_0$ with $\ell, \ell' \in \mathcal{L}$. All these probability rows belong to K_g. Closedness and convexity therefore give $\lambda P_g D \in K_g$. This argument includes u empty and makes no claim that the original controller's word laws factor through G.

If the initial label lies in a closed class C_i, then $e_oE = \mu_i$ and λ is a point mass. Otherwise the k disjoint nonempty closed classes exclude at least the initial label, so $k+1 \le M$. This proves the pure-start charge. For a mixed row η, use ηE instead and retain $k \le M$.

Conversely, a point preparation on X needs only the permutation transitions. For a mixed λ, a further label $\star$ has decoder λD and, on the first actual command a, moves into X with row $\lambda P_{\gamma(a)}$. Subsequent transitions are permutations. Its empty and nonempty word laws are respectively λD and $\lambda P_{\gamma(w)} D$. This gives the required realization and finishes both start conventions. □

For an exact probability table b, let $\kappa_G(b)$ be the least $|X|$ among factorizations $b(g) = \lambda P_g D$. This is its mixed-start minimum by the preceding theorem. The notation $m(b)$ retains the counted pure-start convention; its exact formula and the residual cardinality N are given in Theorem S2.2.

For a target $b : G \to \Delta(C)$ write $\mathrm{TV}(p,q) = \frac{1}{2}\sum_c |p_c - q_c|$ and define

$$\mathcal{E}_M^{\mathrm{p}}(b) = \inf_{\text{pure-start controllers on at most } M \text{ labels}} \sup_{w \in A^*} \mathrm{TV}(b(\gamma(w)), e_o Q_w D_0).$$

Use $\mathcal{E}_M^{\mathrm{m}}$ for the mixed-start convention. For a finite G-set X put

$$d_X(b) = \min_{\lambda, D} \max_{g \in G} \mathrm{TV}(b(g), \lambda P_g D), \qquad d_X^{\mathrm{pt}}(b) = \min_{x, D} \max_{g \in G} \mathrm{TV}(b(g), e_x P_g D).$$

The decoder ranges over stochastic rows; compactness makes both minima attained. Minima over no candidate actions are interpreted as $+\infty$.

Theorem 13.2 (Attained finite optimization for infinite-word error)**.** *For every target b, finite G, and integer $M \ge 1$,*

$$\mathcal{E}_M^{\mathrm{m}}(b) = \min_{|X| \le M} d_X(b), \tag{13.4}$$

$$\mathcal{E}_M^{\mathrm{p}}(b) = \min\left\{ \min_{|X| \le M} d_X^{\mathrm{pt}}(b), \min_{|X| \le M-1} d_X(b) \right\}. \tag{13.5}$$

One representative of each finite G-set is sufficient, so these are finite optimizations and their minima are attained. Their values are 1-Lipschitz functions of b in $\max_g \mathrm{TV}$. If $\mathcal{E}_M^{\mathrm{p}}(b) > 0$, a sufficiently small perturbation of b still cannot be exactly realized on M pure-start labels; the same assertion holds for mixed starts. When the target table is algebraic, the optimum and a realizing finite controller are computable by real quantifier elimination. This is a decidability assertion, not a polynomial-time bound or a claim that quantifier elimination was executed.

Proof. Apply Theorem 13.1 to the closed convex balls $K_g = \{p : \mathrm{TV}(p, b(g)) \le \delta\}$. It sends every controller of error at most δ to an admissible action without increasing the label charge or the error. Conversely every displayed action supplies such a controller. The pure constraint splits precisely into a point preparation with $|X| \le M$ or an arbitrary preparation with $|X| \le M-1$. There are finitely many homomorphisms from the fixed finite group into each symmetric group of degree at most M. For each action the allowed preparations and decoders form compact sets. Hence the finite minima are attained, including limits in which a mixed preparation becomes a point mass. The triangle inequality proves the Lipschitz bound and the perturbation assertion.

For the algorithmic statement, enumerate these actions and introduce variables $\lambda_x, D_{xc}, \delta$ and $u_{gc} \ge 0$ with

$$-u_{gc} \le b(g,c) - (\lambda P_g D)_c \le u_{gc}, \qquad \sum_c u_{gc} \le 2\delta.$$

Together with the stochastic constraints these are polynomial inequalities over the real algebraic numbers. The point-start branch fixes $\lambda = e_x$ and enumerates x. The compact attained minimum is the least real point of the resulting semialgebraic set of feasible δ's. Real quantifier elimination and algebraic sample-point construction, used with the same scope as Theorem S11.2, compute the value and tables. □

The finite optimization concerns synthesis against a supplied finite-group target. It does not solve the cutpoint, equivalence, or isolation problem for a supplied arbitrary probabilistic automaton. Nor does it replace the specification for all words by a check of one representative per group element on the original stochastic controller. Those restrictions would change the problem.

Theorem 13.3 (Robust multiplicity and initialization certificates)**.** *Use the stacked Fourier matrix $A_\rho(b)$ of* (S4.1)*, with singular values in decreasing order, and write $m_\rho(X)$ for the multiplicity of ρ in $\mathbb{C}[X]$. If an action X approximates b to total-variation error at most δ, then*

$$\sigma_{m_\rho(X)+1}(A_\rho(b)) \le \sqrt{2}\,\delta \qquad (\rho \in \widehat{G}). \tag{13.6}$$

A singular value past the number of columns means zero. For a binary response $b = (\beta, 1-\beta)$ the sharper equivalent scalar certificate is

$$\sigma_{m_\rho(X)+1}(\widehat{\beta}(\rho)) \le \delta. \tag{13.7}$$

In particular let r_ρ^δ be the number of singular values of $A_\rho(b)$ strictly exceeding $\sqrt{2}\delta$. Then every mixed-start controller of uniform error at most δ uses at least

$$\min\left\{\sum_H q_H[G:H] : q_H \in \mathbb{Z}_{\ge 0}, \quad \sum_H q_H \ge 1, \quad \sum_H q_H m_\rho(H) \ge r_\rho^\delta \ (\rho \in \widehat{G})\right\} \tag{13.8}$$

labels. The scalar version may be used for binary outputs.

Let N be the number of distinct residual tables $b^h(g) = b(hg)$. If $N > 1$, let

$$\Delta_b = \min_{b^h \ne b^k} \max_g \mathrm{TV}(b(hg), b(kg)) > 0.$$

Whenever $\delta < \Delta_b/2$, a pure-start controller with fewer than N labels must have a projected mixed preparation. Its total budget is therefore at least one plus the lower bound (13.8)*.*

Proof. Put $a(g) = \lambda P_g D$ and $z_c(g) = b(g,c) - a(g,c)$. Since $\sum_c z_c(g) = 0$ and $\frac{1}{2}\sum_c |z_c(g)| \le \delta$, the sums of the positive and negative parts are each at most δ, and $\sum_c |z_c(g)|^2 \le 2\delta^2$. Unitarity gives

$$\left\| \begin{pmatrix} z_{c_1}(g)\rho(g^{-1}) \\ \vdots \\ z_{c_{|C|}}(g)\rho(g^{-1}) \end{pmatrix} \right\|_{\mathrm{op}} = \left(\sum_c |z_c(g)|^2\right)^{1/2} \le \sqrt{2}\delta.$$

Average over g to obtain $\|A_\rho(b) - A_\rho(a)\|_{\mathrm{op}} \le \sqrt{2}\delta$. The Fourier-multiplicity lemma gives $\operatorname{rank} A_\rho(a) \le m_\rho(X)$. For completeness, if B has rank at most m, its nullspace intersects the span of the first $m+1$ right singular vectors of A in a nonzero vector. On a unit vector in that intersection, $\|(A-B)v\| = \|Av\| \ge \sigma_{m+1}(A)$. This proves (13.6); the binary version follows by averaging the scalar error, whose absolute value is at most δ, against unitary matrices. Summing orbit multiplicities and using the mixed-start normal form gives the integer bound.

For the last assertion, suppose the projected preparation were a point x. Its distinct residuals are indexed by points $x \cdot h$ in its orbit. If two target residuals b^h, b^k were represented by the same point, the two approximating residuals would coincide for every future g, and the triangle inequality would give distance at most 2δ between b^h and b^k. This contradicts the definition of Δ_b. The orbit must therefore contain at least N points. With fewer than N total labels this is impossible, so the projected start is mixed and its extra initializer is charged by the normal-form theorem. □

Corollary 13.4 (Uniform error cannot replace a prime orbit)**.** *Let $G = C_p$ with p prime and let $b : G \to \Delta(C)$ be any target. Put $r(b) = \min_{q \in \Delta(C)} \max_g \mathrm{TV}(b(g), q)$. For either start convention,*

$$\mathcal{E}_M(b) = \begin{cases} r(b), & 1 \le M < p, \\ 0, & M \ge p. \end{cases}$$

For binary outputs $r(b) = (\max_g \beta(g) - \min_g \beta(g))/2$.

Proof. Every orbit of C_p has size one or p. Thus any action with fewer than p points is trivial and gives a constant response, regardless of preparation. A one-label controller attains $r(b)$. On p points the regular action with a point preparation and decoder b is exact. For binary outputs the midpoint of the smallest interval containing all values of β is a Chebyshev center of that interval. □

For the symmetric-group target (S4.6), the scalar standard block is $tI/[n(n-1)]$ and $\Delta_{b_t} = 2t/n$. Indeed a permutation moving k points changes a residual by at most kt/n, with equality at an inverse representative; the smallest nonzero k is two. The subgroup-index estimate (S4.12) and the preceding theorem imply

$$\mathcal{E}_M^{\mathrm{p}}(b_t) \geq \frac{t}{n(n-1)} \qquad (n \geq 5,\ 0 < t \leq 1,\ M \leq n(n-1)). \tag{13.9}$$

If the error were smaller, the full standard multiplicity $n-1$ would be forced, and the small-error initialization charge would give at least $n(n-1)+1$ labels; here $n! > n(n-1)$. This explicit lower bound is not asserted to be the optimal error constant. At every exact resource jump the finite optimization also gives a strictly positive robustness radius, without a claimed closed formula for that radius.

14. Shared preparation, invariant polytopes, and noncyclic memory

A device that reproduces one prepared response need not reproduce a family of preparations with the same memory, transitions, and decoder. We now solve the structural problem with this common-device requirement. The preparation is selected before the command word; it is not revealed to the terminal decoder. In particular, one may not choose a different decoder after learning which preparation was selected. This convention is what makes joint realization a resource question.

Fix a finite group G, a surjective command homomorphism $\gamma : A^* \to G$, a finite outcome set C, and a nonempty finite preparation set I. A *shared controller* consists of one family of stochastic matrices Q_a and one stochastic decoder D_0, together with an initial row for each $i \in I$. In the pure convention these rows are prescribed label rows e_{o_i}; different preparations may use the same label when their required laws agree. All labels, including initialization labels, are counted. In the mixed convention the initial rows may be arbitrary probability rows and their external preparation is not counted. The result concerns this explicit distinction, not the bit cost of selecting i or an uncounted label made available to the decoder.

For each i, g let $K_{i,g} \subseteq \Delta(C)$ be closed and convex, possibly empty. The requirement is

$$e_{o_i} Q_w D_0 \in K_{i,\gamma(w)} \qquad (i \in I,\ w \in A^*), \tag{14.1}$$

with arbitrary initial rows in the mixed convention. For probability rows $\Lambda = (\lambda_i)_{i \in I}$ on a finite set put

$$\iota(\Lambda) = \big|\{\lambda_i : i \in I,\ \lambda_i \text{ is not a point mass}\}\big|.$$

Repeated rows are counted once.

Theorem 14.1 (Shared normal form and exact initialization charge)**.** *The specification* (14.1) *has a pure shared controller on at most M labels if and only if there are a finite G-set X, probability rows λ_i on X, and one stochastic decoder D such that*

$$\lambda_i P_g D \in K_{i,g} \quad (i \in I, g \in G), \qquad |X| + \iota(\Lambda) \leq M. \tag{14.2}$$

For mixed preparations the criterion is $|X| \leq M$. The action may have a kernel. Thus a single permutation core suffices for all preparations, and each distinct mixed preparation costs exactly one additional label in this normal form.

Proof. Use the identity-word construction in the proof of Theorem 13.1. Its averaging matrix T, stochastic idempotent $E = \lim_n T^n$, and disjoint closed classes $C_1, \ldots, C_k$ depend only on the

command matrices, not on the initial row. Write μ_x for their stationary probability rows. The sandwiched maps EQ_uE permute these rows and give one action $P_{\gamma(u)}$ on $X = \{1, \dots, k\}$. Put

$$e_{o_i}E = \sum_{x \in X} \lambda_i(x)\mu_x, \qquad D(x, \cdot) = \mu_x D_0.$$

The same limit as in that theorem gives

$$\lambda_i P_g D = \lim_{n\to\infty} e_{o_i} T^n Q_u T^n D_0 \in K_{i,g}, \qquad \gamma(u) = g,$$

because each approximant is a convex combination of laws for words representing g. There is no averaging between different preparations.

If o_i lies in a closed class, then $e_{o_i}E$ is its stationary row, so λ_i is a point mass. Distinct mixed λ_i must therefore arise from distinct initial labels outside the closed classes. These classes use at least k labels, so $\iota(\Lambda) \le M - k$. This argument also permits an outside initial label to project to a point mass, and it does not charge that label when the projected realization no longer needs it. It permits several initial labels to have the same projected mixed row, in which case they are merged. For mixed initial rows the same projection gives $k \le M$ without the additional count.

Conversely, retain the permutation labels X. For every distinct nonpoint row λ among the λ_i, adjoin one label $\star_\lambda$ with decoder λD and, on command a, transition row $\lambda P_{\gamma(a)}$ into X. No transition returns to an adjoined label. A point row uses its own point of X as initial label; a nonpoint row uses $\star_\lambda$. The empty-word law is $\lambda_i D$ and the law after every nonempty word is $\lambda_i P_{\gamma(w)} D$. This establishes (14.2) with the exact displayed count. The mixed case starts directly from λ_i on X. □

For exact targets write $\mathcal{B} = \{b_i : i \in I\}$, where $b_i : G \to \Delta(C)$; duplicate target tables are identified. Let $\kappa_G(\mathcal{B})$ and $m_G(\mathcal{B})$ denote the mixed and pure shared minima. The first equals the least $|X|$ in a simultaneous factorization $b_i(g) = \lambda_i P_g D$; the second includes $\iota(\Lambda)$. In particular

$$\max_i \kappa_G(b_i) \le \kappa_G(\mathcal{B}), \qquad \kappa_G(\mathcal{B}) \le m_G(\mathcal{B}).$$

Neither inequality compares a collection of separately optimized decoders with a single common decoder.

14.1. **A geometric formula with all initialization labels counted.** Let $\mathcal{P}_G(\mathcal{B})$ be the set of nonempty polytopes in the finite-dimensional product simplex

$$\mathcal{Q} = \{q : G \to \Delta(C)\}$$

that contain $\mathcal{B}$ and are invariant under $(q \cdot h)(g) = q(hg)$. The whole product simplex is one such polytope. Write $v(P) = |\operatorname{vert} P|$. The polytope is allowed to extend beyond the linear span of the prescribed responses. Forbidding this extension would impose an additional positive-realization constraint.

Theorem 14.2 (Invariant-polytope resource formulas)**.** *For every nonempty finite target family,*

$$\kappa_G(\mathcal{B}) = \min_{P \in \mathcal{P}_G(\mathcal{B})} v(P), \tag{14.3}$$

$$m_G(\mathcal{B}) = \min_{P \in \mathcal{P}_G(\mathcal{B})} \left(v(P) + |\mathcal{B} \setminus \operatorname{vert} P|\right). \tag{14.4}$$

The minima are attained. Every minimizing polytope constructs an optimal controller: its vertices are permutation labels, and each distinct prescribed nonvertex response receives one transient initialization label.

Proof. Given a simultaneous factorization, associate to each core label the profile $q_x(g) = e_x P_g D$. The set of profiles is invariant under G, and $P = \operatorname{conv}\{q_x : x \in X\}$ belongs to $\mathcal{P}_G(\mathcal{B})$. Every vertex is a profile, so $v(P) \le |X|$. This proves the necessary mixed bound.

For the pure bound, apply the preceding normal form to an arbitrary M-label pure realization. Let $s = \iota(\Lambda)$ and $k = |X|$, so $k + s \le M$. If a target b_i is not a vertex of P, it either equals a nonvertex core profile q_x, or is not any core profile. There are at most $k - v(P)$ distinct profiles of the first kind: at least $v(P)$ core labels are required to represent the different vertices. Every distinct target of the second kind requires a distinct nonpoint λ_i, so there are at most s such targets. Therefore

$$v(P) + |\mathcal{B} \setminus \operatorname{vert} P| \le v(P) + (k - v(P)) + s \le M.$$

This is the step that prevents a nonextreme continuing label used by another preparation from being charged twice.

Conversely, a finite group preserving P permutes its vertices. Use these vertices as X, with $D(q, c) = q(e, c)$. Barycentric coordinates of $b_i \in P$ supply a probability row λ_i with $\lambda_i P_g D = b_i(g)$. If b_i is a vertex choose its point row. If it is not a vertex, every barycentric representation is nonpoint. Choose the same representation for identical target tables. Different target tables have different representations, since their barycentric images differ. The normal-form construction thus costs exactly $v(P) + |\mathcal{B} \setminus \operatorname{vert} P|$ in the pure convention, and $v(P)$ in the mixed convention.

The feasible integer costs form nonempty subsets of the positive integers. Their least elements are realized by some feasible polytope; the two constructions prove the asserted equalities and attainment. □

Invariant cones in positive realization and the relation between symmetry and nonnegative factorizations are classical [48, 54, 67]. Formula (14.3) is a response-polytope form of that structure. The additional resource statement here is the exact shared-device pure-initialization term in (14.4), obtained after the common identity-word projection. It neither assumes that a supplied stochastic controller satisfies the group relations nor identifies linear rank with positive memory.

14.2. **Joint Fourier multiplicity and exact low-contrast synthesis.** Let

$$V_{\mathcal{B}} = \operatorname{span}_{\mathbb{R}}\{b_i \cdot h : i \in I, h \in G\}, \qquad d = \dim V_{\mathcal{B}}.$$

For an irreducible unitary representation ρ of degree d_ρ, use the stacked matrix $A_\rho(b_i)$ from (S4.1) and define the *joint* block

$$\mathcal{A}_\rho(\mathcal{B}) = \big[A_\rho(b_1) \ \cdots \ A_\rho(b_{|I|})\big], \qquad r_\rho(\mathcal{B}) = \operatorname{rank}_{\mathbb{C}} \mathcal{A}_\rho(\mathcal{B}). \tag{14.5}$$

Unlike the single-preparation rank, $r_\rho(\mathcal{B})$ can exceed d_ρ. The same decoder must accommodate the union of the Fourier column spaces, rather than their largest separate rank.

Lemma 14.3 (Joint multiplicity)**.** *The multiplicity of ρ in $V_{\mathcal{B}} \otimes_{\mathbb{R}} \mathbb{C}$ equals $r_\rho(\mathcal{B})$. Consequently every simultaneous factorization on a G-set X satisfies*

$$r_\rho(\mathcal{B}) \le m_\rho(X) \qquad (\rho \in \widehat{G}).$$

Proof. Fourier transformation sends the translate $b_i \cdot h$ to $A_\rho(b_i)\rho(h)$. The group-algebra decomposition used in Lemma S4.1 therefore identifies its ρ-component with

$$\left\{ \sum_{i \in I} A_\rho(b_i) M_i : M_i \in M_{d_\rho}(\mathbb{C}) \right\}.$$

The matrices M_i are independent, since the different generators may have independent group-algebra coefficients. This is the space of all matrices whose columns belong to the sum of the column spaces of the $A_\rho(b_i)$. That sum has dimension $r_\rho(\mathcal{B})$, so the right module consists of precisely that many rows of length d_ρ. For a simultaneous positive factorization, the span of all reachable preparation rows is a submodule of $\mathbb{R}^X$, and evaluation against the common decoder maps it onto $V_{\mathcal{B}}$. Complete reducibility therefore embeds $V_{\mathcal{B}}$ into $\mathbb{R}^X$, proving the bound. □

Write $\nu_G(V_{\mathcal{B}})$ for the smallest size of a permutation module containing $V_{\mathcal{B}}$. For one representative H of each subgroup conjugacy class, Frobenius reciprocity gives

$$\nu_G(V_{\mathcal{B}}) = \min\left\{\sum_H q_H[G:H] : q_H \in \mathbb{Z}_{\geq 0}, \quad \sum_H q_H m_\rho(H) \geq r_\rho(\mathcal{B})\ (\rho \in \widehat{G})\right\}. \tag{14.6}$$

The equality follows by orbit decomposition and the real-descent part of Lemma S4.1; it applies also to complex and quaternionic representation types.

Theorem 14.4 (Exact noncyclic memory at small common contrast)**.** *Suppose that all preparations have the same group-averaged law,*

$$\frac{1}{|G|}\sum_{g\in G} b_i(g) = \pi \qquad (i \in I),$$

and put $b_{i,t}(g) = \pi + t\{b_i(g) - \pi\}$ for $0 \leq t \leq 1$. There is an explicitly positive t_0 such that

$$\kappa_G(\{b_{i,t} : i \in I\}) = \nu_G(V_{\mathcal{B}}) \qquad (0 < t \leq t_0). \tag{14.7}$$

For every $t > 0$ the joint module and its Fourier multiplicities are unchanged. The exact pure resource is the invariant-polytope formula (14.4)*; in particular, if $r = |\mathcal{B}|$ and $N = |\{b_i \cdot g : i \in I, g \in G\}|$, then on this interval*

$$\nu_G(V_{\mathcal{B}}) \leq m_G(\{b_{i,t}\}) \leq \min\{N, \nu_G(V_{\mathcal{B}}) + r\}.$$

At $t = 0$ both resources equal one.

Proof. Remove outcomes with $\pi_c = 0$; positivity makes them identically zero for every preparation and group element. They will be restored as zero decoder columns. Let $\bar{b}$ be the constant table π. Group averaging maps every generating residual to $\bar{b}$, hence

$$V_{\mathcal{B}} = \mathbb{R}\bar{b} \oplus W, \qquad W = \ker\left(|G|^{-1}\sum_{g\in G} L_g\right).$$

The nontrivial part W has no invariant vector and all its elements have zero total output mass. If $W = 0$, all targets equal $\bar{b}$ and the assertion is immediate. Otherwise choose a minimum G-set X with $k = \nu_G(V_{\mathcal{B}})$ and a real equivariant injection $j : W \to \mathbb{R}^X$. Let $F : \mathbb{R}^X \to W$ be orthogonal projection onto $j(W)$ followed by j^{-1}. Since the permutation inner product is invariant, F is equivariant and annihilates every invariant row. Write

$$p = (1/k, \ldots, 1/k), \qquad a_i = j(b_i - \bar{b}), \qquad T(x,c) = F(e_x)(e,c).$$

Then $a_i\mathbf{1} = 0$, $T\mathbf{1} = 0$, $pT = 0$, and $a_i P_g T = b_i(g) - \pi$. Put

$$A = \max_i \|a_i\|_\infty > 0, \quad \epsilon = \frac{1}{2kA}, \quad B = \max_{x,c}\frac{|T(x,c)|}{\pi_c} > 0, \quad t_0 = \min\{1, \epsilon/B\}.$$

The rows $\lambda_i = p + \epsilon a_i$ have all entries at least $1/(2k)$. For $0 \leq t \leq t_0$ the *common* matrix $D_t(x,c) = \pi_c + (t/\epsilon)T(x,c)$ is stochastic, and

$$\lambda_i P_g D_t = \pi + t\{b_i(g) - \pi\} = b_{i,t}(g)$$

for every preparation simultaneously. This proves the matching upper bound, with one decoder and one action.

The equivariant map that is the identity on $\mathbb{R}\bar{b}$ and multiplication by t on W is invertible for $t > 0$ and sends all residuals to their contrasted versions. Thus their joint module and all equality relations are unchanged. Lemma 14.3 and (14.6) give the lower bound. The constructed preparations have at most r distinct nonpoint rows, so their pure initialization costs at most $k + r$. Alternatively, the union of all residuals is a deterministic shared machine with N labels. At zero contrast one common constant decoder and one fixed label suffice. □

The common-mean hypothesis has mathematical content. Distinct constant preparation laws can already have positive-factorization cost larger than their linear rank; contrast does not change them. For example, with the trivial group, the four preparation rows

$$\frac{1}{2}\begin{pmatrix}1&1&0&0\\0&1&1&0\\0&0&1&1\\1&0&0&1\end{pmatrix}$$

have linear rank three but require four shared mixed labels. In any nonnegative rank-one summand, at most one of the four positive diagonal entries can be covered: covering two would make both corresponding cross entries positive, whereas at least one is zero. Four summands are therefore necessary, and the displayed rows supply a four-label realization. This also explains why a separate scalar module bound for each preparation cannot replace joint positive synthesis.

14.3. **Robust joint certificates and sharp resource separations.** For weights $\alpha_i \ge 0$ with $\sum_i \alpha_i = 1$, let

$$\mathcal{A}_{\rho,\alpha}(\mathcal{B}) = [\sqrt{\alpha_1}A_\rho(b_1)\ \cdots\ \sqrt{\alpha_{|I|}}A_\rho(b_{|I|})].$$

Theorem 14.5 (Joint robust multiplicity certificate)**.** *If one mixed shared controller approximates every $b_i(\gamma(w))$ to total-variation error at most δ for every word w, there is a G-set X of size at most its memory such that, for every irreducible ρ and every such weight vector,*

$$\sigma_{m_\rho(X)+1}(\mathcal{A}_{\rho,\alpha}(\mathcal{B})) \le \sqrt{2}\delta. \tag{14.8}$$

Hence the integer cover (14.6)*, with each r_ρ replaced by the number of singular values strictly above $\sqrt{2}\delta$ in any chosen weighted block, is a memory lower bound. Weights may be chosen separately for different irreducibles. The same lower bound applies to pure shared controllers.*

Proof. Apply Theorem 14.1 to the closed convex total-variation balls. It supplies a common permutation action and tables $a_i(g) = \lambda_i P_g D$ with the same error bound. The calculation in Theorem 13.3 gives $\|A_\rho(b_i) - A_\rho(a_i)\|_{\mathrm{op}} \le \sqrt{2}\delta$. For block rows $z = (z_i)_i$, Cauchy–Schwarz therefore gives

$$\left\|\sum_i \sqrt{\alpha_i}\{A_\rho(b_i) - A_\rho(a_i)\}z_i\right\| \le \sqrt{2}\delta\sum_i \sqrt{\alpha_i}\|z_i\| \le \sqrt{2}\delta\left(\sum_i \|z_i\|^2\right)^{1/2}.$$

The weighted joint error has operator norm at most $\sqrt{2}\delta$. The approximating joint block has rank at most $m_\rho(X)$ by Lemma 14.3; zero weights can only reduce that rank. The singular-value nullspace argument used in Theorem 13.3 proves (14.8). Orbit decomposition gives the stated cover. The pure normal form has a core no larger than its total memory. □

Proposition 14.6 (Unbounded joint multiplicity for a fixed group)**.** *For every integer $q \ge 1$ there is a q-preparation experiment on $G = C_2$ with $q+1$ terminal outcomes such that each preparation alone has pure and mixed memory two, while their common-device minima are*

$$\kappa_{C_2}(\mathcal{B}) = m_{C_2}(\mathcal{B}) = 2q.$$

Its joint sign multiplicity is q, although the sign representation has degree one. All target response probabilities may be chosen rational and strictly positive. Moreover, in the explicit family below every common approximation with uniform total-variation error $\delta < \epsilon/\sqrt{2q}$ still requires at least $2q$ labels.

Proof. Take $C = \{0, 1, \dots, q\}$, the uniform law $\pi = (q+1)^{-1}\mathbf{1}$, and $v_i = e_i - e_0$ for $1 \le i \le q$. For any rational $0 < \epsilon < 1/(q+1)$ put

$$b_i(e) = \pi + \epsilon v_i, \qquad b_i(s) = \pi - \epsilon v_i,$$

where s is the nonidentity element of C_2. Each individual response is nonconstant and is realized by a two-cycle with these two decoder rows, so its exact memory is two. For a shared realization the joint sign block is the $(q+1)\times q$ matrix $\epsilon[v_1\ \cdots\ v_q]$, of rank q. Every two-cycle in a C_2-set contributes one sign copy; fixed points contribute none. At least q two-cycles, or $2q$ labels, are necessary. The disjoint union of the q displayed two-cycles attains this bound, with pure preparation i at the positive point of its own cycle.

For equal weights $\alpha_i = 1/q$, the squared nonzero singular values of the joint sign block are the eigenvalues of $(\epsilon^2/q)(I + \mathbf{1}\mathbf{1}^{\mathsf{T}})$. Its least singular value is at least $\epsilon/\sqrt{q}$ (also for $q = 1$). When $\sqrt{2}\delta < \epsilon/\sqrt{q}$, the joint robust certificate still forces q sign copies. This lower bound allows arbitrary stochastic transitions and arbitrary stochastic terminal decoders satisfying the all-word approximation requirement. □

Proposition 14.7 (An unbounded pure initialization cost)**.** *For every integer $r \geq 1$ there is an r-preparation binary experiment on C_2 with*

$$\kappa_{C_2}(\mathcal{B}) = 2, \qquad m_{C_2}(\mathcal{B}) = r + 1,$$

although every preparation separately has pure memory two.

Proof. Choose $0 < t_1 < \cdots < t_r < 1/2$ and set

$$b_i(e) = (1/2 + t_i, 1/2 - t_i), \qquad b_i(s) = (1/2 - t_i, 1/2 + t_i).$$

Use one two-cycle with decoder first coordinates $1/2 + t_r$ and $1/2 - t_r$. Appropriate mixtures of its two states realize every preparation, proving the mixed upper two; nonconstancy proves its lower. The last preparation uses the positive core point, and the other $r - 1$ preparations use distinct transient initialization labels. This gives $r + 1$ pure labels by Theorem 14.1.

For the lower bound, the r distinct empty-word laws require r distinct pure initial labels. If those were all the labels, every decoder row would have first coordinate strictly greater than $1/2$. Every stochastic transition from any of them would then still give a first-coordinate probability strictly greater than $1/2$. This contradicts the required response after s, whose first coordinate is strictly less than $1/2$. At least one further label is necessary. □

Both families are ordinary finite causal experiments, and the constructions also admit diagonal quantum realizations: place the decoder laws on the diagonal, use permutation unitaries for commands, and use the stated diagonal preparations. No quantum advantage is asserted. Their purpose is to separate simultaneous preparation, noncyclic multiplicity, and counted pure initialization from the already solved single-preparation resource. For arbitrary different mean laws, arbitrary quantum preparation sets, or nonreversible commands, the displayed common-mean equality is not asserted.

15. Finite memory certificates without physical decoder assumptions

The physical-dictionary lower bound uses the geometry of density matrices at individual hidden labels. An arbitrary classical simulator need not have that geometry. We now give a different certificate, valid for arbitrary classical decoder tables. Its conclusion is quantitatively weaker, but applies to the original algebraic qubit instrument without an unproved Diophantine hypothesis. More generally it applies whenever experimentally accessible preparations have separated exposing tests.

Nonnegative factorizations, support antichains and their relation to memory and extension complexity are classical. In particular the exact zero-diagonal support bound is already in the literature on nonnegative rank; see [67, Section 4.1 and Corollary 4.13] and [85, Section 3.1]. The argument below supplies an explicit error margin and derives its hypotheses from joint stopped records. It does not treat an approximate zero as an exact zero, or assume that a rare conditional distribution inherits the unconditional approximation error.

15.1. **A robust support certificate.** For $J \geq 2$ define the antichain threshold

$$a(J) = \min\left\{ m \geq 1 : \binom{m}{\lfloor m/2 \rfloor} \geq J \right\}. \tag{15.1}$$

In particular $\lceil \log_2 J \rceil \leq a(J) \leq J$.

Lemma 15.1 (Robust antichain factorization certificate)**.** *Let $\lambda_1, \dots, \lambda_J$ be probability rows on M labels, and let $d_1, \dots, d_J \in [0,1]^M$ be arbitrary columns. If*

$$\lambda_n d_n \leq \epsilon, \qquad \lambda_n d_m \geq b \quad (n \neq m), \qquad 0 < b \leq 1, \qquad \epsilon < \frac{b^2}{2M^2}, \tag{15.2}$$

then $J \leq \binom{M}{\lfloor M/2 \rfloor}$. The conclusion uses no linearity, noncontextuality or quantum representation of the decoder columns.

Proof. Put $S_n = \{i : \lambda_{n,i} \geq b/(2M)\}$. For $n \neq m$, the nonnegative sum $\lambda_n d_m \geq b$ has a summand $\lambda_{n,i} d_m(i) \geq b/M$. Since both factors are at most one,

$$\lambda_{n,i} \geq b/M, \qquad d_m(i) \geq b/M.$$

Thus $i \in S_n$. On the other hand

$$\lambda_{m,i} \leq \epsilon / d_m(i) \leq M\epsilon/b < b/(2M),$$

so $i \notin S_m$. Every ordered pair of distinct sets has this witness, and the S_n form an antichain in the subsets of $[M]$.

For completeness, choose a uniformly random permutation of $[M]$. For a fixed set S of size k, the probability that its elements are the first k elements of this permutation, in any order, is $1/\binom{M}{k}$. The corresponding events for members of an antichain are disjoint, since the initial sets of one permutation form a chain. Hence

$$\frac{J}{\binom{M}{\lfloor M/2 \rfloor}} \leq \sum_{n=1}^{J} \binom{M}{|S_n|}^{-1} \leq 1,$$

as required. □

The combinatorial conclusion is sharp at the level of these hypotheses. For $M \geq 2$, let $k = \lfloor M/2 \rfloor$, index preparations and tests by all k-element subsets S of $[M]$, and put

$$\lambda_S(i) = \frac{\mathbf{1}_{\{i \in S\}}}{k}, \qquad d_T(i) = \mathbf{1}_{\{i \notin T\}}.$$

Then $\lambda_S d_S = 0$ and $\lambda_S d_T \geq 1/k$ for $S \neq T$, with exactly $\binom{M}{k}$ preparations. This construction concerns the abstract preparation–test table. It does not construct a small controller for the qubit's chronological command–report dynamics.

15.2. **An operational certificate from joint records.** Consider any causal experiment with J preparation procedures. The nth procedure either declares failure or ends at an observable herald H_n. Only after that herald is a binary terminal test $m \in [J]$ requested. The choice of m, including a policy's future test code, is not an input to the simulator before the herald. Its terminal decoder is one fixed table for each requested test, independent of the earlier preparation procedure except through the simulator's current counted label. These are the usual causal finite-memory conventions; allowing a decoder to inspect an uncounted copy of the entire past would define a different resource.

Let p_n be the target probability of H_n, and let f_{nm} be the target rejection probability conditional on H_n and test m. Suppose that every one of these preparation–test experiments has target expected cost at most B, and that

$$p_n \geq p_* > 0, \qquad f_{nn} = 0, \qquad f_{nm} \geq s > 0 \quad (n \neq m). \tag{15.3}$$

The terminal tests and preparation procedures may be classical or quantum; only their observable probabilities enter the next theorem.

Theorem 15.2 (Memory lower bound from approximately exposing tests)**.** *Suppose* (15.3) *holds, with $J \ge 2$ and $0 < s \le 1$. An M-label classical simulator that approximates the complete stopped record of every specified experiment to total-variation error at most δ satisfies*

$$\delta \le \frac{p_* s^2}{32M^2} \quad \Longrightarrow \quad J \le \binom{M}{\lfloor M/2 \rfloor}. \tag{15.4}$$

Consequently

$$\delta \le \frac{p_* s^2}{32J^2} \quad \Longrightarrow \quad M \ge a(J). \tag{15.5}$$

Both assertions allow freely mixed or counted pure starts and arbitrary stochastic terminal decoder tables. They require only the finite collection of J^2 displayed preparation–test experiments.

Proof. For the nth procedure let q_n be the simulator's herald probability, and let λ_n be its conditional distribution on its counted labels at that herald. It does not depend on the later requested m. Write $d_m(i) \in [0,1]$ for the rejection probability of test m at label i. Total variation controls both the herald event and the joint herald-and-rejection event, so

$$|q_n - p_n| \le \delta, \qquad |q_n \lambda_n d_m - p_n f_{nm}| \le \delta. \tag{15.6}$$

Under the hypothesis in (15.4), $\delta/p_n \le s^2/(32M^2) \le 1/32$, so $q_n > 0$ and

$$\lambda_n d_n \le \frac{\delta}{p_n - \delta} \le \frac{2\delta}{p_*} \le \frac{s^2}{16M^2}.$$

For $m \ne n$ put $t = \delta/p_n \le s^2/(32M^2) \le s/4$. Equation (15.6) gives

$$\lambda_n d_m \ge \frac{p_n s - \delta}{p_n + \delta} = \frac{s - t}{1 + t} \ge \frac{s}{2}.$$

Apply Lemma 15.1 with $b = s/2$ and $\epsilon = s^2/(16M^2) < b^2/(2M^2)$. This proves (15.4). If $M \ge J$, then $M \ge a(J)$ already. Otherwise the hypothesis in (15.5) implies the one in (15.4), and gives the same conclusion. No conditional error estimate was presumed before (15.6); normalization was performed only after the herald mass was controlled. □

This theorem keeps the cost budget in the target law. It makes no assertion about the simulator's expected duration. The hidden label can have any classical interpretation, and the decoder for one binary POVM can be unrelated to decoders for other POVMs.

15.3. **An unrestricted lower bound for the original algebraic instrument.** For the undamped qubit instrument (S7.1), fix one commanded Bloch angle θ and put $z = e^{i\theta}$ and $\omega = \theta/(2\pi)$. Let $N_{\mathrm{cl}}^{\mathrm{m}}(\delta; B)$ and $N_{\mathrm{cl}}^{\mathrm{p}}(\delta; B)$ be the least label counts approximating every common visible stopping and terminal-POVM policy of target expected calls at most B, with arbitrary classical terminal decoder tables. Transitions are fixed independently of later terminal requests. Thus these classes include the physical dictionaries of Section S9.4, but impose no Born representation on their individual labels.

Fix $B > 0$ and write $\zeta = \min\{1, 7B/8\}$. For $J \ge 2$ and a phase which is not a root of unity put

$$s_J = \min_{1 \le q < J} \frac{1 - \mathrm{Re}(z^q)}{2} > 0. \tag{15.7}$$

Theorem 15.3 (Unrestricted finite certificates and arithmetic rates)**.** *For either start convention, the following statements hold.*

(i) *For every non-root phase and every integer $J \ge 2$,*

$$0 < \delta \le \frac{\zeta\, 8^{-J} s_J^2}{32J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{15.8}$$

This lower bound uses J terminal projectors and finitely many preparation prefixes, with no assumption on individual decoder rows.

(ii) *At the original algebraic phase $z = (3+4i)/5$, the explicit sufficient condition is*

$$0 < \delta \le \frac{25\zeta}{128 \cdot 200^J J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{15.9}$$

No badly-approximable assertion about $\theta/(2\pi)$ is required.

(iii) *If $\theta = 2\pi\omega$ and $\|q\omega\|_{\mathbb{R}/\mathbb{Z}} \ge c/q$ for every $q \ge 1$, then another explicit sufficient condition is*

$$0 < \delta \le \frac{\zeta c^4}{2 \cdot 8^J J^6} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{15.10}$$

In case (ii)*, and also in case* (iii)*,*

$$\liminf_{\delta \downarrow 0} \frac{N_{\mathrm{cl}}^{\mathrm{m}}(\delta; B)}{\log_2 \log_8(1/\delta)} \ge 1, \qquad \liminf_{\delta \downarrow 0} \frac{N_{\mathrm{cl}}^{\mathrm{p}}(\delta; B)}{\log_2 \log_8(1/\delta)} \ge 1. \tag{15.11}$$

The inherited physical constructions give, at the same fixed phase, $N_{\mathrm{cl}}^{\mathrm{m}} \le K_\omega + L - 1$ and $N_{\mathrm{cl}}^{\mathrm{p}} \le K_\omega + L$, where L is (S9.9)*; at $z = (3+4i)/5$ one may take $K_\omega = 7$. The unrestricted lower and upper therefore leave a gap between logarithmic-in-logarithm and logarithmic growth. A matching logarithmic lower is asserted only for the physical-dictionary class under the hypotheses of Theorem S9.4.*

Proof. For $n = 0, \dots, J-1$, begin with probability ζ and otherwise stop with no call. If begun, follow the report prefix $1\,0^n$, stopping immediately on a different report. Success is the herald H_n. The target probability and conditional terminal state are

$$p_n = \zeta 8^{-(n+1)}, \qquad \rho_n = U_\theta^n \tau (U_\theta^*)^n.$$

In the real Bloch plane ρ_n has vector $(\cos(n\theta), \sin(n\theta))$. The target expected number of calls, before the cost-free terminal test, is

$$\zeta \sum_{j=0}^{n} 8^{-j} \le \zeta\, 8/7 \le B.$$

After success request the binary terminal POVM $(\rho_m, I - \rho_m)$, and call its second outcome rejection. Its target conditional rejection probability is

$$f_{nm} = 1 - \mathrm{tr}(\rho_n \rho_m) = \frac{1 - \cos((n-m)\theta)}{2}.$$

Thus $f_{nn} = 0$, $f_{nm} \ge s_J$ for $n \ne m$, and $p_n \ge p_* := \zeta 8^{-J}$. These are the hypotheses of Theorem 15.2, which gives (i). The simulator's terminal projector request is made only after the prefix. Causality is precisely what gives one common label distribution λ_n for all its J possible terminal tests.

For (ii), write

$$z^q = \frac{A_q + iB_q}{5^q}, \qquad A_q, B_q \in \mathbb{Z}.$$

The phase is not a root of unity: otherwise $z + z^{-1} = 6/5$ would be both a rational number and an algebraic integer, hence an integer. Consequently $\mathrm{Re}\, z^q < 1$ for every $q \ge 1$, and $5^q - A_q$ is a positive integer. Therefore

$$\frac{1 - \mathrm{Re}\, z^q}{2} = \frac{5^q - A_q}{2 \cdot 5^q} \ge \frac{1}{2 \cdot 5^q}, \qquad s_J \ge \frac{1}{2 \cdot 5^{J-1}}.$$

Substitution in (i) gives

$$\frac{\zeta 8^{-J}}{32 J^2} \frac{1}{4 \cdot 25^{J-1}} = \frac{25\zeta}{128 \cdot 200^J J^2},$$

which proves the claimed rational certificate. The phase here is the Bloch rotation, as in (S7.1); its unitary uses the half-angle. No angle is squared in this calculation.

For (iii), $\sin(\pi t) \ge 2t$ for $0 \le t \le 1/2$ gives, for $1 \le q < J$,

$$\frac{1-\cos(2\pi q\omega)}{2} = \sin^2(\pi q\omega) \ge 4\|q\omega\|_{\mathbb{R}/\mathbb{Z}}^2 \ge \frac{4c^2}{q^2} \ge \frac{4c^2}{J^2}.$$

Part (i) then yields (15.10).

For fixed B, the sufficient condition in (ii) holds with $J = \log_{200}(1/\delta) - O_B(\log\log(1/\delta))$ chosen as an integer slightly below this expression. The corresponding choice in (iii) is $J = \log_8(1/\delta) - O_{B,c}(\log\log(1/\delta))$. Since $a(J) \ge \log_2 J$, both imply (15.11). The asserted uppers are the unchanged orbit-prefix constructions of Theorem S9.4 and its final algebraic-phase paragraph, which are also admissible in the unrestricted class. □

In fact the exact integer lower $a(J)$ retains a second-order term: the central-binomial estimate $\binom{m}{\lfloor m/2\rfloor} = \Theta(2^m/\sqrt{m})$ gives $a(J) = \log_2 J + \frac{1}{2}\log_2\log_2 J + O(1)$. This is a refinement of the support certificate, not an assertion of the optimal asymptotic simulator size. The lower uses terminal projector requests, whereas the earlier exact infinite-memory obstruction already used report words alone. These are different sets of operational probes. No positive extra initializer charge is deduced for the unrestricted approximation problem.

Corollary 15.4 (Every non-root algebraic phase has the same lower order)**.** *Let z be algebraic with $|z| = 1$ and not a root of unity. Let its degree be D, choose an integer $A \ge 1$ such that Az is an algebraic integer, and enumerate its conjugates as $z = z_1, z_2, \dots, z_D$. Choose any supplied real bound*

$$H \ge \max\left\{2, A^D \prod_{j=2}^{D} \max(1, |z_j|)\right\}.$$

At that fixed phase, for either start convention,

$$0 < \delta \le \frac{\zeta H^4}{32 \cdot 16^D\,(8H^4)^J J^2} \quad\Longrightarrow\quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{15.12}$$

In particular (15.11) *holds. No badly-approximable premise is necessary for this lower order.*

Proof. The number $A^q(z^q - 1) = (Az)^q - A^q$ is a nonzero algebraic integer. Its norm is a nonzero integer, so

$$1 \le A^{qD}|z^q - 1| \prod_{j=2}^{D} |z_j^q - 1| \le 2^{D-1} H^q |z^q - 1|.$$

Because $|z| = 1$,

$$\frac{1 - \operatorname{Re} z^q}{2} = \frac{|1 - z^q|^2}{4} \ge 4^{-D} H^{-2q}.$$

It follows that $s_J \ge 4^{-D} H^{-2(J-1)}$. Substitute this in (15.8) to obtain (15.12). A choice $J = \log_{8H^4}(1/\delta) - O_{B,H,D}(\log\log(1/\delta))$ then gives the claimed lower order. This elementary algebraic norm bound does not supply the stronger $s_J \asymp J^{-2}$ bound used for the sharp physical-dictionary logarithmic law. □

16. The exact cost of initializing a stationary response

The distinction between a freely chosen initial distribution and a prescribed initial label is not peculiar to a periodic example. There is an exact general law. Its proof is an elementary elimination in the space of future response functions, within the classical positive realization framework [48, 47, 76]. It does not require hidden states to lie in the minimal linear realization, or require their unobserved transition matrix to be irreducible.

Let A be a nonempty finite command alphabet, with a finite report set Y_a for each $a \in A$. A finite classical realization consists of nonnegative matrices $T_{a,y}$ such that

$$T_a := \sum_{y \in Y_a} T_{a,y}, \qquad T_a \mathbf{1} = \mathbf{1},$$

and an initial probability row μ. For a command–report word $w = (a_1, y_1) \cdots (a_n, y_n)$ write $T_w = T_{a_1,y_1} \cdots T_{a_n,y_n}$ and $p(w) = \mu T_w \mathbf{1}$. Optional terminal readouts have decoder columns $d_{b,z} \geq 0$, with $\sum_z d_{b,z} = \mathbf{1}$, and response $p(w; b, z) = \mu T_w d_{b,z}$. All following equalities include every specified terminal readout when these are part of the experiment.

Call the prepared response *predictively stationary under* ν if ν is a probability row on A with every $\nu(a) > 0$ and

$$\sum_{a \in A} \nu(a) \sum_{y \in Y_a} p((a, y)w) = p(w) \qquad \text{for every future word } w. \tag{16.1}$$

When terminal readouts are specified, additionally require

$$\sum_{a \in A} \nu(a) \sum_{y \in Y_a} p((a, y)w; b, z) = p(w; b, z) \qquad \text{for all } w, b, z, \tag{16.2}$$

including the empty word w. This is an equality of observable responses; it does not assume $\mu \sum_a \nu(a) T_a = \mu$ in a particular hidden realization. For one command and no terminal readout it is ordinary stationarity. A response has a one-label realization precisely when its report probabilities factor as $p(w) = \prod_j p_{a_j}(y_j)$, and every terminal readout has a fixed distribution independent of the preceding word. For one command this is the i.i.d. case.

If a stationary hidden row is desired, it can be chosen without adding labels. Put $\overline{T} = \sum_a \nu(a) T_a$. Every $\mu \overline{T}^j$ has the same future response by the displayed stationarity identities. A subsequential limit of the Cesàro averages $H^{-1} \sum_{j=0}^{H-1} \mu \overline{T}^j$ is stationary for $\overline{T}$ and retains that response. Thus the freely mixed minimum below agrees with the convention requiring a stationary hidden row for the averaged command, and with the usual stationary HMM convention when there is only one command.

Let $n_{\mathrm{mix}}(p)$ be the minimum number of labels with an arbitrary initial probability row, and let $n_{\mathrm{pure}}(p)$ be the minimum when one prescribed initial label is counted. The minima are $+\infty$ when there is no finite realization.

Theorem 16.1 (Stationary initialization law)**.** *Suppose* (16.1) *holds, together with* (16.2) *when terminal readouts are included. If the response has a one-label realization, both minima are one. Otherwise*

$$n_{\mathrm{pure}}(p) = n_{\mathrm{mix}}(p) + 1, \qquad (+\infty) + 1 := +\infty. \tag{16.3}$$

The lower holds for all real nonnegative transition tables, including reducible and nonminimal linear realizations. The upper processes the first actual command while entering the core; it neither discards a physical call nor inserts an uncounted preliminary randomization.

Proof. Take any finite pure-start realization, with initial label o, and let f_i denote the complete future response from hidden label i. Thus $f_i(w) = e_i^{\mathsf{T}} T_w \mathbf{1}$, with the analogous values for terminal readouts. Put $\overline{T} = \sum_a \nu(a) T_a$. Stationarity gives the functional identity

$$f_o = \sum_j \overline{T}_{oj} f_j. \tag{16.4}$$

If $\overline{T}_{oo} = 1$, nonnegativity and full support of ν imply $(T_a)_{oo} = 1$ for every a. All report transitions from o then return to o. Their probabilities, and the terminal readouts at o, constitute a one-label realization. In the case under consideration this is excluded. Hence

$$\beta_j = \frac{\overline{T}_{oj}}{1 - \overline{T}_{oo}} \quad (j \neq o)$$

is a probability row and $f_o = \sum_{j \neq o} \beta_j f_j$.

Delete label o. On the remaining labels define

$$\widehat{T}_{a,y}(i,j) = T_{a,y}(i,j) + T_{a,y}(i,o)\beta_j, \qquad i,j \neq o, \tag{16.5}$$

and retain their terminal decoders. These matrices are nonnegative and their sums over y are stochastic. The recursion $f_i((a,y)w) = \sum_j T_{a,y}(i,j) f_j(w)$, with the functional identity for f_o substituted on its right side, is exactly the recursion for (16.5). Induction on word length, starting with the empty word and each terminal decoder, proves that every retained f_i is unchanged. Initial row β therefore realizes f_o with one fewer label. This proves $n_{\mathrm{mix}} \leq n_{\mathrm{pure}} - 1$ whenever a finite pure realization exists.

Conversely, given a k-label mixed-start realization $(\mu, T_{a,y})$, append one label o. Its report transition on command a enters the core with row $\mu T_{a,y}$ and never returns to o; the core retains its original transitions. A terminal readout at o has row $\mu d_{b,z}$. The first physical command is already included in this row, so induction gives the original law at every word length, including zero. This uses $k+1$ labels and proves the reverse inequality. A finite pure realization is itself a finite mixed one, so the two finiteness assertions are equivalent. The one-label exception is immediate. □

Corollary 16.2 (Stationary processes)**.** *For every stationary finite-alphabet process that is not i.i.d., the minimum pure-start edge-emitting hidden Markov realization has exactly one more label than the minimum freely mixed realization, with the same infinite-value convention. In particular, the q versus $q+1$ conclusion of Theorem S15.1 is an instance of a general stationary principle, while its phase-dependent q remains a separate spectral conclusion.*

Equality of these complete response functions also preserves the law under every common randomized policy for commands, stopping and terminal readout. For each finite stopped history, multiply the equal response by the same policy probabilities. Summation gives all finite stopping events, and decreasing continuation events give the nontermination probability. Consequently all nonnegative costs measurable from that record have equal expectations, including infinite expectations. These are process-realization label counts. If the external policy has its own persistent memory, that memory is counted separately in a closed-loop implementation; it is not supplied free by the theorem.

17. The causal task-order transfer theorem

Suppose that every terminating state-stopped competitor reaches at least $B_0 \geq 1$ live labels under θ_*. Such a certificate must concern actual physical command compatibility, not printed action names. Define

i	$m_-(M)$	$m_+(M)$
cp	M	$\lfloor M/D_0 \rfloor$
ev	M	$\lfloor M/D_0 \rfloor$
st	$M - B_0$	$\lfloor (M-B)/D_0 \rfloor$.

(17.1)

The event upper also requires $M \geq B$. The trivial state-stopped lower $B_0 = 1$ is valid for a procedure initially waiting to make a call.

Theorem 17.1 (Affine task-order transfer with a paid reference and common construction)**.** *Assume all the prepared-kernel, full-report, task-order, effort, occupation, resource and common-construction hypotheses of Section 2. Fix N, M, put $b_- = N/\kappa_-$, $b_+ = N/\kappa_+$, and suppose $b_- \geq 1$ and $m_-(M) \geq 1$. Set*

$$t_N = (1 - A_N)_+ / u_-.$$

If $t_N > 0$, every one of the three risks satisfies

$$R_i^{\Theta}(N,M) \geq \left[l_- t_N \mathcal{Q}_{m_-(M)}(b_-/t_N) - E_N \right]_+. \tag{17.2}$$

If $t_N = 0$, the asserted lower is zero, with no evaluation of $\mathcal{Q}$ at infinity. An empty legal class still has risk $+\infty$.

Suppose also $m_+(M) \geq 1$, $b_+ \geq 1$, and, for event stopping, $M \geq B$. If

$$\beta_N = \frac{l_+ b_+}{1 + b_+ a_+} \geq 1, \tag{17.3}$$

one common Borel program with actual expected cost at most N in every model attains

$$R_i^{\Theta}(N, M) \leq \frac{u_+(1 + b_+ a_+)}{l_+} \mathcal{Q}_{m_+(M)}(\beta_N) + b_+ e_+. \tag{17.4}$$

It commits almost surely. If $\beta_N < 1$, the always-accept first-attempt construction instead gives $R_i^{\Theta}(N, M) \leq u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$ at cost at most $\kappa_+ \leq N$.

The lower is against every legal competing strategy and uses one actual witness. The upper is one parameter-independent strategy for all models. The two effort constants need not agree. Neither direction assumes that the comparison experiment belongs to Θ or that a normalized predictive update is uniformly contracting.

Corollary 17.2 (Scale-uniform relative transfer)**.** *Under Theorem 17.1, suppose $A_N = E_N = a_+ = e_+ = 0$. If $l_+ b_+ \geq 1$, then*

$$\frac{l_-}{u_-} \mathcal{Q}_{m_-(M)}(u_- b_-) \leq R_i^{\Theta}(N, M) \leq \frac{u_+}{l_+} \mathcal{Q}_{m_+(M)}(l_+ b_+). \tag{17.5}$$

The factors $l_\pm, u_\pm$ may be fixed nonunit constants, independent of budget and resolution. In particular no term proportional to $b_\pm$ times a fixed absolute calibration error is required in this relative-order class.

Proof. Substitute $t_N = 1/u_-$ and $\beta_N = l_+ b_+$ in the theorem. □

Corollary 17.3 (Separate exact checkpoint and autonomous laws)**.** *Suppose the certificates are exact: $l_\pm = u_\pm = 1$, all defects vanish, $\kappa_- = \kappa_+ = \kappa$, and exports are the identity. Then, for $N \geq \kappa$,*

$$R_{\mathrm{cp}}^{\Theta}(N, m) = \mathcal{Q}_m(N/\kappa).$$

The constructed autonomous upper uses $B + m$ state-stopped labels, or $\max(B, m)$ event-stopped labels. If the mandatory live count is sharp, $B_0 = B$, then

$$R_{\mathrm{st}}^{\Theta}(N, B + m) = \mathcal{Q}_m(N/\kappa) \quad (m \geq 1), \tag{17.6}$$

$$R_{\mathrm{ev}}^{\Theta}(N, M) = \mathcal{Q}_M(N/\kappa) \quad (M \geq B). \tag{17.7}$$

The subtraction by B belongs only to the state-stopped model. Nonidentity exports require the expanded dictionary in (17.1)*, unless an optimal dictionary is invariant under all exports. These exact stopping distinctions are inherited from the complete companion and are reproved here to fix the resource semantics.*

Proof. The same lower and upper profile coincide in each asserted interface. Identity exports give $D_0 = 1$. The packing and converse widths are proved in Lemma 18.4. No lower on live states is used to forbid event-stopped reuse. □

The familiar absolute-defect theorem is also a specialization: take $l_\pm = u_\pm = 1$ while retaining A_N, E_N, a_+, e_+. Thus relative and absolute calibration are composed in one statement. The chosen metric is the common scored task throughout; it is not changed between the converse and the construction.

18. Proof of the central theorem

Lemma 18.1 (Profile attainment)**.** *For every finite $m \geq 1$ and $b \geq 1$, the minimum in* (2.2) *is attained. The profile is nonincreasing in m, b and left-continuous in $b > 1$.*

Proof. Write a dictionary as an ordered tuple in the compact set H^m, permitting repetitions. The feasible densities are a weak-star closed subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are each closed under pairing with $L^1(\nu)$. They are therefore weak-star compact. As the tuple varies, its squared-distance function varies uniformly on H, since all points belong to a bounded set. Along a convergent product net, split the objective difference into this uniform variation and the weak-star pairing against the fixed bounded measurable loss at the limit tuple. This proves continuity of the objective and attainment. Nets avoid an unwarranted sequential compactness assumption. Monotonicity follows from inclusion of feasible sets. For left continuity take an optimizer h at $b > 1$ and let $h_\lambda = (1-\lambda)h + \lambda$. It has unit mass and cap $b_\lambda = (1-\lambda)b + \lambda < b$. Its objective tends to the optimum as $\lambda \downarrow 0$, while monotonicity bounds the optimum at b_λ between the optimum at b and this objective. □

Lemma 18.2 (Actual-history comparison measure)**.** *Fix any robustly admissible competitor of cost at most N, with its fixed terminal dictionary C. There are finite nonnegative measures ξ_c on X, $c \in C$, such that $\xi = \sum_c \xi_c$ obeys*

$$\xi \leq (\mathbb{E}_* A_\tau)\nu \leq b_- \nu, \tag{18.1}$$

$$1 \leq u_- \xi(X) + A_N, \tag{18.2}$$

$$\mathbb{E}_* \|Z - \widehat{r}\|^2 \geq l_- \sum_{c \in C} \int d_c \, d\xi_c - E_N. \tag{18.3}$$

The measure ξ is not asserted to be the terminal law of any comparison execution.

Proof. Adjoin an independent uniform random variable to the current report on both sides. This is current-call randomness, not a seed retained between calls; it is sufficient to represent the competitor's randomized output and stopping decision. A finite-output kernel has such a representation by cumulative probabilities. At a fixed reached productive history h and chosen command a, the post-report decision then gives an acceptance indicator $q_h(y)$ and an output $r_h(y) \in C$ on acceptance. Give r_h an arbitrary fixed value on rejection. The notation y now includes the current independent random variable. Integrating the original certificates over that variable preserves their constants. Dependence on the entire earlier history is permitted in these functions.

For each Borel $D \subset X$, set

$$\xi_c(D) = \mathbb{E}_* \sum_{j \leq \tau:\ j \ m\text{productive}} \int \mathbf{1}_{\{\pi_{H_{j-1},a_j}(y) \in D\}} q_{H_{j-1}}(y) \mathbf{1}_{\{r_{H_{j-1}}(y) = c\}} P^0_{H_{j-1},a_j}(dy, H). \tag{18.4}$$

The outer expectation is the actual pre-call occupation at θ_*. All terms are jointly measurable and nonnegative. Tonelli defines the measures, whose total is finite because $\mathbb{E}_* A_\tau \leq b_-$. Summing over c and dropping the acceptance indicator gives (18.1): each comparison row has mark law ν.

The analogous sum with the actual rows instead of P^0 has mass exactly one, because the actual procedure commits exactly once almost surely. Apply (2.4) at each actual reached history, then sum. The occupation certificate yields (18.2). Apply (2.5) in the same way to the disjoint actual commitment events. Their expected losses sum to the actual risk. The comparison contribution is $d_{r_h(y)}(\pi_{h,a} y)$ by (2.3); this identity conditions on the full current report and therefore does not overlook extra information seen by the competitor. The accumulated loss defect is at most E_N, giving (18.3). One may first truncate the number of calls. Monotone convergence for the nonnegative contributions and the finite defect bounds justify the limit.

No actual past is replaced by a comparison past. In particular the competitor need not terminate, or obey a cost bound, if hypothetically run under the comparison experiment. This distinction is why the argument applies when that experiment is not an operating model. □

Remark 18.3 (Why the full report is required). A constant mark can conceal a visible fair bit which also equals the future audit. A comparison using only that mark has $f = 0, w = 1$, but a two-value decoder reading the bit predicts it with zero loss. The hypotheses therefore condition on the full report seen by the competitor; matching a compressed mark marginal alone cannot establish the lower.

Lemma 18.4 (Live roles, output roles and physical compatibility)**.** *A program with ℓ live roles and t output roles has a state-stopped implementation with $\ell + t$ labels and an event-stopped implementation with $\max(\ell, t)$ labels, preserving every path cost and complete stopped scored law. Conversely an event-stopped M-label program has at most M decoder values. A state-stopped M-label program with at least B_0 reached live labels has at most $M - B_0$ terminal decoder values.*

For a physical live-state certificate, consider finitely many mandatory pre-call contexts I with no free context input. Suppose the commands legal at a history of context i are exactly $\mathcal{U}_i$, and every terminating competitor visits every context with positive probability under the same witness. This positive probability may depend on the competitor; no uniform lower bound over competitors is required. Let b_{phys} be the least size of a cover of I by subsets J with $\bigcap_{i\in J} \mathcal{U}_i \neq \varnothing$. Then at least b_{phys} live labels are required. This certificate is invariant under duplicate names and bijective stateless recodings of the same physical commands.

Proof. For state stopping take disjoint copies of the live and terminal role sets and type them zero and one. For event stopping embed the live and terminal role sets separately into one alphabet of size $\max(\ell, t)$. On continuing transitions use the live embedding; on commitment emit the final event and use the terminal embedding. A command row is read only on a continuing branch, while the decoder is read only after commitment. Sharing their label therefore causes no conflict and uses no continuing mode bit. Couple the original and recoded coins. Every finite stopped prefix and every path cost agree, and almost-sure termination gives equality of the stopped scored laws. A fixed decoder has one value per retained label. In state stopping the global halt map excludes each reached live label from the output set, proving its subtraction. Event stopping has no such exclusion.

A live label reached in contexts J must have one command distribution supported on all the corresponding $\mathcal{U}_i$. Since J is finite, the union of the exceptional null command sets is null; hence their intersection supports a probability and is nonempty. The sets of contexts served by the labels cover I, proving the bound. Conversely a nonempty intersection contains a command and hence a Dirac command distribution. Replacing names without changing executed kernels, legal domains, costs and reports does not change these intersections. A translator which itself remembers a hidden phase is not such a recoding; its memory must be charged. □

Proof of Theorem 17.1. Fix a competing program. Lemma 18.4 bounds its decoder width by $m_-(M)$. Let $s = \xi(X)$ in Lemma 18.2. If $t_N > 0$, then (18.2) gives $s \geq t_N$. Scale the measure to $\xi' = (t_N/s)\xi$. It has mass t_N and remains bounded above by $b_-\nu$. Thus $h = (d\xi'/d\nu)/t_N$ has unit mass and cap b_-/t_N. Nearest-dictionary projection and nonnegativity imply

$$\sum_{c\in C} \int d_c \, d\xi_c \geq \int [w + \operatorname{dist}(f, C)^2] \, d\xi \geq t_N \mathcal{Q}_{m_-(M)}(b_-/t_N).$$

Together with (18.3) this proves the asserted lower for this competitor. Its worst-model risk dominates its risk at θ_*. Taking the infimum over competitors gives (17.2), without exchanging an infimum and a supremum. Nonnegative loss gives the zero truncation and the stated case $t_N = 0$.

For the upper assume (17.3). Choose an attained optimizer (C, h) at width $m_+(M)$ and cap β_N. Let $r_C(x)$ be the first nearest point to $f(x)$ in a fixed ordering of C. After each attempt of $\mathcal{G}$, accept x with probability $h(x)/\beta_N$, export $T_J r_C(x)$ and commit; otherwise return to the start routing label. All these operations are Borel. The choice J is used within the current attempt; any part retained thereafter is charged through the live role or exported decoder, not a free seed. The exports are assumed jointly measurable and take at most $D_0|C|$ distinct values, so their finite decoder and transitions are Borel as well.

Conditional on every surviving attempt history, the acceptance probability is at least

$$p = \frac{l_+}{\beta_N} - a_+ = \frac{1}{b_+} > 0. \tag{18.5}$$

Consequently the probability of reaching attempt $j+1$ is at most $(1-p)^j$, by induction on conditional survival probabilities. In particular commitment is almost sure and the expected number of attempts is at most $1/p = b_+$. Each begun attempt has conditional expected cost at most κ_+, not merely a cost conditional on a favorable outcome. Summing the predictable attempt costs gives actual expected cost at most $\kappa_+ b_+ = N$.

The conditional unnormalized loss contributed by an attempt is at most $u_+ \mathcal{Q}_{m_+(M)}(\beta_N)/\beta_N + e_+$. Sum its contributions on the disjoint terminal events, using the same survival bound. The result is at most

$$b_+ \left[\frac{u_+}{\beta_N} \mathcal{Q}_{m_+(M)}(\beta_N) + e_+ \right],$$

which is (17.4). The program has B live roles and at most $D_0 m_+(M)$ terminal roles. Lemma 18.4 gives exactly the sufficient widths in (17.1). A checkpoint counts only the terminal width. No physical renewal law was used: the conditional hazard controls survival even if the hidden state after rejection differs arbitrarily from its former state.

If $\beta_N < 1$, commit after the first attempt, and use an optimizer at cap one. The loss certificate with $q = 1$ gives $u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$, and the whole-attempt cost is at most κ_+. This proves all remaining assertions. □

19. Scope and further mathematical questions

The general stability theorem concerns finite-dimensional instruments with mandatory success stopping, a fixed hidden model, a common positive mean cost floor, and a cost guarantee for every initial density matrix. It allows noncommuting failure maps and changing reachable supports. The restarted controller has the same command rows and label budget; its physical state is not reset. The square-root exponent is optimal on this general class by Theorem 5.1; the exact leading coefficient belongs to its displayed scalar family. No optimality of the numerical constant in the general upper bound is inferred. The exact support certificates, finite-model synthesis and compactness argument reuse inherited theory. The resource-preserving perturbation argument extends the complete cost frontier uniformly across all finite budgets. Its sharpness proof uses a positive-polynomial boundary inequality, quantitative rigidity and a matched upper and lower asymptotic over all six controller parameters. Neither the golden-ratio example nor semialgebraic effectivity is a closed classification of every instrument. The unrestricted-history value is the limit of finite-memory values; attainment of that limit at finite memory is characterized here for every finite number of imperfect search locations with arbitrary positive costs. That characterization does not extend by assertion to arbitrary instruments.

For finite-location imperfect search, rational commensurability of all failure logarithms completely distinguishes finite saturation from strict gaps at every finite memory. The greedy ordering, interior preparation saddle, consistent-priority mixtures and common-power cycles are classical search facts. The projected class-word obstruction rules out every stochastic finite controller when more than one commensurability class is present; it does not infer rational frequencies for general finite

Markov chains. Counted pure-start initialization, a primitive least-favourable prior bound, an exponential memory upper rate and an effective rational synthesis cap complete the implementation chain. Arbitrary positive command costs are allowed. The supplement retains sharper normalized two-location constants. Neither rate is a matching lower bound or a minimal-label formula. The arithmetic criterion concerns exact supplied primitives; finite decimal orbit samples cannot decide it. Positive command costs and strictly imperfect detection are actual assumptions of this search classification. Rational nonsaturating examples do not imply transcendence of the unrestricted optimum.

The shared-preparation normal form applies to every finite group, every finite preparation and outcome alphabet, and closed convex wordwise specifications. The input controller need not satisfy group relations. The invariant polytope may extend beyond the target response span. The exact low-contrast formula requires a common mean law; the displayed nonnegative-rank counterexample explains why this hypothesis cannot simply be removed. These are shared-device results with one fixed decoder, not a claim that independently optimized devices can always share their memory at no cost.

The reset frontier now allows positive semidefinite failure effects and classifies every live-label budget, with arbitrary ties. Its normal form uses the all-initial-state quantifier, a fixed row program, reported resetting, a positive cost floor, and mandatory success stopping. It does not apply unchanged to a restricted initial prior, nonreset back-action, extra observation ports or optional commitment. Positive exposure remains necessary for the separate simple transversal formula; the general frontier correctly includes its semidefinite counterexample. The old first-residual certificate remains a lower bound, and the exact hierarchy exhibits a strict factor-two difference from the full cost gap. Algebraic computability is not an efficient algorithm, and the inherited covering reduction prevents a general polynomial-time memory-synthesis claim unless the usual complexity obstruction is resolved.

The approximation lower permits arbitrary classical terminal decoder tables but requires delayed terminal test requests and no uncounted access to the preparation history. At fixed positive target budget and the original algebraic phase it leaves a gap between logarithmic-in-logarithm and logarithmic memory growth. The sharper matching logarithmic physical-dictionary law retains its Born-readout and badly-approximable hypotheses. Neither law is the original general-M squared-sign minimax-risk frontier. The original squared-sign risk frontier and the intermediate phase-type timing frontier remain distinct mathematical problems. The present general cost theorem admits nonreset feedback and fixed unknown models, but does not supply a closed risk formula, optional commitment, or a finite-memory saturation theorem for every experiment.

The inherited mean–contrast law is exact for its displayed growing symmetric-group family, and the general finite-group error principle is an attained finite optimization. No closed formula for every group or universal polynomial-time synthesis is inferred. The return-polytope construction concerns explicitly verified conditional branches, not average mixing. Its target-cost bound does not control the simulator's unbounded expected duration. The stationary-initialization theorem concerns exact predictively stationary laws; an approximate truncation does not automatically inherit its extra-label lower bound.

All original causal, predictive-state, information-compression, finite resource, finite-order measurement, two-stage reset, rank-constrained discrimination, communication, dual, spectral and geometric developments retain their hypotheses. Local geometry remains local. Independent A1, A2 and further Theta construction problems require their own arguments. Program bits, arithmetic precision, working storage and sampling work are not silently identified with persistent label counts. The paired supplement contains complete technical and independent proofs; the separately compilable R58 companion retains its own scope.

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